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Lightly Intrusive Reduced-Order Modeling of Darcy Flow with Corrective Source Term Approach

Master's thesis in Physics and Mathematics
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June 2026

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ABSTRACT

This thesis investigates reduced-order modeling techniques for parameterized transient Darcy flow problems. The objective is to construct computationally efficient surrogate models capable of approximating high-fidelity finite element solutions at significantly reduced computational cost.

A reduced basis framework based on Proper Orthogonal Decomposition (POD) and Galerkin projection is developed for both affine and non-affine parameter-dependent problems. For non-affine systems, a lightly intrusive polynomial approximation based on an L^2-fit is employed to recover an approximate affine structure. In addition, parameter-domain splitting is investigated to improve approximation accuracy in problems with strongly varying parameter dependence. The thesis also investigates the Corrective Source Term Approach (CoSTA), where reduced-order models are combined with data-driven corrections.

Numerical experiments are presented for two parameterized Darcy flow problems involving geometric and permeability-induced non-affinities in both stationary and transient settings. The results demonstrate that the proposed reduced-order models achieve substantial reductions in computational cost while maintaining good approximation accuracy. Furthermore, parameter-domain splitting and CoSTA-based corrections are shown to significantly improve the reduced-order approximations for challenging non-affine problems.

SAMMENDRAG

TODO: Norsk sammendrag

PREFACE

This thesis was written during the spring semester of 2026 as part of the five-year Master of Science programme in Physics and Mathematics at Norwegian University of Science and Technology. The work was carried out under the supervision of Trond Kvamsdal (NTNU & SINTEF), with co-supervision from Eivind Fonn (SINTEF).

The thesis builds upon the author's specialization project completed in autumn 2025 and continues the study of reduced-order modeling techniques. Parts of the theoretical background and methodology are based on and further developed from this previous work.

I would like to thank my supervisors for valuable guidance, discussions, and feedback throughout the project. I would also like to thank my father, Bjørn Kåre, and brother, Torstein, for proofreading parts of the thesis and for valuable feedback.

Henning Hegstad
Trondheim, June 2026

TABLE OF CONTENTS

Abstract **i**

Preface **iii**

Contents **v**

Abbreviations **vi**

1 Introduction* **1**

1.1 Project description 2

1.2 Relation to specialization project 2

2 The Finite Element Method **3**

2.1 Model Problem: Transient Darcy Flow 3

2.2 Weak Formulation 4

2.3 Finite Element Discretization 6

2.4 Temporal Discretization 7

2.5 Manufactured solution 7

3 Reduced-Order Modeling* **9**

3.1 Parameterized Full-Order System 9

3.2 Reduced Basis Method 10

3.2.1 Offline stage 10

3.2.2 Online Stage 14

3.2.3 Error Analysis 15
3.3 Affinization of Reduced Basis Methods 16
3.3.1 L^2-fit then reduce 17
3.4 Parameter Domain Splitting 20

4 Corrective Source Term Approach **21**
4.1 Conceptual Formulation 21
4.2 Discrete Formulation 22
4.2.1 Integration with the Reduced-Order Model 24

5 Numerical Experiments **27**
5.1 Manufactured Solution 28
5.1.1 Convergence Results 28
5.2 Square with moving corner 33
5.2.1 Finding discretization 34
5.2.2 Stationary 34
5.2.3 Transient 41
5.3 Square with Gaussian permeability 47
5.3.1 Finding discretization 48
5.3.2 Stationary 49
5.3.3 Transient 66

6 Conclusions **83**

References **85**

Appendices: **89**

Chapters marked with an asterisk (*) contain substantial material adapted and further developed from the author's specialization project completed in autumn 2025.

ABBREVIATIONS

- **CoSTA** Corrective Source Term Approach
- **DNN** Deep Neural Network
- **DDM** Data Driven Model(ing)
- **DoFs** Degrees of Freedom
- **EIM** Empirical Interpolation Method
- **EOC** Estimated Order of Convergence
- **FEM** Finite Element Method
- **FOM** Full-Order Model
- **HAM** Hybrid Analysis and Model(ing)
- **HF** High-Fidelity
- **LU** Lower–Upper
- **NTNU** Norwegian University of Science and Technology
- **PBM** Physics-Based Model
- **PDE** Partial Differential Equation
- **POD** Proper Orthogonal Decomposition
- **RB** Reduced Basis
- **ROM** Reduced-Order Model(ing)
- **SVD** Singular Value Decomposition

CHAPTER

ONE

INTRODUCTION*

The solution of parameterized partial differential equations (PDEs) using classical high-fidelity (HF) methods, such as the finite element method (FEM), can be prohibitively expensive when repeated solutions are required for multiple values of the input parameters. For example, each solve of a large-scale FEM system (often with thousands of degrees of freedom) may take substantial time, making tasks real-time simulation, optimization, or uncertainty quantification impractical. Reduced-order modeling (ROM) addresses this challenge by constructing a low-dimensional surrogate that approximates the PDE solution at significantly reduced computational cost. In particular, the reduced basis (RB) method exploits an *offline stage* (basis construction done only once) and an *online stage* (rapid evaluation), enabling much faster solves for new parameter values [1, 2, 3, 4]. In many-query or real-time contexts, the expensive offline cost becomes negligible compared to the fast online evaluation [5]. Overall, the RB method can achieve substantial computational savings relative to solving the high-fidelity (HF), or full-order model (FOM), for each parameter instance.

In this thesis, the FEM solution is used as the full-order model, and a ROM is constructed from it. The model problem is parameterized transient Darcy flow, which is a standard model for flow in porous media and arises in applications such as groundwater flow and petroleum reservoir simulation; see, e.g., [6]. Reduced basis methods have previously been applied to Darcy and related porous-media flow problems, including saddle-point formulations for groundwater flow [7] and coupled Stokes–Darcy systems [8]. (TODO: Flere kilder her? Hvilke skal jeg ha med? Alle? Mass conservation [9]. ROM for transient groundwater flow [10]. "Reduced-Order Modeling of Subsurface Multi-phase Flow Models Using Deep Residual Recurrent Neural Networks" - [11])

We address the technical issue of non-affine parameter dependence in Section 3.3. Specifically, when the parameter dependence of the system matrices is non-affine, additional techniques are required. In this work, we will present and apply a L^2-projection method from [12]. In this method, only access to the system matrices and solutions is required.

We also investigate the Corrective Source Term Approach (CoSTA) introduced in [13, 14]. CoSTA is a Hybrid Analysis and Modeling (HAM) approach that combines physics-based modeling (PBM) with data-driven modeling (DDM). In this work, CoSTA is applied to improve the accuracy of the reduced-order approximations, particularly in the presence of strongly non-affine parameter dependence.

Numerical experiments are presented for two parameterized Darcy flow problems involving geometric and permeability-induced non-affinities. We investigate approximation errors, convergence behavior, computational speed-up, and the influence of parameter-domain splitting and CoSTA corrections on the reduced-order models.

1.1 Project description

The objective of this thesis is to develop and investigate reduced-order modeling techniques for parameterized transient Darcy flow problems. The work combines finite element discretization, reduced basis methods, and lightly intrusive polynomial approximation techniques for non-affine parameter dependence.

Particular emphasis is placed on the approximation of problems with geometric and permeability-induced non-affinities. In addition, the Corrective Source Term Approach (CoSTA) is investigated as a data-driven correction strategy for improving reduced-order approximations.

The proposed methods are evaluated through numerical experiments with respect to approximation accuracy, convergence behavior, and computational efficiency.

1.2 Relation to specialization project

This thesis builds upon the author's specialization project completed in autumn 2025, which focused on reduced-order modeling techniques for parameterized PDEs. Several parts of the theoretical background and reduced basis methodology presented in chapter 1 and chapter 3 are based on and further developed from this previous work.

The present thesis extends the specialization project in several directions. Most notably, the focus has shifted from stationary elliptic model problems to parameterized transient Darcy flow formulated as a mixed finite element problem. In addition, the thesis includes the treatment of non-affine parameter dependence through parameter-domain splitting techniques, as well as the investigation of the Corrective Source Term Approach (CoSTA) for improving reduced-order approximations.

The numerical experiments, implementations, and analyses presented in this thesis are substantially expanded and constitute new work beyond the specialization project. Parts of the computational framework and implementation developed during the specialization project have also been reused and further extended in the present work.

CHAPTER

TWO

THE FINITE ELEMENT METHOD

Although the finite element method (FEM) is not strictly required for reduced-order modeling, it is widely used due to its flexibility, robustness, and solid theoretical foundation for the discretization of partial differential equations. In the context of reduced-order modeling, FEM typically serves as the high-fidelity, or full-order model (FOM), on which reduced-order models are constructed.

The purpose of this chapter is to introduce the finite element formulation of the governing equations and to establish the notation used throughout the thesis. The presentation follows standard references on finite element methods and mixed formulations; see, e.g., [15, 16, 17].

In short, the finite element method is based on reformulating a partial differential equation from its strong (differential) form into a weak (variational) form. This is achieved by testing the equations with suitable functions and applying integration by parts, which reduces the regularity requirements on the solution. The resulting weak formulation is expressed in terms of bilinear forms and linear functionals. Finally, the infinite-dimensional solution spaces are approximated by finite-dimensional subspaces, leading to a system of linear equations.

2.1 Model Problem: Transient Darcy Flow

We consider transient Darcy flow in a porous medium as a model problem. This system describes the evolution of pressure and velocity in porous materials and arises in, e.g., groundwater flow and subsurface transport [18].

Let $\Omega \subset \mathbb{R}^2$ be a bounded Lipschitz domain with boundary $\partial\Omega = \Gamma_p \cup \Gamma_{\mathbf{u}}$, where Γ_p and $\Gamma_{\mathbf{u}}$ are disjoint and Γ_p has positive measure.

The transient Darcy equations in mixed form read

$$\mathbf{K}^{-1}\mathbf{u} + \nabla p = 0 \qquad \text{in } \Omega \times (0, T], \tag{2.1}$$

$$S\,\partial_t p + \nabla \cdot \mathbf{u} = f \qquad \text{in } \Omega \times (0, T]. \tag{2.2}$$

Here $\mathbf{u}$ denotes the velocity, p the pressure, $f \in L^2(\Omega)$ a source term, $\mathbf{K}$ the permeability tensor, and $S > 0$ the storage coefficient. We assume that $\mathbf{K}$ is symmetric positive definite and uniformly bounded.

We impose the boundary conditions

$$p = p_D \quad \text{on } \Gamma_p \times (0,T], \qquad \mathbf{u} \cdot \mathbf{n} = 0 \quad \text{on } \Gamma_{\mathbf{u}} \times (0,T],$$

together with the initial condition

$$p(\mathbf{x}, 0) = p_0(\mathbf{x}) \quad \text{in } \Omega.$$

The boundary condition on Γ_p prescribes the pressure, while the condition on $\Gamma_{\mathbf{u}}$ corresponds to an impermeable (no-flow) boundary. In the homogeneous case, $p_D = 0$.

The mixed formulation treats both velocity and pressure as primary unknowns, allowing direct approximation of the velocity field.

2.2 Weak Formulation

To derive the weak formulation of the Darcy flow equations, we reformulate the problem so that it is satisfied in a weak sense, allowing solutions with reduced regularity. This is achieved by multiplying the equations with a test function and applying integration by parts, which transfers derivatives from the solution to the test function.

In this setting, the trial space consists of admissible solutions of the PDE, while the test space contains functions against which the equation is tested in the weak formulation. For the Darcy flow problem with homogeneous Dirichlet boundary conditions, both spaces naturally coincide. In the case of inhomogeneous Dirichlet conditions, the solution space would be an affine shift of the test space.

We therefore introduce the trial (and test) spaces

$$\begin{aligned}
V &:= \{\mathbf{v} \in H(\mathrm{div}; \Omega) : \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \Gamma_{\mathbf{u}}\} \\
&= \left\{\mathbf{v} \in [L^2(\Omega)]^2 : \nabla \cdot \mathbf{v} \in L^2(\Omega),\ \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \Gamma_{\mathbf{u}}\right\}, \\
Q &:= L^2(\Omega).
\end{aligned}$$

Here, the no-flow boundary condition $\mathbf{u} \cdot \mathbf{n} = 0$ on $\Gamma_{\mathbf{u}}$ is imposed strongly in the velocity space. The space V is equipped with the norm

$$\|\mathbf{v}\|^2_{H(\mathrm{div};\Omega)} = \|\mathbf{v}\|^2_{L^2(\Omega)} + \|\nabla \cdot \mathbf{v}\|^2_{L^2(\Omega)}.$$

Multiplying (2.1) by a test function $\mathbf{v} \in V$ and integrating over Ω, we obtain

$$\int_\Omega \mathbf{K}^{-1}\mathbf{u} \cdot \mathbf{v}\, dx + \int_\Omega \nabla p \cdot \mathbf{v}\, dx = 0.$$

Applying integration by parts to the pressure term yields

$$\int_\Omega \mathbf{K}^{-1}\mathbf{u} \cdot \mathbf{v}\, dx - \int_\Omega p\, \nabla \cdot \mathbf{v}\, dx + \int_{\partial\Omega} p\, \mathbf{v} \cdot \mathbf{n}\, ds = 0.$$

Since $\mathbf{v} \cdot \mathbf{n} = 0$ on $\Gamma_{\mathbf{u}}$ and $p = p_D$ on Γ_p, the boundary term reduces to

$$\int_{\partial\Omega} p\,\mathbf{v} \cdot \mathbf{n}\,ds = \int_{\Gamma_p} p_D\,\mathbf{v} \cdot \mathbf{n}\,ds.$$

Hence,

$$\int_{\Omega} \mathbf{K}^{-1}\mathbf{u} \cdot \mathbf{v}\,dx - \int_{\Omega} p\,\nabla \cdot \mathbf{v}\,dx = -\int_{\Gamma_p} p_D\,\mathbf{v} \cdot \mathbf{n}\,ds.$$

Similarly, multiplying (2.2) by $q \in Q$ and integrating over Ω gives

$$\int_{\Omega} S\,\partial_t p\,q\,dx + \int_{\Omega} q\,\nabla \cdot \mathbf{u}\,dx = \int_{\Omega} f q\,dx.$$

We define the bilinear forms

$$\begin{aligned} a(\mathbf{u}, \mathbf{v}) &\coloneqq \int_{\Omega} \mathbf{K}^{-1}\mathbf{u} \cdot \mathbf{v}\,dx, \\ b(\mathbf{v}, q) &\coloneqq \int_{\Omega} q\,\nabla \cdot \mathbf{v}\,dx, \\ m(p, q) &\coloneqq \int_{\Omega} pq\,dx, \end{aligned}$$

and the linear functional

$$\ell(q) \coloneqq \int_{\Omega} f q\,dx.$$

We also define the boundary functional

$$g_D(\mathbf{v}) \coloneqq -\int_{\Gamma_p} p_D\,\mathbf{v} \cdot \mathbf{n}\,ds.$$

The weak formulation reads: Find $(\mathbf{u}(t), p(t)) \in V \times Q$ such that

$$\begin{aligned} a(\mathbf{u}, \mathbf{v}) - b(\mathbf{v}, p) &= g_D(\mathbf{v}) \quad \forall \mathbf{v} \in V, \\ S\,m(\partial_t p, q) + b(\mathbf{u}, q) &= \ell(q) \quad \forall q \in Q. \end{aligned}$$

This formulation is referred to as a mixed formulation, since both the velocity and pressure are treated as unknowns.

The well-posedness of the mixed formulation follows from standard results for saddle-point problems. In particular, the bilinear forms $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$ are continuous, and $a(\cdot, \cdot)$ is coercive on $\ker(b)$.

A key requirement for stability is the inf-sup (Ladyzhenskaya–Babuška–Brezzi) condition: there exists a constant $\beta > 0$ such that

$$\inf_{q \in Q} \sup_{\mathbf{v} \in V} \frac{b(\mathbf{v}, q)}{\|\mathbf{v}\|_{H(\mathrm{div};\Omega)} \|q\|_{L^2(\Omega)}} \geqslant \beta.$$

2.3 Finite Element Discretization

To obtain a numerical approximation, we introduce finite-dimensional subspaces

$$V_h \subset V, \qquad Q_h \subset Q.$$

As in the continuous formulation, the no-flow condition $\mathbf{u} \cdot \mathbf{n} = 0$ on $\Gamma_{\mathbf{u}}$ is imposed strongly in the discrete velocity space V_h. The finite element approximation is obtained by a mixed Galerkin method, where the weak formulation is restricted to the subspaces V_h and Q_h.

The discrete solution (u_h, p_h) is sought in $V_h \times Q_h$. The finite element spaces must be chosen such that a discrete inf-sup condition holds, ensuring stability of the numerical approximation. A common choice is to use Raviart–Thomas spaces for the velocity and discontinuous piecewise polynomial spaces for the pressure; see, e.g., [17].

Let $\{\boldsymbol{\phi}_i\}_{i=1}^{N_{\mathbf{u}}}$ and $\{\psi_j\}_{j=1}^{N_p}$ denote bases for the velocity and pressure spaces. The discrete solutions can be expanded as

$$\mathbf{u}_h = \sum_{i=1}^{N_{\mathbf{u}}} u_i \boldsymbol{\phi}_i, \qquad p_h = \sum_{j=1}^{N_p} p_j \psi_j.$$

We define the coefficient vectors

$$\mathbf{u} = [u_1, \ldots, u_{N_{\mathbf{u}}}]^T, \qquad \mathbf{p} = [p_1, \ldots, p_{N_p}]^T.$$

Substituting these expansions into the weak formulation yields the semi-discrete system

$$\begin{aligned} \mathbf{M}_h \mathbf{u} - \mathbf{B}_h^T \mathbf{p} &= \mathbf{G}_h, \\ \mathbf{B}_h \mathbf{u} + S \mathbf{M}_{p,h} \dot{\mathbf{p}} &= \mathbf{F}_h. \end{aligned}$$

Here the matrices are defined as

$$\begin{aligned} [\mathbf{M}_h]_{ij} &\coloneqq a(\boldsymbol{\phi}_j, \boldsymbol{\phi}_i), \\ [\mathbf{B}_h]_{ij} &\coloneqq b(\boldsymbol{\phi}_j, \psi_i), \\ [\mathbf{M}_{p,h}]_{ij} &\coloneqq m(\psi_j, \psi_i), \\ [\mathbf{F}_h]_i &\coloneqq \ell(\psi_i), \\ [\mathbf{G}_h]_i &\coloneqq g_D(\boldsymbol{\phi}_i). \end{aligned}$$

To apply the finite element method, the domain Ω is discretized into a mesh of non-overlapping elements. In two dimensions, the mesh typically consists of triangular elements, although quadrilateral or mixed meshes may also be used.

The finite element spaces V_h and Q_h are constructed by associating basis functions with a set of degrees of freedom (DoFs), which correspond to the unknown coefficients of the discrete problem. The precise location and interpretation of these DoFs depend on the choice of finite element spaces. For example, in standard nodal finite elements they are associated with mesh vertices, whereas in mixed finite element methods, such as those using Raviart–Thomas elements, they are typically associated with normal fluxes across element interfaces.

The quality of the mesh, in particular the element size and shape regularity, directly influences both the accuracy of the approximation and the efficiency of the numerical solution. The characteristic mesh size is denoted by h, defined as the maximum diameter of the elements in the mesh. The dimensions of the discrete spaces are denoted by $N_{\mathbf{u}} = \dim(V_h)$ and $N_p = \dim(Q_h)$, and the total number of degrees of freedom is $N_h \coloneqq N_{\mathbf{u}} + N_p$.

2.4 Temporal Discretization

To obtain a fully discrete problem, the time interval is partitioned into N_t steps with time levels $t^n = n\Delta t$, where Δt denotes the time step.

Applying the implicit Euler method [19] to the semi-discrete system

$$\begin{aligned} \mathbf{M}_h \mathbf{u} - \mathbf{B}_h^T \mathbf{p} &= \mathbf{G}_h, \\ \mathbf{B}_h \mathbf{u} + S\mathbf{M}_{p,h} \dot{\mathbf{p}} &= \mathbf{F}_h, \end{aligned}$$

and approximating the time derivative as

$$\dot{\mathbf{p}}^{n+1} \approx \frac{\mathbf{p}^{n+1} - \mathbf{p}^n}{\Delta t},$$

we obtain

$$\begin{aligned} \mathbf{M}_h \mathbf{u}^{n+1} - \mathbf{B}_h^T \mathbf{p}^{n+1} &= \mathbf{G}_h^{n+1}, \\ \mathbf{B}_h \mathbf{u}^{n+1} + \frac{S}{\Delta t} \mathbf{M}_{p,h} \mathbf{p}^{n+1} &= \mathbf{F}_h^{n+1} + \frac{S}{\Delta t} \mathbf{M}_{p,h} \mathbf{p}^n. \end{aligned}$$

This system can be written in block form as

$$\mathbf{A}_h \mathbf{x}^{n+1} = \mathbf{f}^{n+1},$$

where

$$\mathbf{x}^n = \begin{bmatrix} \mathbf{u}^n \\ \mathbf{p}^n \end{bmatrix}, \qquad \mathbf{A}_h = \begin{bmatrix} \mathbf{M}_h & -\mathbf{B}_h^T \\ \mathbf{B}_h & \mathbf{C}_h \end{bmatrix}, \qquad \mathbf{C}_h = \frac{S}{\Delta t} \mathbf{M}_{p,h},$$

and

$$\mathbf{f}^{n+1} = \begin{bmatrix} \mathbf{G}_h^{n+1} \\ \mathbf{F}_h^{n+1} + \mathbf{C}_h \mathbf{p}^n \end{bmatrix}.$$

This discrete saddle-point system constitutes the high-fidelity model used throughout this work.

2.5 Manufactured solution

To verify the correctness of the finite element implementation, we employ the method of manufactured solutions. In this approach, analytical expressions for the solution variables are prescribed, and the source term is constructed such that the governing equations are satisfied exactly. This enables a systematic verification of the numerical method independently of modeling assumptions.

A manufactured solution allows direct comparison between the numerical solution $(\mathbf{u}_h, p_h)$ and the exact solution $(\mathbf{u}_{\text{MAN}}, p_{\text{MAN}})$. Errors are measured in the L^2-norm for the pressure and in both the L^2- and $H(\text{div})$-norms for the velocity.

For Raviart–Thomas mixed finite elements of order k with discontinuous pressure approximation of degree $k-1$, the standard a priori estimates are well known; see, e.g., [17].

$$\|p_{\text{MAN}} - p_h\|_{L^2(\Omega)} \leqslant C h_{\max}^{k+1},$$
$$\|\mathbf{u}_{\text{MAN}} - \mathbf{u}_h\|_{L^2(\Omega)} \leqslant C h_{\max}^{k+1},$$
$$\|\mathbf{u}_{\text{MAN}} - \mathbf{u}_h\|_{H(\text{div};\Omega)} \leqslant C h_{\max}^{k}.$$

There is no direct a priori bound written only in terms of total DoFs N_h. For quasi-uniform meshes in d dimensions, we use

$$N_h \sim h_{\max}^{-d} \quad \Longrightarrow \quad h_{\max} \sim N_h^{-1/d}.$$

This gives the DoF-rates

$$\|p_{\text{MAN}} - p_h\|_{L^2(\Omega)} \leqslant C N_h^{-\frac{k+1}{d}},$$
$$\|\mathbf{u}_{\text{MAN}} - \mathbf{u}_h\|_{L^2(\Omega)} \leqslant C N_h^{-\frac{k+1}{d}},$$
$$\|\mathbf{u}_{\text{MAN}} - \mathbf{u}_h\|_{H(\text{div};\Omega)} \leqslant C N_h^{-\frac{k}{d}}.$$

Thus, in 2D the expected slopes versus N_h are $(k+1)/2$ in L^2 and $k/2$ in $H(\text{div})$.

In the transient setting, the errors are evaluated over the full time interval. In particular, we consider discrete approximations of the $L^2(0,T;L^2(\Omega))$ and $L^2(0,T;H(\text{div};\Omega))$ norms. The spatial convergence rates discussed above are therefore interpreted as the expected asymptotic rates with respect to the spatial discretization parameter for these time-integrated norms.

For the implicit Euler method, first-order convergence in time is expected; see, e.g., [20].

CHAPTER

THREE

REDUCED-ORDER MODELING*

In the following, we first present the idea of parameterized PDEs, then the standard RB framework under the assumption of affine parameter dependence, i.e., when the system operators and right-hand sides admit an affine decomposition with respect to the parameters. Later, in Section 3.3, we extend the discussion to non-affine problems and introduce techniques for recovering an approximate affine structure.

3.1 Parameterized Full-Order System

We consider a parameter-dependent model, where the governing equations depend on a parameter vector

$$\boldsymbol{\mu} \in \mathscr{P} \subset \mathbb{R}^d,$$

with $\mathscr{P}$ denoting the parameter domain. The parameter $\boldsymbol{\mu}$ may represent, for example, material properties, source terms, or other inputs to the system.

After spatial and temporal discretization, the finite element model derived in Section 2 leads to a parameter-dependent algebraic system of the form

$$\mathbf{A}_h(\boldsymbol{\mu})\mathbf{x}^{n+1}(\boldsymbol{\mu}) = \mathbf{f}_h^{n+1}(\boldsymbol{\mu}),$$

for each time step t^{n+1}, with $n \in [0, 1, \ldots, N_t]$.

The state vector is given by

$$\mathbf{x}^n(\boldsymbol{\mu}) = \begin{bmatrix} \mathbf{u}^n(\boldsymbol{\mu}) \\ \mathbf{p}^n(\boldsymbol{\mu}) \end{bmatrix} \in \mathbb{R}^{N_h},$$

where $\mathbf{u}^n$ and $\mathbf{p}^n$ are the coefficient vectors corresponding to the velocity and pressure, respectively, and $N_h = N_{\mathbf{u}} + N_p$ denotes the dimension of the full-order system.

The matrix $\mathbf{A}_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h \times N_h}$ inherits the saddle-point structure introduced in the previous section, and depends on the parameter through the underlying coefficients of the model.

In many applications, the system must be solved repeatedly for different parameter values $\boldsymbol{\mu} \in \mathscr{P}$. Since the dimension N_h is typically large, this leads to a significant computational cost. The goal of reduced-order modeling is therefore to construct a low-dimensional approximation of the solution map

$$\boldsymbol{\mu} \longmapsto \mathbf{x}(\boldsymbol{\mu}),$$

that can be evaluated efficiently for new parameter values.

3.2 Reduced Basis Method

The reduced basis method is widely used for parameterized PDEs [1, 2, 3]. In such problems, the solution $\mathbf{x}(\boldsymbol{\mu})$ depends on a parameter vector $\boldsymbol{\mu} \in \mathscr{P}$, where $\mathscr{P}$ denotes the parameter domain, as described in the section above. Throughout this section, we assume that the parameter dependence is affine, meaning that the system operators and right-hand sides can be expressed as finite sums of parameter-dependent scalar functions multiplied by parameter-independent operators.

The central idea of the RB method is to construct a low-dimensional approximation space from a carefully selected set of high-fidelity solutions, known as *snapshots*. Once this reduced space has been built, the solution corresponding to a new parameter value can be computed efficiently by projecting the high-fidelity system matrices and load vectors onto this reduced space. This leads naturally to an *offline–online* decomposition. In the offline stage, which is only done once, expensive tasks like generating snapshots and assembling the reduced basis are carried out. The online stage consists of inexpensive reduced solves whose computational cost is independent of the dimension of the full-order model. An overview of the offline–online workflow is shown in Figure 3.2.1. The RB methodology employed in this work follows the framework described in [1].

3.2.1 Offline stage

The offline stage is executed only once and its results are reused during the online phase. Consequently, computational cost is less critical at this stage. In this stage, we solve the high-fidelity problem for a set of parameter samples and construct the reduced basis by projecting these solutions, a procedure that can be computationally expensive.

We assume affine parameter dependence of the form

$$\mathbf{A}_h(t, \boldsymbol{\mu}) = \sum_{q=1}^{Q_A} \theta_A^q(t, \boldsymbol{\mu}) \mathbf{A}_h^q,$$

and

$$\mathbf{f}_h(t, \boldsymbol{\mu}) = \sum_{q=1}^{Q_f} \theta_f^q(t, \boldsymbol{\mu}) \mathbf{f}_h^q + \sum_{q=1}^{Q_D} \theta_D^q(t, \boldsymbol{\mu}) \mathbf{g}_h^q,$$

where the coefficient functions θ_A^q, θ_f^q, and θ_D^q may depend on both time and parameters, while the matrices $\mathbf{A}_h^q$ and vectors $\mathbf{f}_h^q$, $\mathbf{g}_h^q$ are independent of both. In the present implementation, any time dependence is assumed to be separable and is contained only

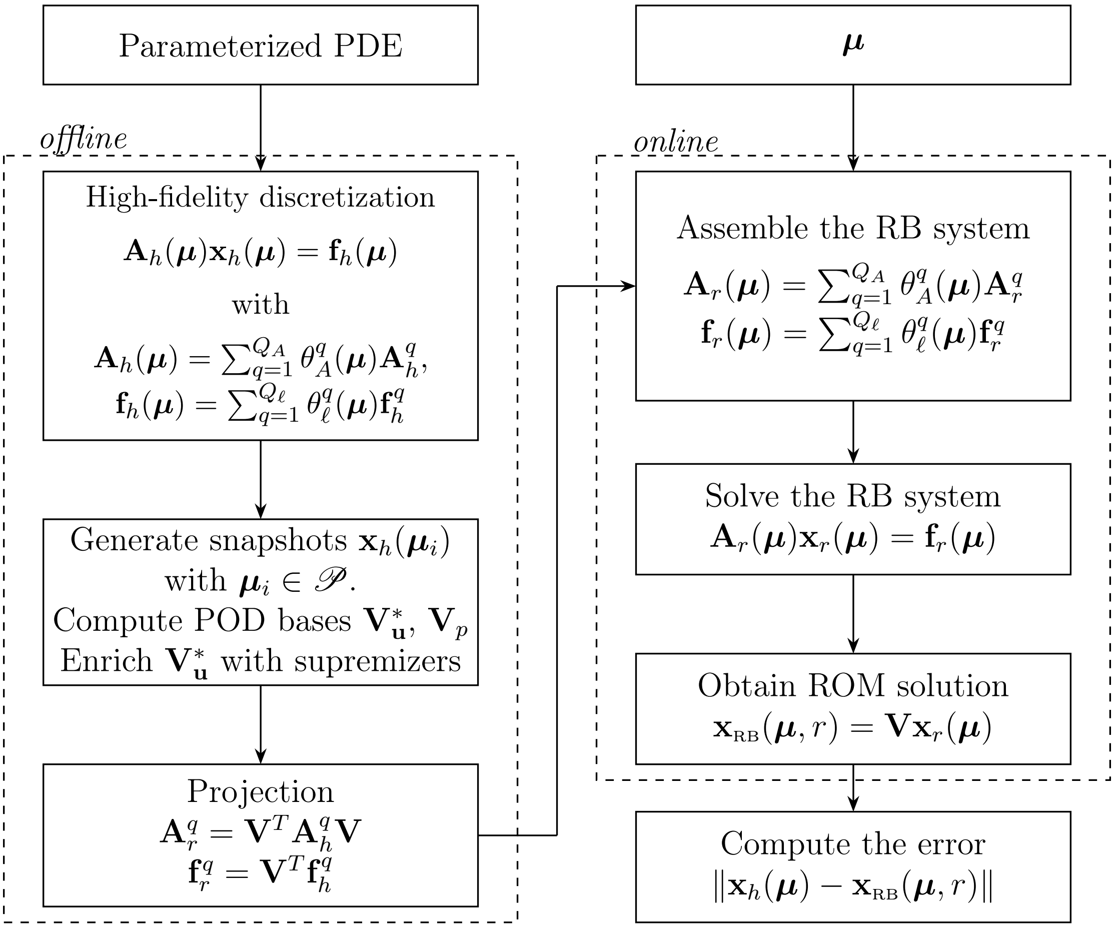

Figure 3.2.1: Offline–Online workflow of the reduced basis method. Illustration inspired by [1].

in these scalar coefficient functions. Here, $\mathbf{g}_h^q$ denotes contributions arising from the non-homogeneous Dirichlet boundary condition.

For notational simplicity, the explicit time dependence will be omitted in the following whenever no ambiguity arises.

We now aim to construct a reduced basis for the high-fidelity solutions. In this work, we employ the Proper Orthogonal Decomposition (POD), which derives a low-dimensional approximation space from a prescribed set of solution snapshots; see, e.g., [1, 21, 22, 23]. POD is particularly well suited for problems where representative solution snapshots can be obtained efficiently, as it provides a reduced space with optimal global approximation properties for the given snapshot set.

However, if solving the high-fidelity problem is very expensive, generating a sufficiently rich snapshot set for POD may become prohibitive. In such cases, an alternative and widely used approach is the Greedy algorithm, which iteratively enriches the reduced space by selecting new parameter samples based on a posteriori error estimators; see, e.g., [1, 24, 25, 26, 27].

Since our focus here is on POD, we describe this procedure in detail. We begin by sampling the parameter domain $\mathscr{P}$ at points $\{\boldsymbol{\mu}_i\}_{i=1}^m$ and solving the high-fidelity model for each sample. The corresponding FEM coefficient vectors are stored in the weighted snapshot matrix

$$
\begin{aligned}
\mathbf{S}_p =& \big[\sqrt{w_1}\mathbf{p}_h^1(\boldsymbol{\mu}_1), \ldots, \sqrt{w_1}\mathbf{p}_h^{N_t}(\boldsymbol{\mu}_1), \ldots, \sqrt{w_m}\mathbf{p}_h^1(\boldsymbol{\mu}_m), \ldots, \sqrt{w_m}\mathbf{p}_h^{N_t}(\boldsymbol{\mu}_m)\big] \\
\mathbf{S}_\mathbf{u} =& \big[\sqrt{w_1}\mathbf{u}_h^1(\boldsymbol{\mu}_1), \ldots, \sqrt{w_1}\mathbf{u}_h^{N_t}(\boldsymbol{\mu}_1), \ldots, \sqrt{w_m}\mathbf{u}_h^1(\boldsymbol{\mu}_m), \ldots, \sqrt{w_m}\mathbf{u}_h^{N_t}(\boldsymbol{\mu}_m)\big]
\end{aligned}
$$

where $\mathbf{S}_p \in \mathbb{R}^{N_p \times (mN_t)}$, $\mathbf{S}_\mathbf{u} \in \mathbb{R}^{N_\mathbf{u} \times (mN_t)}$. The weights $w_i > 0$ may be used to emphasize certain regions of the parameter domain. Each column corresponds to a snapshot at a given parameter sample and time level. Several sampling strategies may be employed to generate the snapshot set, including random sampling, equispaced sampling, and sampling based on quadrature points over the parameter domain. In this work, however, we restrict ourselves to equispaced sampling for simplicity.

The goal of POD is to find a low-dimensional subspace that best approximates the snapshot set in a least-squares sense. This is achieved by solving the optimization problem of finding a low-dimensional subspace that minimizes the projection error of the snapshots. The POD basis is obtained by computing a singular value decomposition (SVD) of the snapshot matrix,

$$\mathbf{S}_p = \mathbf{W}_p \boldsymbol{\Sigma}_p \mathbf{Z}_p^T.$$

Let $s_p = \min(N_p, mN_t)$. Then $\mathbf{W}_p \in \mathbb{R}^{N_p \times s_p}$ contains the left singular vectors, $\boldsymbol{\Sigma}_p = \mathrm{diag}(\sigma_1^{(p)}, \ldots, \sigma_{s_p}^{(p)}) \in \mathbb{R}^{s_p \times s_p}$ contains the singular values in non-increasing order, and $\mathbf{Z}_p \in \mathbb{R}^{s_p \times s_p}$ contains the right singular vectors. The columns of $\mathbf{W}_p$ form an orthonormal basis with respect to the inner product used to construct $\mathbf{S}_p$. Computing the SVD of the snapshot matrix can be computationally expensive, particularly when both the number of degrees of freedom N_p and the number of snapshots m are large. In practice, the SVD is computed using standard numerical linear algebra routines. The same construction is applied separately to the velocity and pressure snapshot matrices, yielding preliminary reduced bases $\mathbf{V}_\mathbf{u}^*$ and $\mathbf{V}_p$. We denote the corresponding singular values by $\{\sigma_i^{(u)}\}$ and $\{\sigma_i^{(p)}\}$ for the velocity and pressure snapshot matrices, respectively.

The choice of inner product determines in which norm the POD error is minimized. For PDE problems, the energy inner product

$$(u, v)_E := a_{\mathrm{ref}}(u, v) \tag{3.1}$$

is often the most natural, as it reflects the physical structure of the operator. However, since the bilinear form $a(\cdot, \cdot; \boldsymbol{\mu})$ depends on the parameter $\boldsymbol{\mu}$, one must select a representative form for the POD construction. A practical approach is to define an averaged or weighted energy inner product over the chosen training set, using the same weights w_i that appear in the snapshot matrix.

We truncate the SVD by retaining only the r_p largest singular values and the corresponding left and right singular vectors, yielding the rank-r_p approximation

$$\mathbf{S}_p \approx \widetilde{\mathbf{S}}_p = \widetilde{\mathbf{W}}_p \widetilde{\boldsymbol{\Sigma}}_p \widetilde{\mathbf{Z}}_p^T,$$

where $\widetilde{\mathbf{W}}_p \in \mathbb{R}^{N_p \times r_p}$, $\widetilde{\boldsymbol{\Sigma}}_p \in \mathbb{R}^{r_p \times r_p}$, and $\widetilde{\mathbf{Z}}_p \in \mathbb{R}^{s_p \times r_p}$. The columns of $\widetilde{\mathbf{W}}_p$ form an orthonormal basis for the reduced subspace.

This reduced space is optimal among all r_p-dimensional subspaces and minimizes the projection error of the snapshot matrix $\mathbf{S}_p$ with respect to the chosen norm. This follows from the Eckart–Young–Mirsky theorem (see, e.g., [28, 29]), which states that the POD basis obtained from the first r_p left singular vectors yields the best rank-r_p approximation of $\mathbf{S}_p$ in both the Frobenius norm and the operator 2-norm. Specifically, if $\sigma_1^{(p)} \geqslant \sigma_2^{(p)} \geqslant \cdots \geq \sigma_{s_p}^{(p)} \geqslant 0$ are the singular values of $\mathbf{S}_p$, then

$$\|\mathbf{S}_p - \widetilde{\mathbf{S}}_p\|_F^2 = \sum_{i=r_p+1}^{s_p} (\sigma_i^{(p)})^2, \qquad \|\mathbf{S}_p - \widetilde{\mathbf{S}}_p\|_2 = \sigma_{r_p+1}^{(p)}.$$

Consequently, no other r_p-dimensional subspace can represent the snapshots with a smaller projection error.

For the pressure, the reduced basis projection operator is defined as

$$\mathbf{V}_p \coloneqq \widetilde{\mathbf{W}}_p.$$

Applying the same SVD construction to the velocity snapshot matrix $\mathbf{S}_{\mathbf{u}}$ yields a preliminary velocity basis

$$\mathbf{V}_{\mathbf{u}}^* \coloneqq \widetilde{\mathbf{W}}_{\mathbf{u}},$$

with reduced dimension $r_{\mathbf{u}}$.

Since the full-order problem has a saddle-point structure, the reduced velocity and pressure spaces obtained from POD do not automatically satisfy a discrete inf-sup condition. To ensure stability, the velocity space is enriched with so-called supremizer functions associated with the pressure basis [30]. Without stabilization, the reduced spaces may violate a discrete inf-sup condition, leading to spurious pressure modes and instability.

Let $\{\zeta_i^p\}_{i=1}^{r_p}$ denote the pressure POD basis, i.e., the columns of $\mathbf{V}_p$. For each ζ_i^p, we define a supremizer $s_i \in V_h$ as the solution of

$$a_{\text{ref}}(s_i, v_h) = -\,b(v_h, \zeta_i^p) \qquad \forall v_h \in V_h,$$

where $a_{\text{ref}}(\cdot,\cdot)$ is a reference velocity inner product.

Collecting the supremizer vectors in the matrix

$$\mathbf{S} \coloneqq \left[\mathbf{s}_1, \ldots, \mathbf{s}_{r_p}\right],$$

the stabilized reduced velocity basis is defined as

$$\mathbf{V}_{\mathbf{u}} \coloneqq \operatorname{orth}\left(\left[\mathbf{V}_{\mathbf{u}}^*\ \mathbf{S}\right]\right),$$

where orth$(\cdot)$ denotes an orthonormalization procedure (e.g., via Gram–Schmidt or QR factorization). We denote the added dimensions from the stabilized velocity space by $r_s \coloneqq \dim(\mathbf{V}_{\mathbf{u}}) - r_{\mathbf{u}}$. We also define the total reduced dimension $r \coloneqq r_p + r_{\mathbf{u}} + r_s$.

Due to the saddle-point structure, the projection is performed separately in the velocity and pressure spaces. Introducing the block-diagonal reduced basis matrix

$$\mathbf{V} \coloneqq \begin{bmatrix} \mathbf{V}_{\mathbf{u}} & \mathbf{0} \\ \mathbf{0} & \mathbf{V}_p \end{bmatrix},$$

the reduced system can be written compactly as the Galerkin projection

$$\mathbf{A}_r(\boldsymbol{\mu}) = \mathbf{V}^T \mathbf{A}_h(\boldsymbol{\mu}) \mathbf{V}, \qquad \mathbf{f}_r(\boldsymbol{\mu}) = \mathbf{V}^T \mathbf{f}_h(\boldsymbol{\mu}).$$

Using the block structure of the full-order matrix

$$\mathbf{A}_h(\boldsymbol{\mu}) = \begin{bmatrix} \mathbf{M}_h(\boldsymbol{\mu}) & -\mathbf{B}_h^T(\boldsymbol{\mu}) \\ \mathbf{B}_h(\boldsymbol{\mu}) & \mathbf{C}_h(\boldsymbol{\mu}) \end{bmatrix},$$

this yields

$$\mathbf{A}_r(\boldsymbol{\mu}) = \begin{bmatrix} \mathbf{V}_{\mathbf{u}}^T \mathbf{M}_h(\boldsymbol{\mu}) \mathbf{V}_{\mathbf{u}} & -\mathbf{V}_{\mathbf{u}}^T \mathbf{B}_h^T(\boldsymbol{\mu}) \mathbf{V}_p \\ \mathbf{V}_p^T \mathbf{B}_h(\boldsymbol{\mu}) \mathbf{V}_{\mathbf{u}} & \mathbf{V}_p^T \mathbf{C}_h(\boldsymbol{\mu}) \mathbf{V}_p \end{bmatrix}.$$

In the presence of non-homogeneous pressure Dirichlet boundary conditions, the full-order right-hand side contains both volumetric and boundary contributions. The fully discrete right-hand side is therefore written as

$$\mathbf{f}_h^{n+1}(\boldsymbol{\mu}) = \begin{bmatrix} \mathbf{G}_h^{n+1}(\boldsymbol{\mu}) \\ \mathbf{F}_h^{n+1}(\boldsymbol{\mu}) + \mathbf{C}_h(\boldsymbol{\mu}) \mathbf{p}^n(\boldsymbol{\mu}) \end{bmatrix},$$

where the boundary vector is defined by

$$[\mathbf{G}_h^{n+1}(\boldsymbol{\mu})]_i := -\int_{\Gamma_p} p_D(\mathbf{x}, t^{n+1}, \boldsymbol{\mu})\, \boldsymbol{\phi}_i \cdot \mathbf{n}\, ds.$$

The reduced right-hand side is obtained by Galerkin projection,

$$\mathbf{f}_r^{n+1}(\boldsymbol{\mu}) = \mathbf{V}^T \mathbf{f}_h^{n+1}(\boldsymbol{\mu}),$$

which yields

$$\mathbf{f}_r^{n+1}(\boldsymbol{\mu}) = \begin{bmatrix} \mathbf{V}_{\mathbf{u}}^T \mathbf{G}_h^{n+1}(\boldsymbol{\mu}) \\ \mathbf{V}_p^T \left(\mathbf{F}_h^{n+1}(\boldsymbol{\mu}) + \mathbf{C}_h(\boldsymbol{\mu}) \mathbf{p}^n(\boldsymbol{\mu}) \right) \end{bmatrix}.$$

3.2.2 Online Stage

In the online stage, the reduced system is assembled using the quantities precomputed in the offline stage. For a given parameter value $\boldsymbol{\mu} \in \mathscr{P}$, we define

$$\begin{aligned} \mathbf{A}_r(\boldsymbol{\mu}) &:= \mathbf{V}^T \mathbf{A}_h(\boldsymbol{\mu}) \mathbf{V} = \sum_{q=1}^{Q_A} \theta_A^q(\boldsymbol{\mu}) \mathbf{V}^T \mathbf{A}_h^q \mathbf{V} = \sum_{q=1}^{Q_A} \theta_A^q(\boldsymbol{\mu}) \mathbf{A}_r^q, \\ \mathbf{f}_r^{n+1}(\boldsymbol{\mu}) &:= \mathbf{V}^T \mathbf{f}_h^{n+1}(\boldsymbol{\mu}) \\ &= \sum_{q=1}^{Q_f} \theta_f^q(\boldsymbol{\mu}) \mathbf{V}^T \mathbf{f}_h^{q,n+1} + \sum_{q=1}^{Q_D} \theta_D^q(\boldsymbol{\mu}) \mathbf{V}^T \mathbf{g}_h^{q,n+1} \\ &= \sum_{q=1}^{Q_f} \theta_f^q(\boldsymbol{\mu}) \mathbf{f}_r^{q,n+1} + \sum_{q=1}^{Q_D} \theta_D^q(\boldsymbol{\mu}) \mathbf{g}_r^{q,n+1}. \end{aligned}$$

This yields the reduced system

$$\mathbf{A}_r(\boldsymbol{\mu}) \mathbf{x}_r^{n+1}(\boldsymbol{\mu}) = \mathbf{f}_r^{n+1}(\boldsymbol{\mu}),$$

where $\mathbf{x}_r^{n+1}(\boldsymbol{\mu}) \in \mathbb{R}^r$ and $r \ll N_h$.

Once the reduced coefficients have been computed, the approximation in the full-order space is reconstructed as

$$\mathbf{x}_{\mathrm{RB}}^{n+1}(\boldsymbol{\mu}) = \mathbf{V} \mathbf{x}_r^{n+1}(\boldsymbol{\mu}).$$

Hence, the online stage requires only the assembly and solution of a low-dimensional system, whose size is independent of the full-order dimension N_h. See Figure 3.2.2 for an illustration of the reduction.

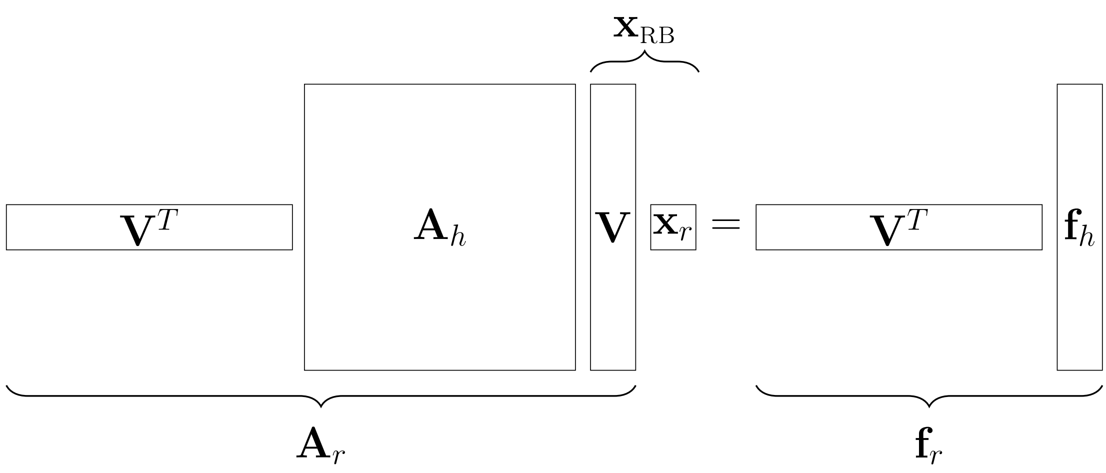

Figure 3.2.2: Illustration of the reduction we achieve from the RB method. The final system becomes a small $r \times r$ system instead of a large $N_h \times N_h$ system.

3.2.3 Error Analysis

To assess the accuracy of the reduced-order model, we compare the reduced solution with the corresponding full-order solution over all time steps. Since the problem is time dependent, we measure the error over the entire time interval using relative $L^2(0,T)$ norms.

Let

$$e_p(\boldsymbol{\mu}) \coloneqq p_h(\boldsymbol{\mu}) - p_{\text{RB}}(\boldsymbol{\mu}),$$

and

$$\mathbf{e}_{\mathbf{u}}(\boldsymbol{\mu}) \coloneqq \mathbf{u}_h(\boldsymbol{\mu}) - \mathbf{u}_{\text{RB}}(\boldsymbol{\mu}).$$

The relative pressure error is defined by

$$\varepsilon_p(\boldsymbol{\mu}) \coloneqq \frac{\|e_p(\boldsymbol{\mu})\|_{L^2(0,T;L^2(\Omega))}}{\|p_h(\boldsymbol{\mu})\|_{L^2(0,T;L^2(\Omega))}}. \tag{3.2}$$

For the velocity, we consider both the $L^2(\Omega)$ norm and the energy norm induced by

$$\|\mathbf{v}\|_E^2 \coloneqq a_{\text{ref}}(\mathbf{v}, \mathbf{v}).$$

The corresponding relative errors are defined as

$$\varepsilon_{\mathbf{u},L^2}(\boldsymbol{\mu}) \coloneqq \frac{\|\mathbf{e}_{\mathbf{u}}(\boldsymbol{\mu})\|_{L^2(0,T;L^2(\Omega))}}{\|\mathbf{u}_h(\boldsymbol{\mu})\|_{L^2(0,T;L^2(\Omega))}}, \tag{3.3}$$

and

$$\varepsilon_{\mathbf{u},E}(\boldsymbol{\mu}) \coloneqq \frac{\|\mathbf{e}_{\mathbf{u}}(\boldsymbol{\mu})\|_{L^2(0,T;E(\Omega))}}{\|\mathbf{u}_h(\boldsymbol{\mu})\|_{L^2(0,T;E(\Omega))}}. \tag{3.4}$$

Choosing an appropriate reduced dimension is a key aspect of the reduced basis method. Since separate POD constructions are performed for the velocity and pressure variables, the decay of the corresponding singular values is examined individually.

Let $\{\sigma_i^{(\mathbf{u})}\}_{i=1}^{s_{\mathbf{u}}}$ and $\{\sigma_i^{(p)}\}_{i=1}^{s_p}$ denote the singular values obtained from the POD of the velocity and pressure snapshot matrices, respectively. We define the relative discarded POD energy for velocity and pressure as

$$\varepsilon_{\text{DIS},\mathbf{u}}(r_{\mathbf{u}}) = \left(\frac{\sum_{i=r_{\mathbf{u}}+1}^{s_{\mathbf{u}}} (\sigma_i^{(\mathbf{u})})^2}{\sum_{i=1}^{s_{\mathbf{u}}} (\sigma_i^{(\mathbf{u})})^2} \right)^{1/2}, \tag{3.5}$$

$$\varepsilon_{\text{DIS},p}(r_p) = \left(\frac{\sum_{i=r_p+1}^{s_p} (\sigma_i^{(p)})^2}{\sum_{i=1}^{s_p} (\sigma_i^{(p)})^2} \right)^{1/2}. \tag{3.6}$$

For the snapshot sets used to construct the reduced bases, the POD spaces of dimensions $r_{\mathbf{u}}$ and r_p minimize the projection error, and the discarded energies provide a direct measure of the truncation error. Under standard assumptions on the smoothness of the parameter-to-solution map and sufficiently representative sampling of the parameter domain, this quantity is commonly used as an indicator for the approximation quality of the reduced space.

In particular, one often observes that the reduced-order error decreases at a rate comparable to the decay of the discarded POD energy. Hence, choosing tolerances for $\varepsilon_{\text{DIS},\mathbf{u}}(r_{\mathbf{u}})$ and $\varepsilon_{\text{DIS},p}(r_p)$ typically leads to reduced-order errors of similar magnitude, at least for parameter values close to the training set. However, this relationship is not rigorous in general, and the discarded POD energies are therefore used here primarily as a heuristic indicator.

In practice, the reduced dimensions $r_{\mathbf{u}}$ and r_p are chosen such that

$$\varepsilon_{\text{DIS},\mathbf{u}}(r_{\mathbf{u}}) < \tau_{\mathbf{u}}, \qquad \varepsilon_{\text{DIS},p}(r_p) < \tau_p.$$

for prescribed tolerances $\tau_{\mathbf{u}}$ and τ_p. In many cases, the same tolerance is used for both variables, i.e., $\tau_{\mathbf{u}} = \tau_p = \tau$, and this is the choice used in the numerical experiments.

3.3 Affinization of Reduced Basis Methods

In the previous sections, we considered only the case of affine parameter dependence in both the stiffness matrix and the load vector. In this section, we extend the RB method to problems with non-affine parameter dependence.

One commonly used approach for treating non-affine parameter dependence is the Empirical Interpolation Method (EIM) [1, 31, 32, 33]. While effective, EIM typically requires intrusive access to the underlying problem implementation, which may be undesirable in practice. Instead, we adopt the lightly intrusive approach proposed in [12], which relies solely on access to the system matrices and corresponding high-fidelity solutions.

The method proposed in [12] combines a POD of the solution snapshots with an L^2-based least-squares fit of the parameter-dependent system matrices, yielding an approximate affine representation.

For simplicity, we assume a two-dimensional parameter space $\mathscr{P} \subset \mathbb{R}^2$ sampled at parameter values $\{\boldsymbol{\mu}_i\}_{i=1}^m$. Furthermore, we assume that the system matrices $\mathbf{A}_h(\boldsymbol{\mu}_i)$, load vectors $\mathbf{f}_h(\boldsymbol{\mu}_i)$ for $i = 1, \dots, m$, and the POD projection matrix $\mathbf{V}$ have already been computed. The method extends naturally to higher-dimensional parameter spaces.

For simplicity, we first describe the affinization procedure for a parameter-dependent operator $\mathbf{A}_h(\boldsymbol{\mu})$. In the transient setting considered in this work, the time dependence is assumed to be separable and represented through known scalar coefficient functions. The affinization procedure is therefore applied only with respect to the parameter dependence on $\boldsymbol{\mu}$. To keep the notation concise, the explicit time dependence is omitted in the following.

3.3.1 L^2-fit then reduce

We will show the details for the stiffness matrix $\mathbf{A}_h(\boldsymbol{\mu})$. The treatment of the load vector $\mathbf{f}_h(\boldsymbol{\mu})$ follows analogously. Since we cannot assume that the bilinear form can be written in a parameter-separable form, we construct an approximate affine decomposition. We seek coefficient functions $\{\theta_A^q(\boldsymbol{\mu})\}_{q=1}^{Q_A}$ and parameter-independent operators $\{\widetilde{\mathbf{A}}_h^q\}_{q=1}^{Q_A}$ such that

$$\mathbf{A}_h(\boldsymbol{\mu}) \approx \widetilde{\mathbf{A}}_h(\boldsymbol{\mu}) = \sum_{q=1}^{Q_A} \theta_A^q(\boldsymbol{\mu})\, \widetilde{\mathbf{A}}_h^q.$$

Here, the matrices $\widetilde{\mathbf{A}}_h^q$ are parameter-independent coefficient matrices associated with the scalar coefficient functions $\theta_A^q(\boldsymbol{\mu})$. In this way, each entry of $[\widetilde{\mathbf{A}}_h(\boldsymbol{\mu})]_{i,j}$ is a function that approximates the corresponding entry $[\mathbf{A}_h(\boldsymbol{\mu})]_{i,j}$.

We determine the coefficients as the minimizers of

$$E_{L^2}^2 = \int_{\mathscr{P}} \|\widetilde{\mathbf{A}}_h(\boldsymbol{\mu}) - \mathbf{A}_h(\boldsymbol{\mu})\|_{\mathrm{F}}^2 \,\mathrm{d}\boldsymbol{\mu},$$

where $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm. Instead of minimizing the continuous problem, we minimize the discrete problem, discretized with some quadrature rule as

$$E_{L^2}^2 \approx Q_{L^2}^2 = \sum_{i=1}^{m} w_i \|\widetilde{\mathbf{A}}_h(\boldsymbol{\mu}_i) - \mathbf{A}_h(\boldsymbol{\mu}_i)\|_{\mathrm{F}}^2,$$

where w_i are the weights of the quadrature rule and $\boldsymbol{\mu}_i$ are the same parameters that we used to compute $\mathbf{A}_h(\boldsymbol{\mu})$ initially. Since the objective is to approximate an integral over the parameter domain, it is natural to employ sampling strategies associated with accurate quadrature rules. In particular, quadrature points such as Gauss–Lobatto nodes may provide improved accuracy compared with uniformly distributed samples.

For this explanation, we choose $\mathscr{P} \subset \mathbb{R}^2$ and the coefficient functions θ_a^q to be the tensor product of Legendre polynomials L_q, where q denotes the polynomial degree. Since Legendre polynomials are orthogonal only in the interval $[-1, 1]$, we have to scale the parameters accordingly. For a parameter component $\mu_j \in [\mu_{j,\min}, \mu_{j,\max}]$, we introduce the affine rescaling

$$\hat{\mu}_j = \frac{2(\mu_j - \mu_{j,\min})}{\mu_{j,\max} - \mu_{j,\min}} - 1, \qquad \hat{\mu}_j \in [-1, 1].$$

All evaluations $L_\ell(\mu_j)$ below should be understood as $L_\ell(\hat{\mu}_j)$.

Using the tensor product of the Legendre polynomials as coefficient functions, the form becomes

$$\widetilde{\mathbf{A}}_h(\boldsymbol{\mu}) = \sum_{k=0}^{P}\sum_{q=0}^{P} L_q(\mu_1)L_k(\mu_2)\widetilde{\mathbf{A}}_h^{(q,k)}.$$

Note that in general, in a d dimensional parameter space, the form becomes

$$\widetilde{\mathbf{A}}_h(\boldsymbol{\mu}) = \sum_{p_1=0}^{P}\sum_{p_2=0}^{P}\cdots\sum_{p_d=0}^{P}\left(L_{p_1}(\mu_1)L_{p_2}(\mu_2)\dots L_{p_d}(\mu_d)\widetilde{\mathbf{A}}_h^{(p_1,p_2,\dots,p_d)}\right).$$

We see that the size of the polynomials grows rapidly as a function of the dimension d of the parameter space.

To simplify notation, we introduce

$$\boldsymbol{\theta}(\boldsymbol{\mu}) := \begin{bmatrix} L_0(\mu_1)L_0(\mu_2) \\ L_1(\mu_1)L_0(\mu_2) \\ \vdots \\ L_P(\mu_1)L_0(\mu_2) \\ \vdots \\ L_P(\mu_1)L_P(\mu_2) \end{bmatrix} \in \mathbb{R}^{(P+1)^2}, \qquad [\boldsymbol{\theta}(\boldsymbol{\mu})]_{(P+1)k+q} = L_q(\mu_1)L_k(\mu_2),$$

where $k, q = 0, \dots, P$. The design and weight matrices are given as

$$\mathbf{B} := \begin{bmatrix} \boldsymbol{\theta}(\boldsymbol{\mu}_1)^T \\ \boldsymbol{\theta}(\boldsymbol{\mu}_2)^T \\ \vdots \\ \boldsymbol{\theta}(\boldsymbol{\mu}_m)^T \end{bmatrix} \in \mathbb{R}^{m\times(P+1)^2}, \qquad \mathbf{W} := \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_m \end{bmatrix} \in \mathbb{R}^{m\times m}.$$

Minimizing $Q_{L^2}^2$ produces the weighted normal equations as

$$\mathbf{B}^T\mathbf{W}\mathbf{B}[\widetilde{\mathbf{A}}_h^{(0,0)}, \widetilde{\mathbf{A}}_h^{(1,0)}, \dots, \widetilde{\mathbf{A}}_h^{(P,0)}, \dots, \widetilde{\mathbf{A}}_h^{(P,P)}]^T = \mathbf{B}^T\mathbf{W}[\mathbf{A}_h(\boldsymbol{\mu}_1), \dots, \mathbf{A}_h(\boldsymbol{\mu}_m)]^T,$$

which corresponds to solving a weighted L^2-least-squares problem for the coefficients $\widetilde{\mathbf{A}}_h^{(q,k)}$. For a proof, see, e.g. [12, Appendix A].

For practical implementation it is convenient to consider the problem entrywise. Define, for each matrix entry (i, j)

$$\mathbf{a}_{i,j} := \begin{bmatrix} [\mathbf{A}_h(\boldsymbol{\mu}_1)]_{i,j} \\ [\mathbf{A}_h(\boldsymbol{\mu}_2)]_{i,j} \\ \vdots \\ [\mathbf{A}_h(\boldsymbol{\mu}_m)]_{i,j} \end{bmatrix} \in \mathbb{R}^m,$$

and each element of the coefficient matrices is then

$$[\widetilde{\mathbf{A}}_h^{(q,k)}]_{i,j} = [(\mathbf{B}^T\mathbf{W}\mathbf{B})^{-1}\mathbf{B}^T\mathbf{W}\mathbf{a}_{i,j}]_{(P+1)k+q}.$$

Conceptually, this corresponds to solving a weighted least-squares problem with a fixed left-hand side for each matrix entry (i, j), resulting in up to N_h^2 right-hand sides in the general case. Note that this number can be significantly reduced in practice by exploiting the symmetry and/or sparsity of the high-fidelity stiffness matrix.

Forming and solving with $\mathbf{B}^T\mathbf{W}\mathbf{B}$ repeatedly is computationally inefficient [34]. Let $\mathbf{x}_{i,j} \in \mathbb{R}^{(P+1)^2}$ denote the weighted least-squares coefficient vector associated with the matrix entry (i,j). Then the entries of the coefficient matrices are given by

$$(\widetilde{\mathbf{A}}_h^{(q,k)})_{i,j} = (\mathbf{x}_{i,j})_{(P+1)k+q}.$$

Using the QR factorization $\mathbf{W}^{1/2}\mathbf{B} = \mathbf{Q}\mathbf{R}$, the vector $\mathbf{x}_{i,j}$ is obtained by solving

$$\mathbf{R}\mathbf{x}_{i,j} = \mathbf{Q}^T\mathbf{W}^{1/2}\mathbf{a}_{i,j}.$$

The QR factorization of $\mathbf{W}^{1/2}\mathbf{B}$ is computationally expensive, but it is performed only once and reused for all matrix entries (i,j).

Assuming that we have done the same procedure for $\widetilde{\mathbf{f}}_h(\boldsymbol{\mu})$, the FEM coefficients can then be approximated as

$$\widetilde{\mathbf{x}}_h(\boldsymbol{\mu}) = \widetilde{\mathbf{A}}_h^{-1}(\boldsymbol{\mu})\widetilde{\mathbf{f}}_h(\boldsymbol{\mu}),$$

and the corresponding global approximation $\widetilde{\mathbf{x}}_h(\boldsymbol{\mu})$ is obtained from these coefficients. Note that [12, Theorem 1] states that this inverse indeed exists.

Since we have found an affine representation of the stiffness matrix and load vector, we can now obtain a reduced system. Applying the projection matrix $\mathbf{V}$, we obtain

$$\widetilde{\mathbf{A}}_r^{(q,k)} = \mathbf{V}^T\widetilde{\mathbf{A}}_h^{(q,k)}\mathbf{V} \qquad \widetilde{\mathbf{f}}_r^{(q,k)} = \mathbf{V}^T\widetilde{\mathbf{f}}_h^{(q,k)}.$$

Using the linearity of summing, we obtain

$$\begin{aligned}
\widetilde{\mathbf{A}}_r(\boldsymbol{\mu}) &:= \mathbf{V}^T\widetilde{\mathbf{A}}_h(\boldsymbol{\mu})\mathbf{V} = \sum_{q=0}^{P}\sum_{k=0}^{P} L_q(\mu_1)L_k(\mu_2)\widetilde{\mathbf{A}}_r^{(q,k)},\\
\widetilde{\mathbf{f}}_r(\boldsymbol{\mu}) &:= \mathbf{V}^T\widetilde{\mathbf{f}}_h(\boldsymbol{\mu}) = \sum_{q=0}^{P}\sum_{k=0}^{P} L_q(\mu_1)L_k(\mu_2)\widetilde{\mathbf{f}}_r^{(q,k)},
\end{aligned}$$

and the reduced system becomes

$$\widetilde{\mathbf{A}}_r(\boldsymbol{\mu})\widetilde{\mathbf{x}}_r(\boldsymbol{\mu}) = \widetilde{\mathbf{f}}_r(\boldsymbol{\mu}).$$

Let $\widetilde{u}_h(\boldsymbol{\mu})$ and $\widetilde{p}_h(\boldsymbol{\mu})$ denote the velocity and pressure components of the affinized approximation $\widetilde{\mathbf{x}}_h(\boldsymbol{\mu})$. We are interested in estimating the error of the operator and the solution. The relative error of the operator is computed as

$$\widetilde{\varepsilon}_{\mathrm{OP}}^2 = \frac{\displaystyle\int_{\mathscr{P}} \|\widetilde{\mathbf{A}}_h(\boldsymbol{\mu}) - \mathbf{A}_h(\boldsymbol{\mu})\|_{\mathrm{F}}^2 \,\mathrm{d}\boldsymbol{\mu}}{\displaystyle\int_{\mathscr{P}} \|\mathbf{A}_h(\boldsymbol{\mu})\|_{\mathrm{F}}^2 \,\mathrm{d}\boldsymbol{\mu}} \approx \frac{\displaystyle\sum_{i=1}^{M} \omega_i \|\widetilde{\mathbf{A}}_h(\boldsymbol{\mu}_i) - \mathbf{A}_h(\boldsymbol{\mu}_i)\|_{\mathrm{F}}^2}{\displaystyle\sum_{i=1}^{M} \omega_i \|\mathbf{A}_h(\boldsymbol{\mu}_i)\|_{\mathrm{F}}^2}. \tag{3.7}$$

The relative solution errors are computed separately for the velocity and pressure variables. In the transient setting, the errors are measured using the L^2-in-time norm, i.e. the $L^2(0,T;\cdot)$ norm.

For the velocity, we define

$$\widetilde{\varepsilon}^2_{\mathrm{SOL},\mathbf{u}} = \frac{\displaystyle\int_{\mathscr{P}} \|\mathbf{u}_h(\boldsymbol{\mu}) - \widetilde{\mathbf{u}}_h(\boldsymbol{\mu})\|^2_{L^2(0,T;U)}\,\mathrm{d}\boldsymbol{\mu}}{\displaystyle\int_{\mathscr{P}} \|\mathbf{u}_h(\boldsymbol{\mu})\|^2_{L^2(0,T;U)}\,\mathrm{d}\boldsymbol{\mu}} \approx \frac{\displaystyle\sum_{i=1}^{M} \omega_i \,\|\mathbf{u}_h(\boldsymbol{\mu}_i) - \widetilde{\mathbf{u}}_h(\boldsymbol{\mu}_i)\|^2_{L^2(0,T;U)}}{\displaystyle\sum_{i=1}^{M} \omega_i \,\|\mathbf{u}_h(\boldsymbol{\mu}_i)\|^2_{L^2(0,T;U)}}, \tag{3.8}$$

and for the pressure,

$$\widetilde{\varepsilon}^2_{\mathrm{SOL},p} = \frac{\displaystyle\int_{\mathscr{P}} \|p_h(\boldsymbol{\mu}) - \widetilde{p}_h(\boldsymbol{\mu})\|^2_{L^2(0,T;L^2(\Omega))}\,\mathrm{d}\boldsymbol{\mu}}{\displaystyle\int_{\mathscr{P}} \|p_h(\boldsymbol{\mu})\|^2_{L^2(0,T;L^2(\Omega))}\,\mathrm{d}\boldsymbol{\mu}} \approx \frac{\displaystyle\sum_{i=1}^{M} \omega_i \,\|p_h(\boldsymbol{\mu}_i) - \widetilde{p}_h(\boldsymbol{\mu}_i)\|^2_{L^2(0,T;L^2(\Omega))}}{\displaystyle\sum_{i=1}^{M} \omega_i \,\|p_h(\boldsymbol{\mu}_i)\|^2_{L^2(0,T;L^2(\Omega))}}. \tag{3.9}$$

Here, M is the number of quadrature points, and $\boldsymbol{\mu}_i$ and ω_i are the corresponding quadrature points and weights.

3.4 Parameter Domain Splitting

For problems with strongly varying parameter dependence, constructing a single global reduced basis may require a relatively large reduced dimension in order to achieve satisfactory accuracy over the entire parameter domain. A strategy for addressing this issue is to partition the parameter domain into smaller subdomains and construct separate local reduced bases.

The idea of local reduced basis spaces has been explored in, e.g., [35], where the partitioning is guided by a posteriori error estimators. In the present work, however, we do not employ the adaptive partitioning strategy proposed there. Instead, we consider a simpler approach based on predefined grid-based partitions of the parameter domain.

For each subdomain, a separate snapshot set and corresponding reduced basis are constructed. This allows the reduced spaces to better capture local solution behavior while keeping the reduced dimensions moderate.

CHAPTER

FOUR

CORRECTIVE SOURCE TERM APPROACH

While numerical models based on governing equations provide a principled description of physical systems, their accuracy may deteriorate when important physical effects are not fully captured or when discretization errors become significant. To mitigate these limitations, Hybrid Analysis and Modeling (HAM) approaches that combine physics-based models with data-driven corrections have received increasing attention in recent years.

In this work, we employ the Corrective Source Term Approach (CoSTA), introduced by Blakseth et al. [13, 14] and studied further for elasticity problems by [36]. CoSTA belongs to a class of HAM that combine equation-based modeling with data-driven modeling. The goal of such approaches is to retain the interpretability and structure of the physics-derived governing equations while leveraging data-driven techniques to compensate for modeling and discretization errors.

The central idea of CoSTA is to augment the governing equations with a corrective source term that accounts for discrepancies (in the residual) between the model and reference data. In the original formulation, this corrective term is generated by a deep neural network (DNN) trained on experimental measurements. In this thesis, however, we utilize ridge regression.

4.1 Conceptual Formulation

Consider a physical system described by the partial differential equation

$$\mathcal{L}(u) = f,$$

where $\mathcal{L}$ denotes the exact governing differential operator, u the exact solution, and f a source term.

In practice, the governing equations are not solved exactly, but rather approximated by a physics-based model. This can be interpreted as replacing the true operator $\mathcal{L}$ by a perturbed operator $\widetilde{\mathcal{L}}$, which accounts for modeling assumptions and discretization errors. The resulting approximate solution $\widetilde{u}$ therefore satisfies

$$\widetilde{\mathcal{L}}(\widetilde{u}) = f.$$

In general, $\widetilde{u}$ does not satisfy the true governing equation. Inserting $\widetilde{u}$ into the exact operator yields the residual

$$\widetilde{r} = f - \mathcal{L}(\widetilde{u}),$$

which quantifies the discrepancy between the approximate model and the true dynamics. In this work we assume linear operators for simplicity, although the method can be extended to non-linear operators as well. Using $\mathcal{L}(u) = f$ and linearity of $\mathcal{L}$, the residual can be expressed in terms of the solution error $\widetilde{e} = u - \widetilde{u}$ as

$$\widetilde{r} = \mathcal{L}(u) - \mathcal{L}(\widetilde{u}) = \mathcal{L}(u - \widetilde{u}) = \mathcal{L}(\widetilde{e}).$$

The CoSTA framework introduces a corrective source term σ that compensates for this discrepancy, leading to the corrected model

$$\widetilde{\mathcal{L}}(u_{\text{COSTA}}) = f + \sigma.$$

Since the exact operator $\mathcal{L}$ is not available in practice, we instead introduce the residual with respect to the perturbed operator,

$$\hat{\widetilde{r}} := \widetilde{\mathcal{L}}(\widetilde{e}), \qquad \widetilde{e} = u - \widetilde{u}.$$

If the corrective term is chosen as $\sigma = \hat{\widetilde{r}}$, then

$$\widetilde{\mathcal{L}}(u_{\text{COSTA}}) = f + \hat{\widetilde{r}} = \widetilde{\mathcal{L}}(\widetilde{u}) + \widetilde{\mathcal{L}}(u - \widetilde{u}) = \widetilde{\mathcal{L}}(u),$$

which implies $u_{\text{COSTA}} = u$ provided that $\widetilde{\mathcal{L}}$ is invertible. In practice, σ is approximated from data, yielding a corrected solution that improves the accuracy of the physics-based model while preserving its mathematical structure.

4.2 Discrete Formulation

After spatial and temporal discretization, the physics-based model typically yields a system of algebraic equations of the form

$$\mathbf{A}\mathbf{x}^{n+1} = \mathbf{b}(\mathbf{x}^n),$$

where $\mathbf{x}$ denotes the discrete state vector of the system and n the time step index. Within the CoSTA framework, this system is augmented by a corrective source term,

$$\mathbf{A}\mathbf{x}^{n+1}_{\text{COSTA}} = \mathbf{b}(\mathbf{x}^n_{\text{COSTA}}) + \boldsymbol{\sigma}^{n+1}.$$

To generate training data for the corrective model, reference solutions obtained from high-fidelity simulations or measurements are used. Let $\mathbf{x}_{\text{ref}}$ denote such a reference solution. The corrective source term is defined as the residual of the discrete model evaluated along the reference trajectory,

$$\boldsymbol{\sigma}^{n+1}_{\text{ref}} = \mathbf{A}\mathbf{x}^{n+1}_{\text{ref}} - \mathbf{b}(\mathbf{x}^n_{\text{ref}}).$$

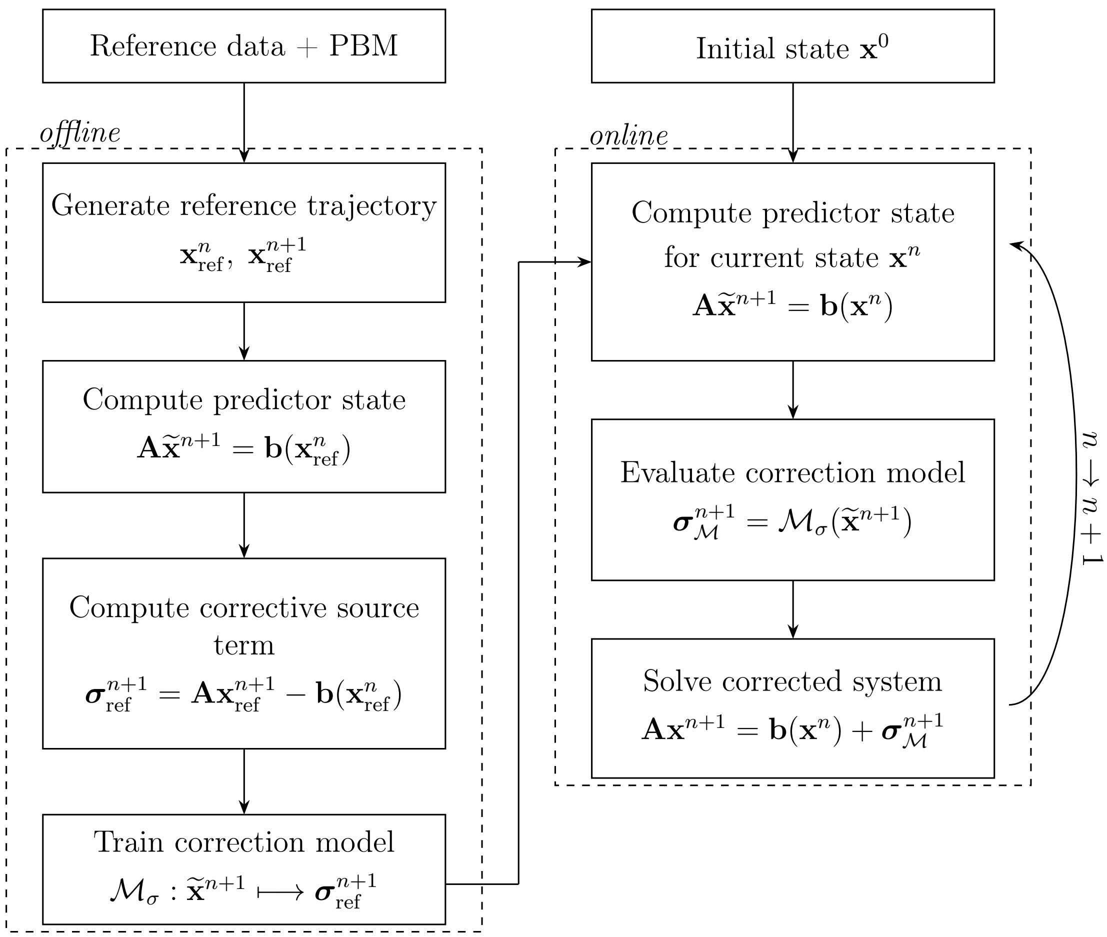

Figure 4.2.1: Offline–online workflow of the Corrective Source Term Approach (CoSTA).

To construct training inputs, a predictor state $\widetilde{\mathbf{x}}^{n+1}$ is obtained by evaluating the physics-based model along the reference trajectory,

$$\mathbf{A}\widetilde{\mathbf{x}}^{n+1} = \mathbf{b}(\mathbf{x}_{\mathrm{ref}}^{n}).$$

Using this relation, the corrective source term can equivalently be expressed as

$$\boldsymbol{\sigma}_{\mathrm{ref}}^{n+1} = \mathbf{A}\mathbf{x}_{\mathrm{ref}}^{n+1} - \mathbf{A}\widetilde{\mathbf{x}}^{n+1} = \mathbf{A}\left(\mathbf{x}_{\mathrm{ref}}^{n+1} - \widetilde{\mathbf{x}}^{n+1}\right),$$

which highlights its interpretation as the residual induced by the prediction error.

A data-driven model is then trained to approximate the mapping

$$\widetilde{\mathbf{x}}^{n+1} \longmapsto \boldsymbol{\sigma}_{\mathrm{ref}}^{n+1}.$$

In the original works, this mapping was approximated using a DNN. In this work, we instead employ linear ridge regression (see e.g., [37]). The ridge regression training problem is defined as

$$\mathbf{M}_{\sigma} \coloneqq \arg\min_{\mathbf{M}\in\mathbb{R}^{d_\sigma \times d_x}} \sum_{k=1}^{N} \left\| \boldsymbol{\sigma}_{\mathrm{ref},k} - \mathbf{M}\widetilde{\mathbf{x}}_k \right\|_{\ell^2}^2 + \lambda \left\| \mathbf{M} \right\|_F^2 .$$

Here, λ is a regularization parameter that must be tuned for the given problem. The resulting map is linear, computationally efficient, and regularized through an ℓ^2-penalty to reduce overfitting and improve numerical stability.

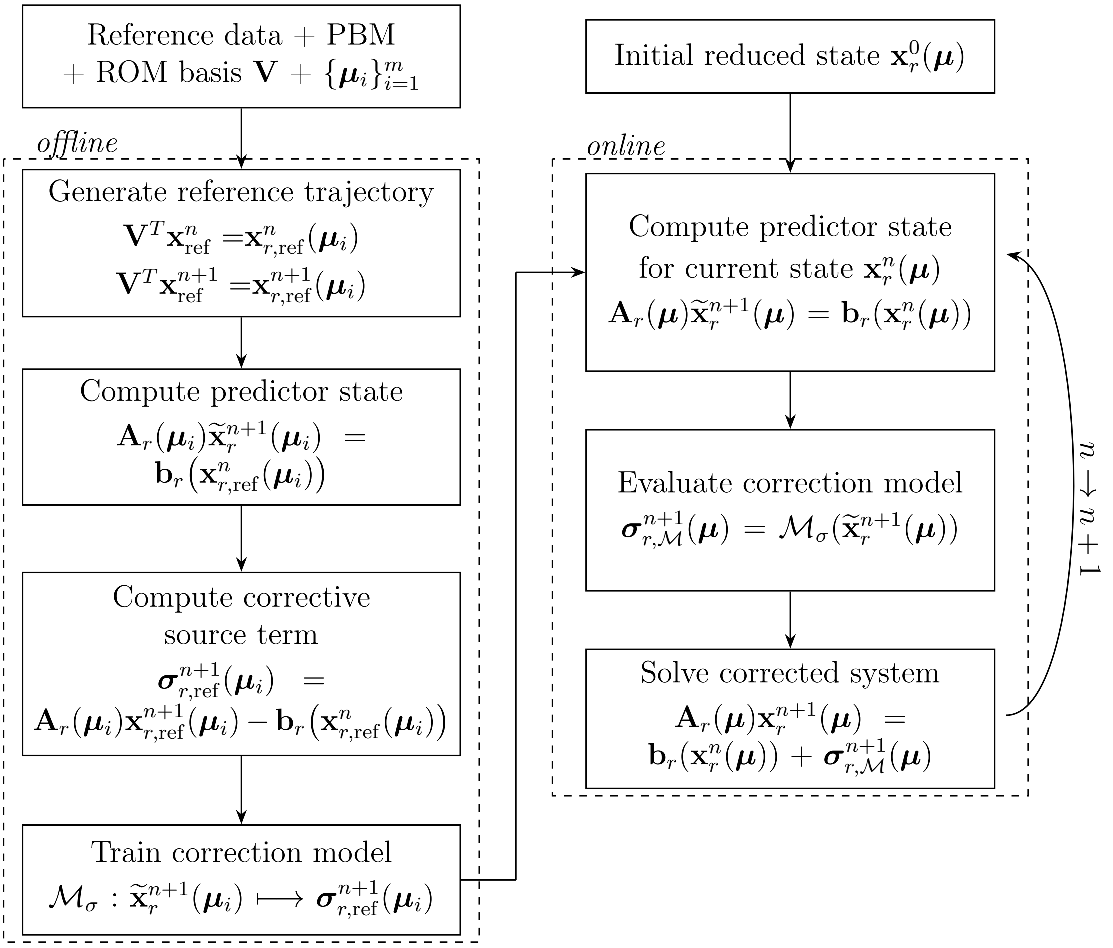

Figure 4.2.2: Offline–online workflow of CoSTA integrated with ROM. The $\boldsymbol{\mu}$ in the online stage is an arbitrary parameter from the parameter space $\mathscr{P}$.

The learned correction mapping is therefore given, for any input state $\mathbf{x}$, by

$$\mathcal{M}_\sigma(\mathbf{x}) = \mathbf{M}_\sigma \mathbf{x}.$$

During online simulation, the predictor is evaluated using the current state $\mathbf{x}^n$, and the learned correction is computed as

$$\boldsymbol{\sigma}^{n+1}_{\mathcal{M}} = \mathcal{M}_\sigma(\widetilde{\mathbf{x}}^{n+1}).$$

The correction is then inserted into the system,

$$\mathbf{A}\mathbf{x}^{n+1} = \mathbf{b}(\mathbf{x}^n) + \boldsymbol{\sigma}^{n+1}_{\mathcal{M}},$$

yielding a hybrid model that combines the physics-based solver with data-driven error correction.

4.2.1 Integration with the Reduced-Order Model

In this work, the CoSTA framework is combined with the reduced-order modeling approach described in chapter 3. For each parameter value $\boldsymbol{\mu}$, projection onto a reduced basis yields

$$\mathbf{A}_r(\boldsymbol{\mu})\,\mathbf{x}^{n+1}_r(\boldsymbol{\mu}) = \mathbf{b}_r(\mathbf{x}^n_r(\boldsymbol{\mu}); \boldsymbol{\mu})\,,$$

where $\mathbf{x}_r(\boldsymbol{\mu})$ denotes the reduced state vector.

Training data for the corrective model are generated from full-order model (FOM) simulations, treated as reference solutions. In principle, experimental or field measurements could also be used, but such data are not available in this work.

Let $\mathbf{x}_{\text{ref}}(\boldsymbol{\mu})$ denote the full-order reference state and let $\mathbf{V}$ be the POD basis matrix. The corresponding reduced reference coefficients are obtained by

$$\mathbf{x}_{r,\text{ref}}(\boldsymbol{\mu}) = \mathbf{V}^T \mathbf{x}_{\text{ref}}(\boldsymbol{\mu}).$$

Using the projected reference trajectory, the reduced corrective source term is defined as

$$\boldsymbol{\sigma}_{r,\text{ref}}^{n+1}(\boldsymbol{\mu}) = \mathbf{A}_r(\boldsymbol{\mu})\, \mathbf{x}_{r,\text{ref}}^{n+1}(\boldsymbol{\mu}) - \mathbf{b}_r\left(\mathbf{x}_{r,\text{ref}}^{n}(\boldsymbol{\mu}); \boldsymbol{\mu}\right),$$

which is the residual of the reduced physics-based model evaluated along the reduced reference trajectory.

To construct inputs for the correction model, we introduce the reduced predictor state

$$\mathbf{A}_r(\boldsymbol{\mu})\, \widetilde{\mathbf{x}}_r^{n+1}(\boldsymbol{\mu}) = \mathbf{b}_r\left(\mathbf{x}_{r,\text{ref}}^{n}(\boldsymbol{\mu}); \boldsymbol{\mu}\right).$$

Hence,

$$\boldsymbol{\sigma}_{r,\text{ref}}^{n+1}(\boldsymbol{\mu}) = \mathbf{A}_r(\boldsymbol{\mu})\, \mathbf{x}_{r,\text{ref}}^{n+1}(\boldsymbol{\mu}) - \mathbf{A}_r(\boldsymbol{\mu})\, \widetilde{\mathbf{x}}_r^{n+1}(\boldsymbol{\mu}) = \mathbf{A}_r(\boldsymbol{\mu}) \Big(\mathbf{x}_{r,\text{ref}}^{n+1}(\boldsymbol{\mu}) - \widetilde{\mathbf{x}}_r^{n+1}(\boldsymbol{\mu}) \Big).$$

A correction mapping is then trained to approximate

$$\widetilde{\mathbf{x}}_r^{n+1}(\boldsymbol{\mu}) \longmapsto \boldsymbol{\sigma}_{r,\text{ref}}^{n+1}(\boldsymbol{\mu}).$$

During online simulation, the reduced predictor is first computed from the current reduced state,

$$\mathbf{A}_r(\boldsymbol{\mu})\, \widetilde{\mathbf{x}}_r^{n+1}(\boldsymbol{\mu}) = \mathbf{b}_r(\mathbf{x}_r^{n}(\boldsymbol{\mu}); \boldsymbol{\mu})$$

then the correction is evaluated from the learned mapping

$$\boldsymbol{\sigma}_{r,\mathcal{M}}^{n+1}(\boldsymbol{\mu}) = \mathcal{M}_\sigma\left(\widetilde{\mathbf{x}}_r^{n+1}(\boldsymbol{\mu})\right),$$

finally, the corrected system is solved

$$\mathbf{A}_r(\boldsymbol{\mu})\, \mathbf{x}_r^{n+1}(\boldsymbol{\mu}) = \mathbf{b}_r(\mathbf{x}_r^{n}(\boldsymbol{\mu}); \boldsymbol{\mu}) + \boldsymbol{\sigma}_{r,\mathcal{M}}^{n+1}(\boldsymbol{\mu}).$$

In this way, the resulting hybrid model retains ROM structure and efficiency while introducing data-driven corrections for reduction and discretization errors.

Note that in practice, a factorization of $\mathbf{A}_r$, such as an LU (lower–upper) factorization, is precomputed, making the second solve inexpensive.

CHAPTER

FIVE

NUMERICAL EXPERIMENTS

In this chapter, we first verify the finite element implementation for Darcy flow using a manufactured solution in Section 5.1.1. The convergence study confirms that the spatial and temporal discretizations exhibit the expected theoretical rates of convergence.

Two parameterized Darcy flow problems are then considered. In Section 5.2, the parameter controls a geometric deformation of the computational domain through displacement of the upper-right corner of the square. In Section 5.3, the parameter instead defines a localized region of increased permeability.

The first example provides a relatively smooth parameter dependence with weaker non-affinity, allowing the effects of geometry-induced parameterization to be studied in both stationary and transient settings. The second example introduces a more localized and strongly non-affine parameter dependence, making it significantly more challenging for the reduced-order approximation.

Reduced-order model errors are measured with $L^2(0,T)$ in time, and L^2 and the energy norm in space, as defined in (3.2), (3.3), and (3.4). The L^2-fit operator error is defined in (3.7), while the corresponding solution errors for velocity and pressure are given in (3.8) and (3.9), respectively.

For both examples, the velocity error is measured in both the L^2 norm and the energy norm defined in (3.1). For the moving-corner example, the permeability is constant and the energy norm reduces to the standard $H(\text{div})$ norm. For the Gaussian permeability example, the permeability varies spatially, and the energy norm is therefore evaluated using a fixed reference parameter and denoted by E.

The numerical experiments presented in this chapter are performed using the finite element library *NGSolve* [38] together with the model reduction framework *pyMOR* [39].

5.1 Manufactured Solution

To verify the implementation and assess the expected convergence behavior, we consider the manufactured-solution framework introduced in Section 2.5.

We consider the unit square $\Omega = (0,1)^2$, with permeability tensor $\mathbf{K} = \mathbf{I}$ and storage coefficient $S > 0$. We define manufactured solution as

$$p_{\mathrm{MAN}}(x,y,t) = t^2 e^{-t^2} \left(x^5 - 10x^3y^2 + 5xy^4\right),$$

which is smooth in both space and time. The corresponding velocity field is obtained from Darcy's law,

$$\mathbf{u}_{\mathrm{MAN}} = -\nabla p_{\mathrm{MAN}},$$

yielding

$$\mathbf{u}_{\mathrm{MAN}}(x,y,t) = -t^2 e^{-t^2} \begin{pmatrix} 5x^4 - 30x^2y^2 + 5y^4 \\ -20x^3y + 20xy^3 \end{pmatrix}.$$

By construction, the velocity field is divergence-free,

$$\nabla \cdot \mathbf{u}_{\mathrm{MAN}} = 0.$$

Substituting the manufactured solution into the transient mass conservation equation,

$$S\,\partial_t p + \nabla \cdot \mathbf{u}_{\mathrm{MAN}} = f,$$

we obtain the corresponding source term

$$f(x,y,t) = S\,\partial_t p_{\mathrm{MAN}}(x,y,t) = -2St(1-t^2)e^{-t^2}\left(x^5 - 10x^3y^2 + 5xy^4\right).$$

The boundary conditions are chosen consistently with the manufactured solution by prescribing the exact pressure on Γ_p and the corresponding normal flux on $\Gamma_{\mathbf{u}}$.

5.1.1 Convergence Results

The spatial convergence results are presented in Table 5.1.1 and Figure 5.1.1. The estimated orders of convergence (EOC) for both pressure and velocity in the L^2 norm agree well with the theoretical rates discussed in Section 2.5, for order $k = 2$. In contrast, the observed convergence in the $H(\mathrm{div})$ norm is higher than expected. A plausible explanation is that the manufactured velocity field is divergence-free, which suppresses the contribution from the divergence term in the $H(\mathrm{div})$ norm. As a result, the error is dominated by the L^2 component, leading to an apparently higher convergence rate.

Temporal convergence results are shown in Table 5.1.2 and Figure 5.1.2. The estimated order of convergence is consistently close to one across all error measures, in agreement with the expected first-order accuracy of the implicit Euler scheme.

Visualizations of the numerical solutions together with the computational mesh are provided in Figure 5.1.3 and Figure 5.1.4. For the pressure field, noticeable mesh-dependent artifacts appear for coarse meshes, particularly near element boundaries. These artifacts diminish as the mesh is refined. For the velocity field, the differences between discretizations are less visually pronounced, although the quantitative error analysis confirms consistent convergence.

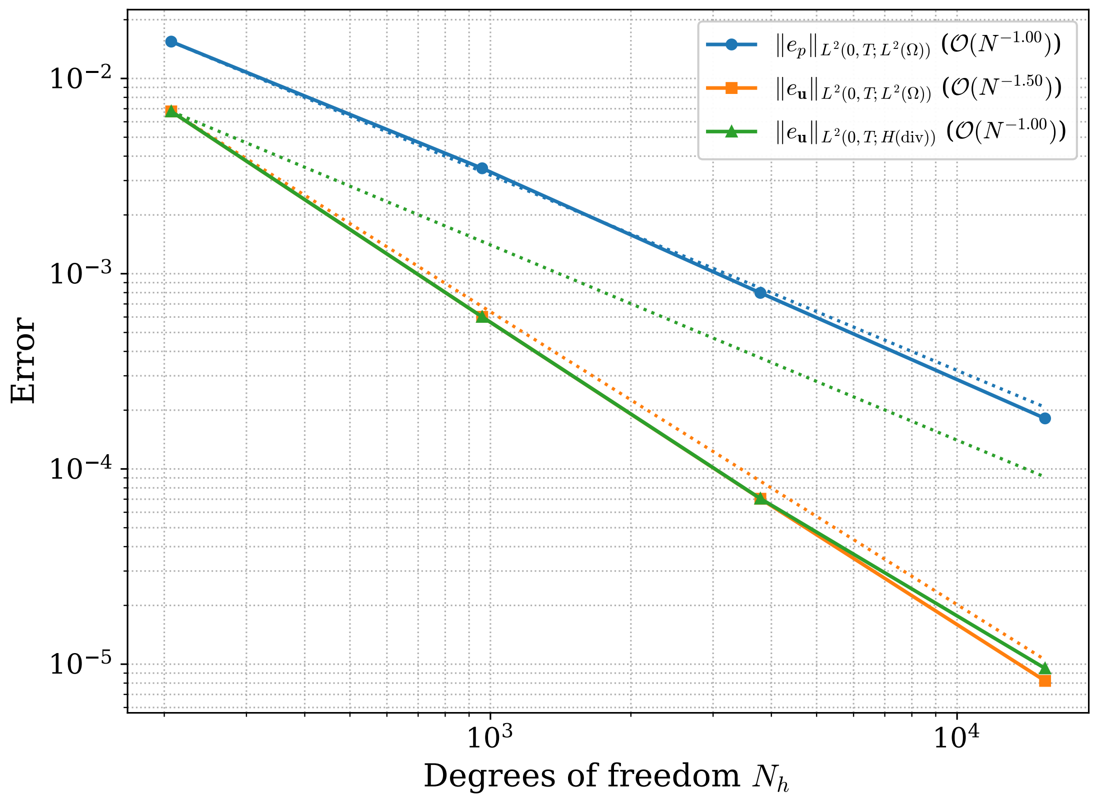

Figure 5.1.1: *Transient Darcy flow - Convergence Test.* Spatial convergence of the finite element discretization for transient Darcy flow. The errors are shown as functions of the number of degrees of freedom. The dashed lines indicate the theoretical convergence rates in the respective norms.

Table 5.1.1: *Transient Darcy flow - Convergence Test.* Spatial convergence of the finite element discretization for transient Darcy flow. Estimated orders of convergence (EOC) are given in parentheses.

h	N_p	N_u	$\|e_p\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;H(\mathrm{div};\Omega))}$
0.32	54	153	1.5×10^{-2} (n/a)	6.8×10^{-3} (n/a)	6.8×10^{-3} (n/a)
0.16	264	696	3.5×10^{-3} (0.9)	6.0×10^{-4} (1.6)	6.0×10^{-4} (1.6)
0.08	1062	2727	8.0×10^{-4} (1.1)	7.0×10^{-5} (1.6)	7.1×10^{-5} (1.6)
0.04	4368	11070	1.8×10^{-4} (1.0)	8.2×10^{-6} (1.5)	9.5×10^{-6} (1.4)

Table 5.1.2: *Transient Darcy flow - Convergence Test.* Temporal convergence of the finite element discretization using the implicit Euler method for transient Darcy flow. Estimated orders of convergence (EOC) are given in parentheses.

Δt	N_t	$\|e_p\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;H(\mathrm{div};\Omega))}$
0.040	25	2.1×10^{-4} (n/a)	1.6×10^{-3} (n/a)	1.9×10^{-2} (n/a)
0.020	50	1.1×10^{-4} (1.0)	8.1×10^{-4} (1.0)	9.6×10^{-3} (1.0)
0.010	100	5.5×10^{-5} (1.0)	4.1×10^{-4} (1.0)	4.8×10^{-3} (1.0)
0.005	200	2.8×10^{-5} (1.0)	2.0×10^{-4} (1.0)	2.4×10^{-3} (1.0)

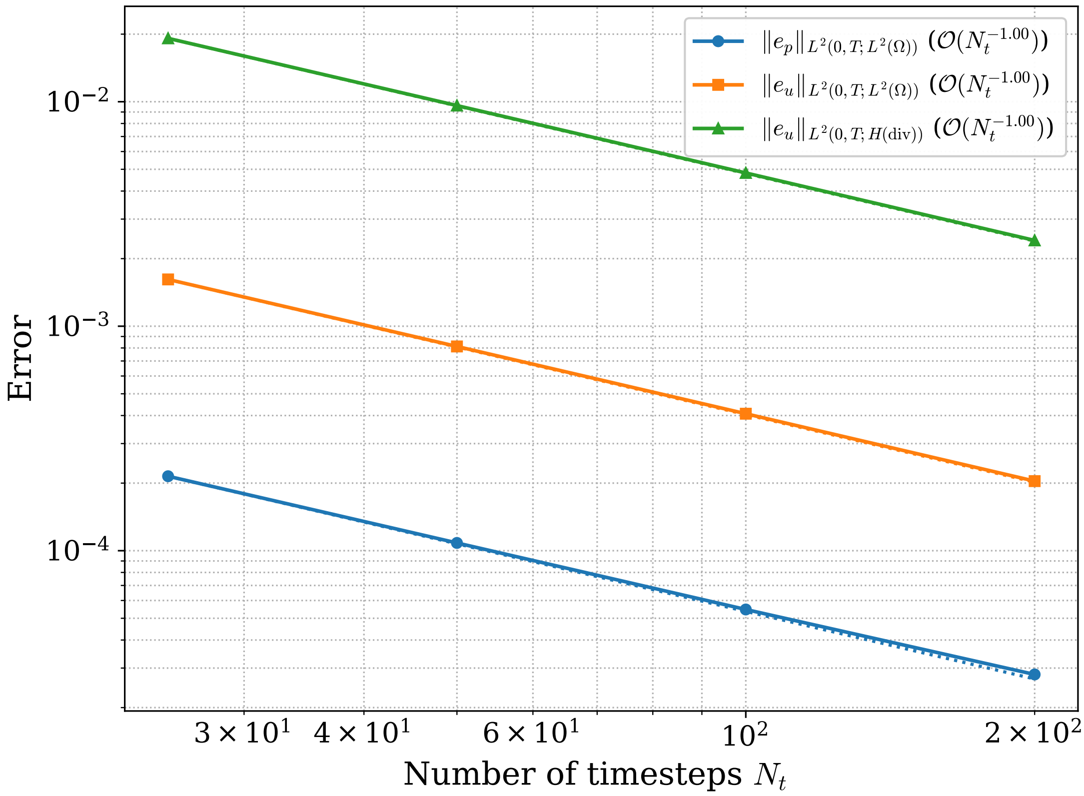

Figure 5.1.2: *Transient Darcy flow - Convergence Test.* Temporal convergence of the implicit Euler time discretization for transient Darcy flow. The errors are shown as functions of the number of time steps. The dashed lines indicate the expected first-order convergence rate.

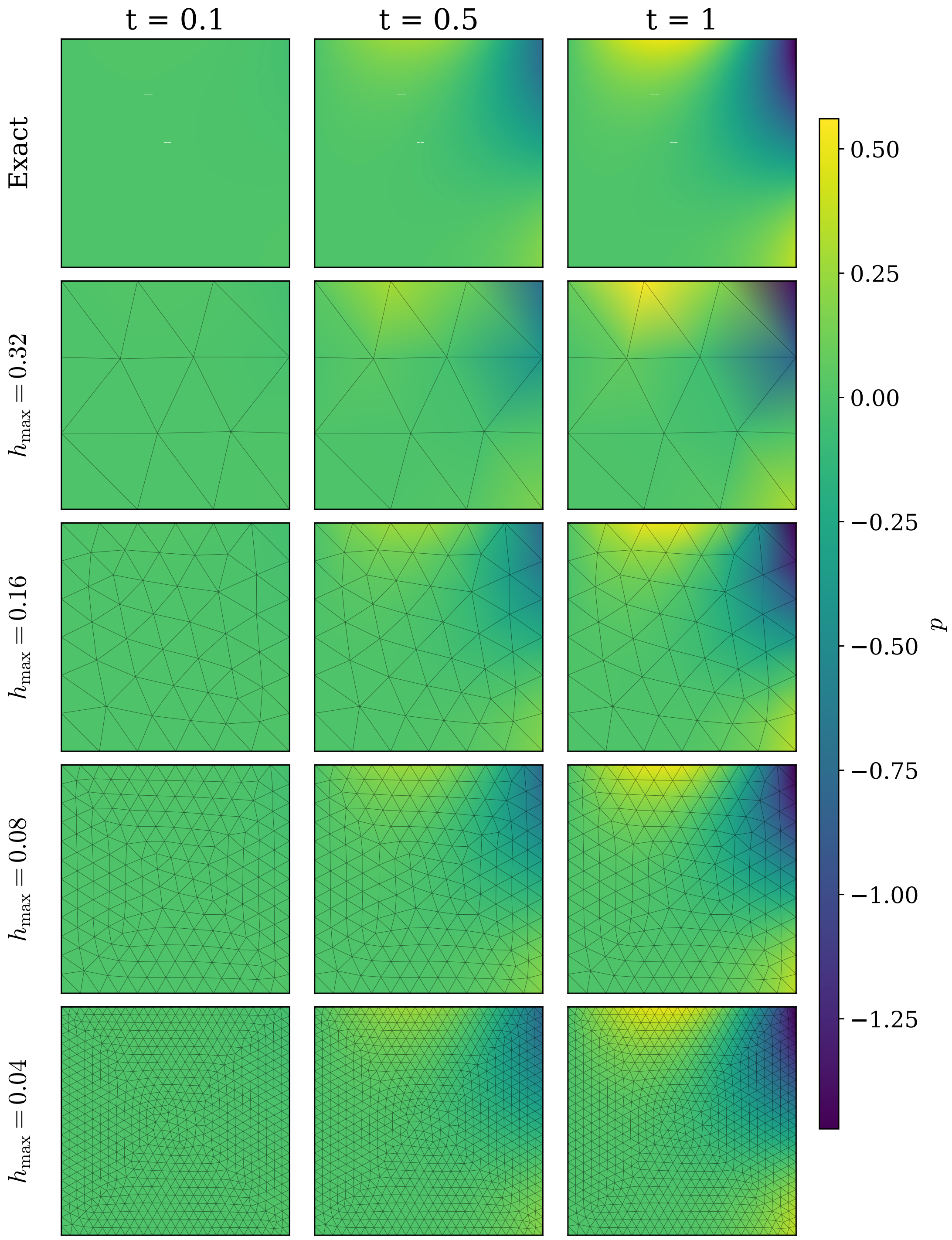

Figure 5.1.3: *Transient Darcy flow - Convergence Test.* Exact pressure solution and corresponding finite element approximations for different spatial discretizations. The computational mesh is overlaid to illustrate the influence of mesh refinement on the numerical approximation.

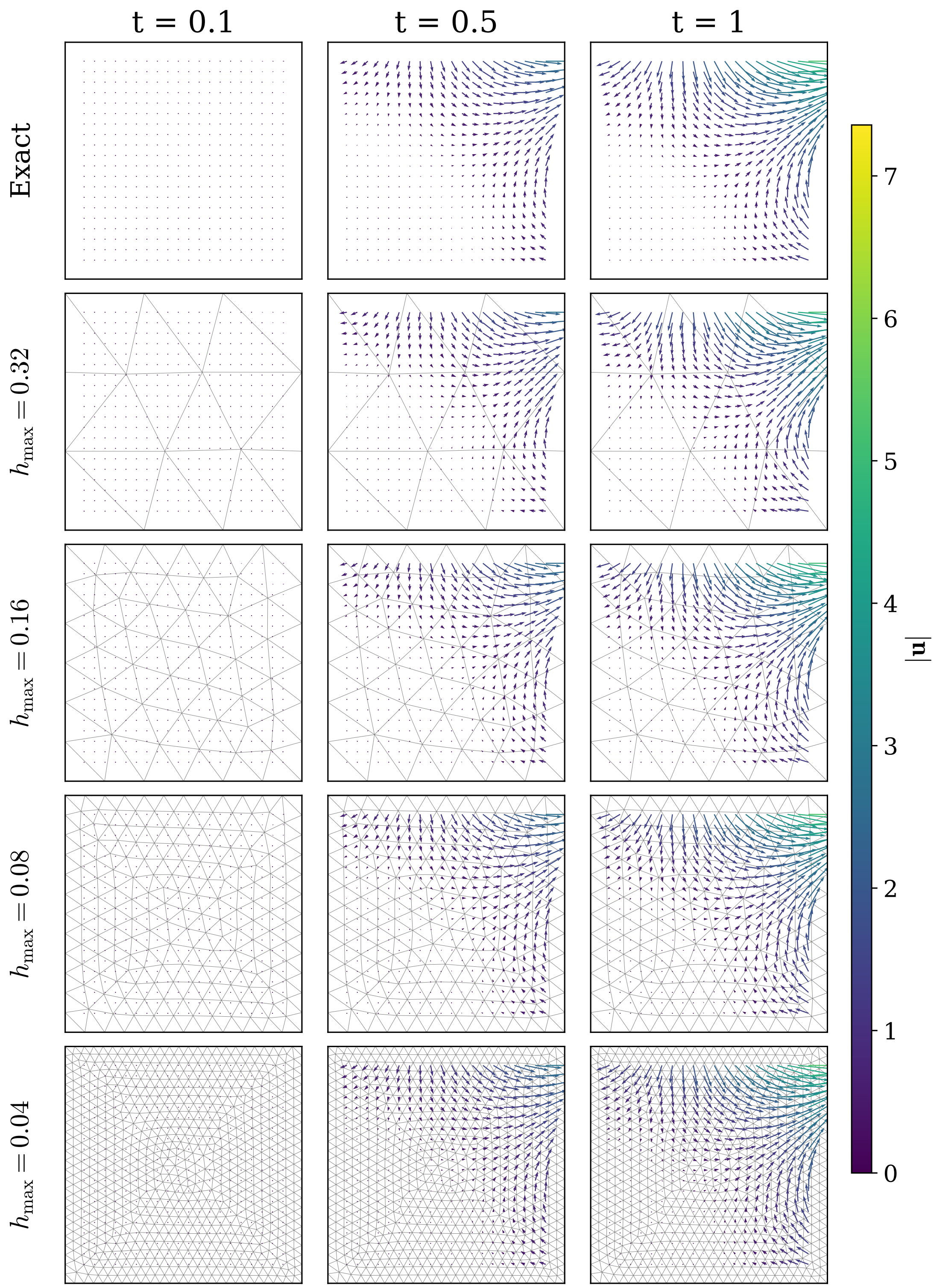

Figure 5.1.4: *Transient Darcy flow - Convergence Test.* Exact velocity field and corresponding finite element approximations for different spatial discretizations. The computational mesh is overlaid to illustrate the effect of mesh refinement on the velocity approximation.

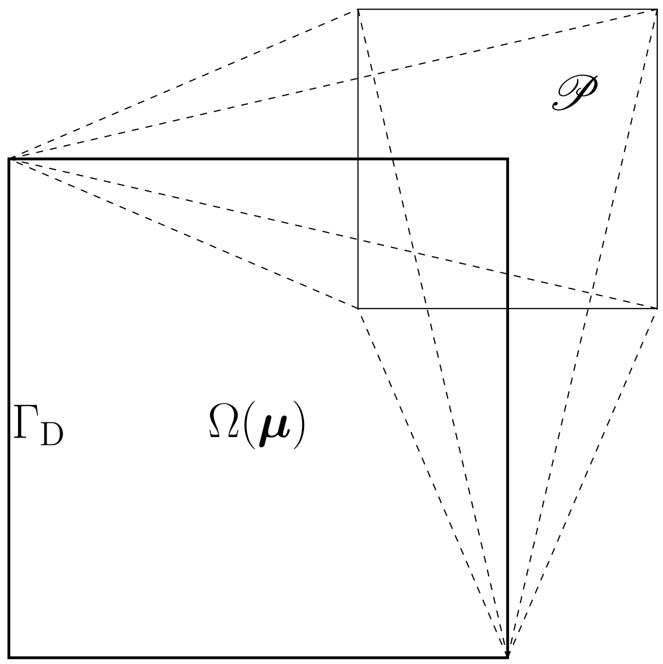

Figure 5.2.1: *Square with moving corner.* Parameter-dependent domain $\Omega(\boldsymbol{\mu})$ obtained by displacing the upper-right corner of the unit square. The parameter domain $\mathscr{P}$ is illustrated, and Dirichlet boundary conditions are imposed on Γ_{D}.

5.2 Square with moving corner

This example is adapted from [12], replacing quasi-static elasticity with Darcy flow as described in (2.1) and (2.2). Let $\Omega(\boldsymbol{\mu}) \subset \mathbb{R}^2$ denote a parameter-dependent domain, where $\boldsymbol{\mu} \in \mathscr{P} = [-0.3, 0.3]^2$ controls the displacement of the upper-right corner of the unit square, as illustrated in Figure 5.2.1.

We consider the manufactured solution

$$p(x, y, t) = \alpha(t)(x^2 - y^2),$$

with

$$\alpha(t) = 1 - e^{-t/t_r}, \qquad t_r = 0.25.$$

The permeability is taken as $\mathbf{K} = \mathbf{I}$ and the source term is $f = 0$. Dirichlet boundary conditions are prescribed as

$$p_D(x, y, t) = \alpha(t)(x^2 - y^2) \quad \text{on } \partial\Omega(\boldsymbol{\mu}) \times (0, T].$$

and initial condition

$$p(\cdot, 0) = 0.$$

To perform computations on this problem, we want to solve the system in a reference domain $\widehat{\Omega}$. To achieve this, we define the mapping

$$\varphi_{\boldsymbol{\mu}} : \widehat{\Omega} \to \Omega(\boldsymbol{\mu}).$$

We let $(\xi, \eta) \in \widehat{\Omega} = [0, 1]^2$ and define

$$\varphi_{\boldsymbol{\mu}}(\xi, \eta) = (\xi + \xi\eta\mu_1,\ \eta + \xi\eta\mu_2).$$

The Jacobian of the mapping is

$$J_{\varphi_{\boldsymbol{\mu}}}(\xi,\eta) = \begin{pmatrix} 1+\eta\mu_1 & \xi\mu_1 \\ \eta\mu_2 & 1+\xi\mu_2 \end{pmatrix}, \qquad \det(J_{\varphi_{\boldsymbol{\mu}}}(\xi,\eta)) = 1 + \xi\mu_2 + \eta\mu_1.$$

Because each entry of $J_{\varphi_{\boldsymbol{\mu}}}$ and $\det J_{\varphi_{\boldsymbol{\mu}}}$ contains products of the form $\xi\mu_1$, $\eta\mu_2$, and mixed terms in (ξ,η) and $\boldsymbol{\mu}$, the mapping depends non-affinely on the parameters. As a consequence, the transformed bilinear form involves $J_{\varphi_{\boldsymbol{\mu}}}^{-1}$, $J_{\varphi_{\boldsymbol{\mu}}}^{-T}$, and $\det J_{\varphi_{\boldsymbol{\mu}}}$, all of which inherit this non-affine dependence. Therefore, the stiffness matrix $\mathbf{A}_h(\boldsymbol{\mu})$ cannot be expressed as an affine combination of parameter-independent matrices $\mathbf{A}_h^q$. This geometric non-affinity motivates the use of the L^2-based affinization techniques introduced in Section 3.3.

In the stationary case, the ramp function is removed, yielding a time-independent solution. In this setting, the velocity field is divergence-free, consistent with the absence of a source term.

5.2.1 Finding discretization

To determine suitable discretization parameters for the numerical experiments, a discretization study was performed using the parameter value $\boldsymbol{\mu} = (0.3, 0.3)$. The resulting relative errors are reported in Table 5.2.1, where the errors are computed with respect to a reference solution obtained using $h_{\max} = 0.04$ and $\Delta t = 0.00125$.

Based on these results, we choose $h_{\max} = 0.16$ and $\Delta t = 0.005$, which yield relative errors below 1% in the considered norms. Note that this study is intended as a practical discretization assessment rather than a fully rigorous convergence analysis, since the comparison is performed only for a single parameter value.

5.2.2 Stationary

In this section, we present numerical results for stationary Darcy flow for square with moving corner. The reduced-order models are trained using a 16×16 equidistant sampling of the parameter domain.

The relative L^2 fit errors for both the operator and the resulting solutions are presented in Figure 5.2.2. The operator approximation converges clearly with increasing polynomial degree P, reflecting the improved accuracy of the polynomial representation of the parameter dependence in the stiffness matrix. In contrast, the solution errors stagnate beyond approximately $P = 4$. This indicates that, although the operator approximation continues to improve, the induced error in the solution does not decrease correspondingly. The results therefore suggest that the solution is relatively insensitive to further improvements in the operator approximation, and that the dominant approximation error is already captured at moderate polynomial degrees.

The reduced-order model errors for pressure and velocity are shown in Figure 5.2.3 and Figure 5.2.4, respectively. In both cases, a clear interaction between the polynomial degree P and the number of retained modes r is observed, where reducing the error requires increasing both quantities. Increasing the total number of modes from $r = 12$ to $r = 18$ yields only marginal improvement, and similarly, little additional reduction is

Table 5.2.1: *Square with moving corner.* Relative discretization errors for different spatial and temporal resolutions, computed with respect to a reference solution obtained using $h_{\max} = 0.04$ and $\Delta t = 0.00125$. The comparison is performed for the parameter value $\boldsymbol{\mu} = (0.3, 0.3)$.

$h_{\max}$	Δt	$\|e_p\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;H(\mathrm{div};\Omega))}$
0.32	0.04	2.4×10^{-2}	5.2×10^{-3}	3.6×10^{-2}
0.32	0.02	2.3×10^{-2}	3.9×10^{-3}	2.6×10^{-2}
0.32	0.01	2.3×10^{-2}	3.1×10^{-3}	2.2×10^{-2}
0.32	0.005	2.3×10^{-2}	2.9×10^{-3}	2.1×10^{-2}
0.16	0.04	5.7×10^{-3}	4.6×10^{-3}	3.5×10^{-2}
0.16	0.02	5.2×10^{-3}	3.0×10^{-3}	2.3×10^{-2}
0.16	0.01	5.0×10^{-3}	1.8×10^{-3}	1.5×10^{-2}
0.16	0.005	4.9×10^{-3}	1.1×10^{-3}	8.9×10^{-3}
0.08	0.04	3.2×10^{-3}	4.5×10^{-3}	3.5×10^{-2}
0.08	0.02	2.2×10^{-3}	2.9×10^{-3}	2.3×10^{-2}
0.08	0.01	1.5×10^{-3}	1.7×10^{-3}	1.4×10^{-2}
0.08	0.005	1.3×10^{-3}	8.3×10^{-4}	7.9×10^{-3}

obtained for polynomial degrees above $P = 4$. This suggests that the approximation error is limited by a combination of basis truncation and operator approximation accuracy.

The relative discarded POD energy as a function of the number of retained modes is shown in Figure 5.2.5. The energy decays rapidly, indicating that most of the snapshot information is captured by a small number of modes. This rapid decay is reflected in the ROM error, which suggests that the discarded energy is a useful predictor of the final approximation accuracy for this problem.

Visual comparisons between the full-order and reduced-order solutions for pressure and velocity are provided in Figure 5.2.6 and Figure 5.2.7, respectively. Both reduced models capture the main features of the solution well. The case with $r = 9$ is nearly indistinguishable from the full-order solution at the plotted scale, while the $r = 6$ case shows slightly larger but still localized discrepancies.

The online speed-up of the reduced-order model is shown in Figure 5.2.8. A clear dependence on the polynomial degree P is observed, with higher degrees leading to reduced speed-up due to the increased cost of evaluating the affine decomposition. Increasing the number of modes r also leads to a slight reduction in speed-up, as the size of the reduced system increases. However, since the reduced dimensions remain small, this effect is relatively minor.

Overall, the results demonstrate that the proposed reduced-order model accurately approximates the stationary moving-corner problem using relatively small reduced dimensions and moderate polynomial degrees. The approximation errors decrease consistently with increasing reduced dimension and polynomial degree, although diminishing returns are observed beyond approximately $P = 4$. The rapid decay of the POD energy further indicates that the solution manifold is well suited for reduced basis approximation, while substantial computational speed-ups are achieved relative to the full-order model.

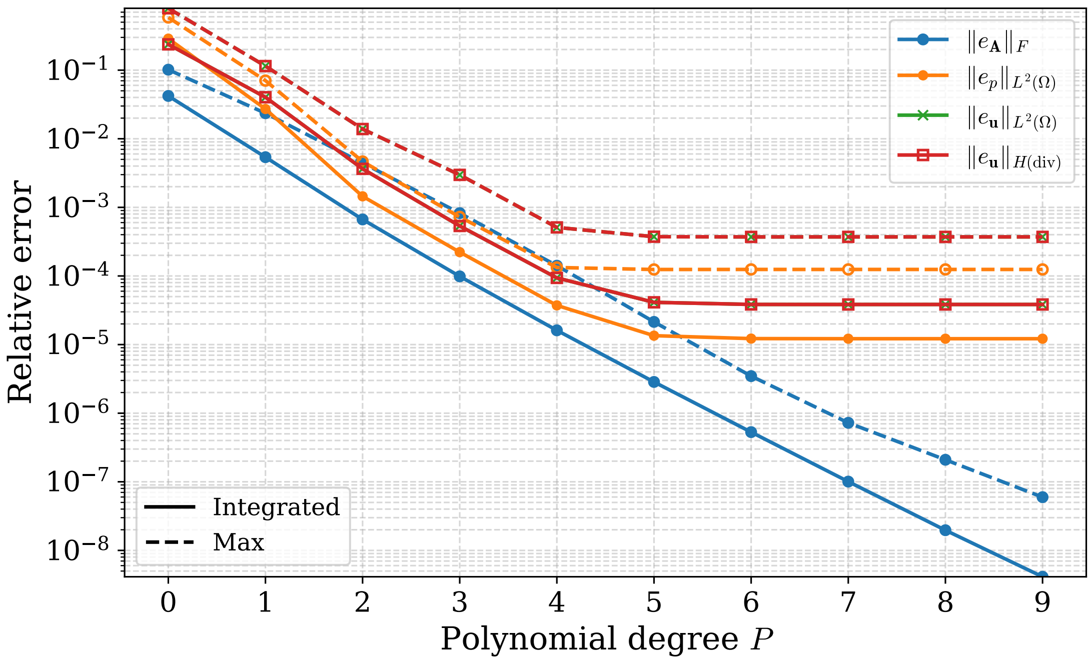

Figure 5.2.2: *Square with moving corner (stationary).* Relative operator and solution errors obtained from the L^2-fit as functions of the polynomial degree P. Errors are computed using Gauss–Lobatto quadrature with 256 integration points over the parameter domain.

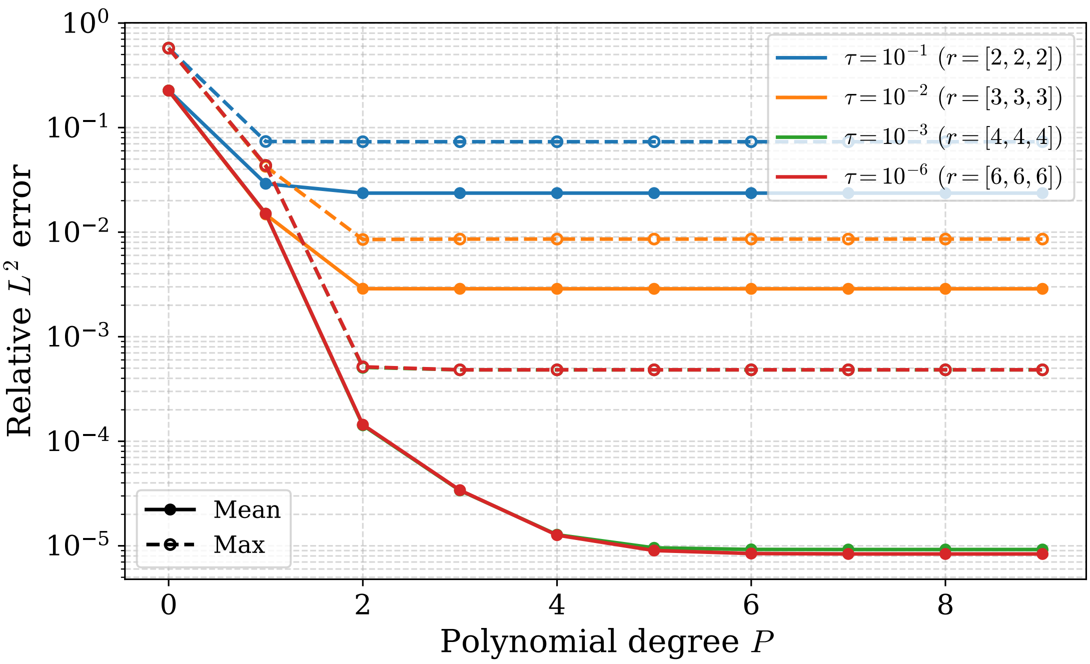

Figure 5.2.3: *Square with moving corner (stationary).* Relative pressure error in the L^2 norm as a function of the polynomial degree P and reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$. The model is trained using 256 parameter samples and evaluated on 256 randomly sampled test points. No significant improvement is observed when decreasing the tolerance from $\tau = 10^{-3}$ to $\tau = 10^{-6}$.

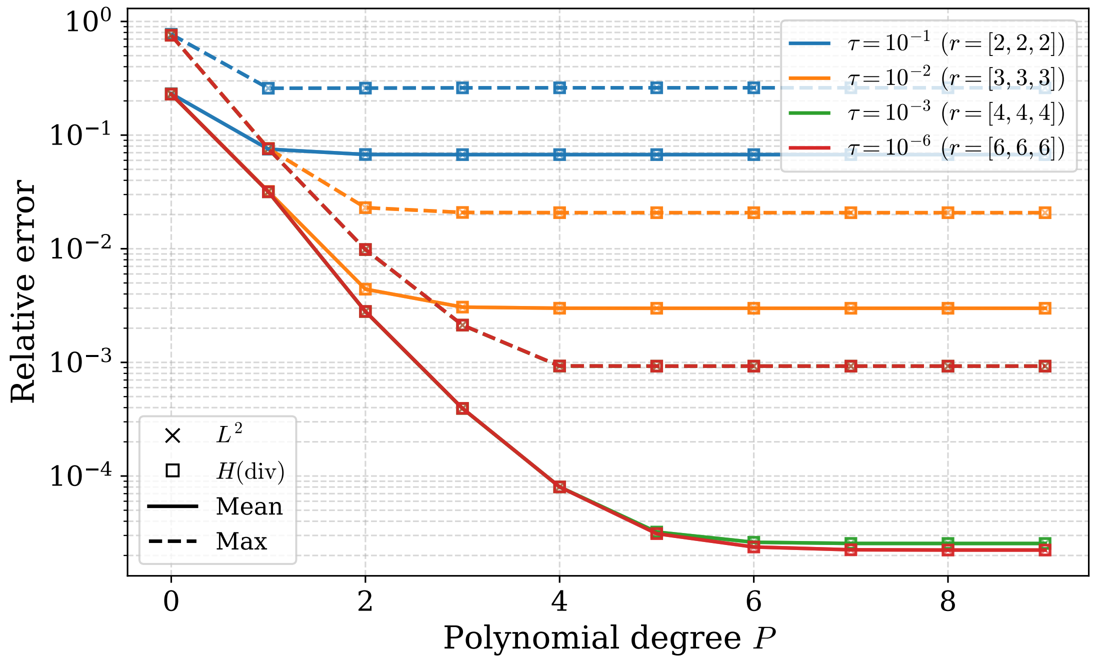

Figure 5.2.4: *Square with moving corner (stationary).* Relative velocity error in the L^2 and $H(\mathrm{div})$ norms as a function of the polynomial degree P and reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$. Errors are evaluated on 256 random test points. The L^2 and $H(\mathrm{div})$ errors coincide almost exactly, indicating a negligible contribution from the divergence term.

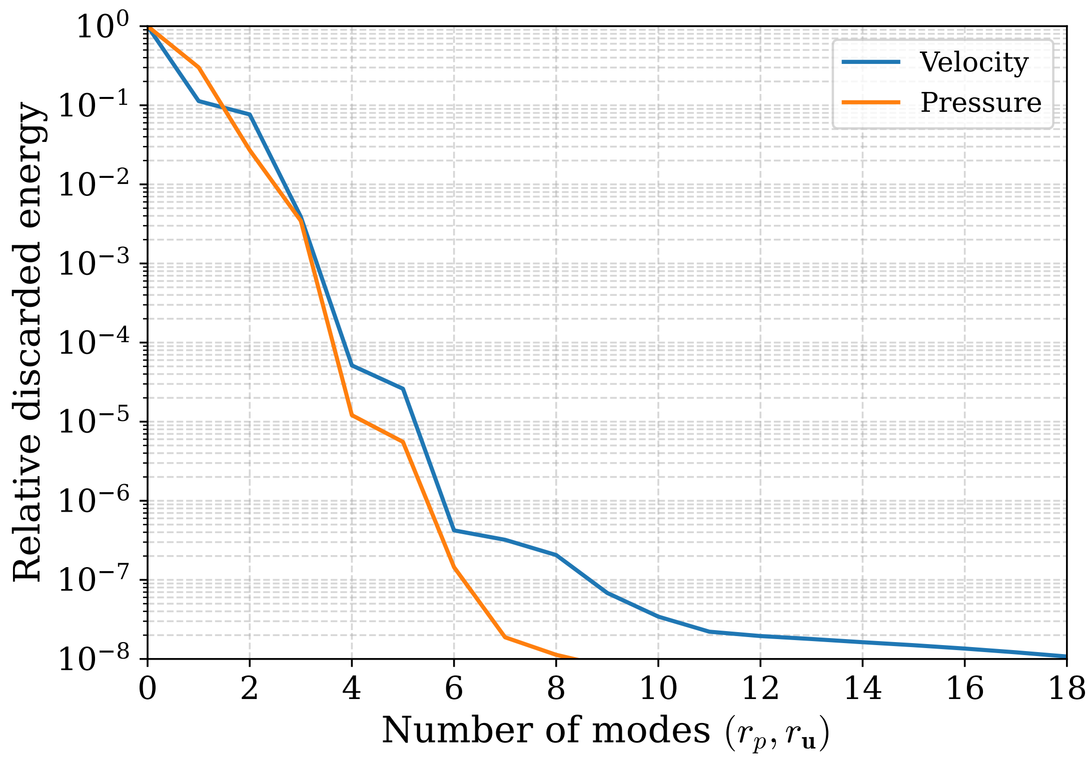

Figure 5.2.5: *Square with moving corner (stationary).* Relative discarded POD energy as a function of the number of retained modes r, computed from 256 training samples.

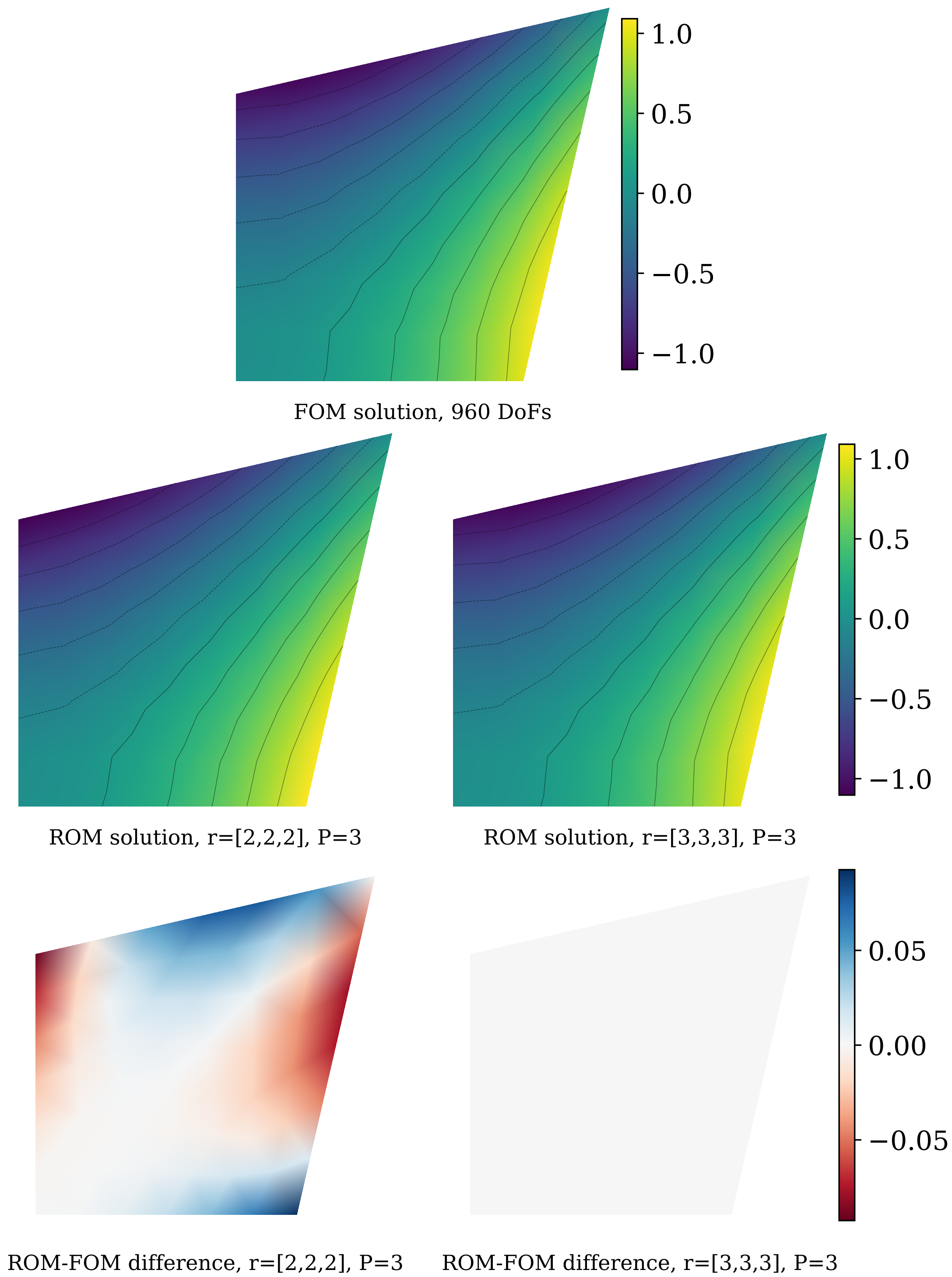

Figure 5.2.6: *Square with moving corner (stationary).* Comparison between the full-order and reduced-order pressure solutions. The top row shows the full-order solution, the middle row shows the reduced-order approximations, and the bottom row shows the pointwise differences between them. Results are shown for two different reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$ at polynomial degree $P = 3$.

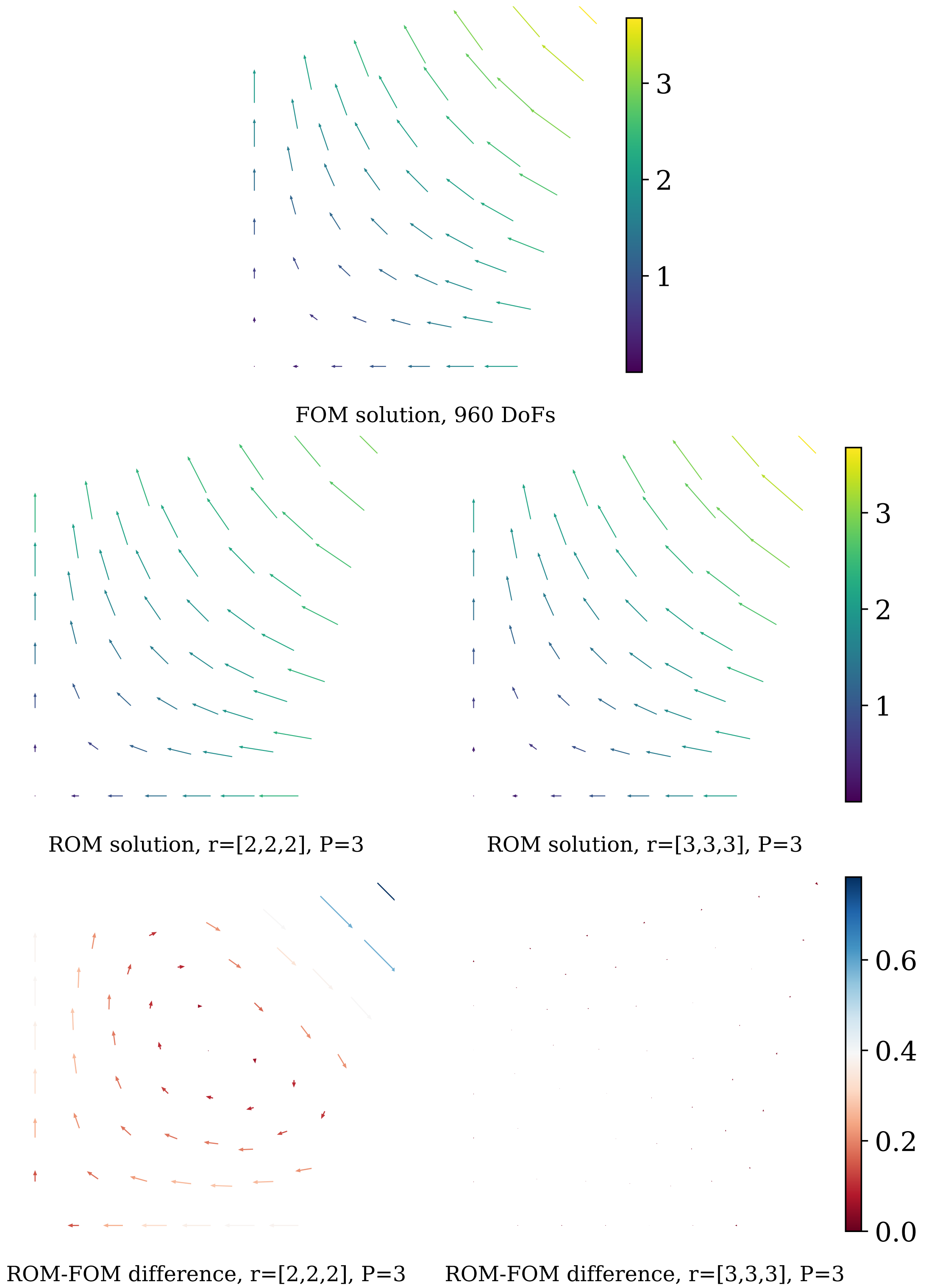

Figure 5.2.7: *Square with moving corner (stationary).* Comparison between the full-order and reduced-order velocity fields. The top row shows the full-order solution, the middle row shows the reduced-order approximations, and the bottom row shows the pointwise differences between the full-order and reduced-order solutions. Results are shown for two different reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$ at polynomial degree $P = 3$.

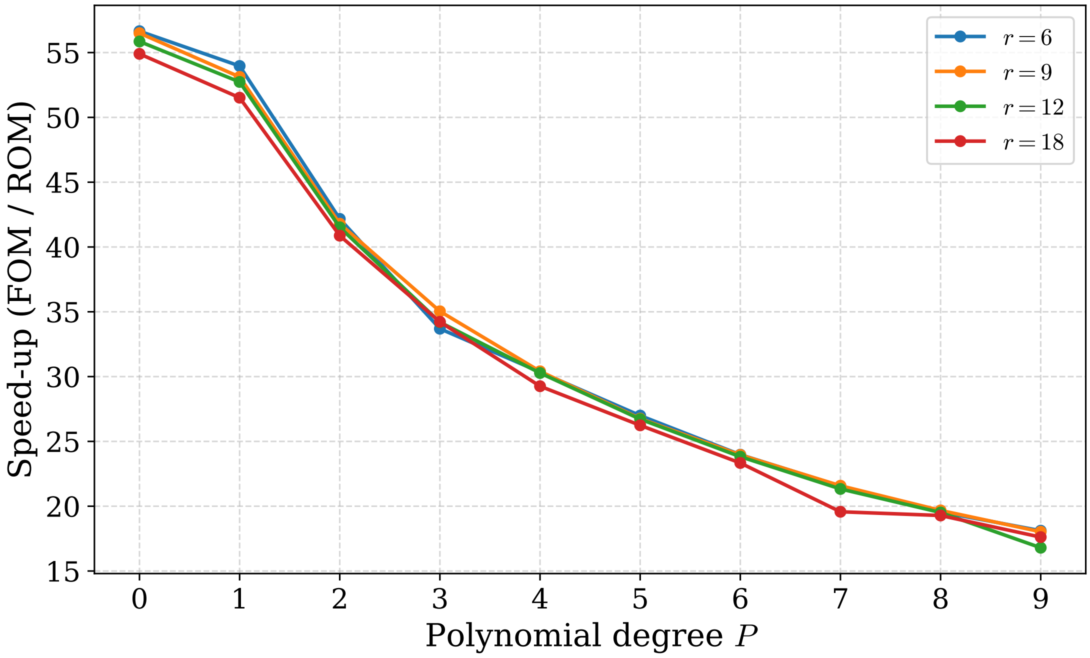

Figure 5.2.8: *Square with moving corner (stationary).* Computational speed-up of the affinized reduced-order model relative to the full-order model

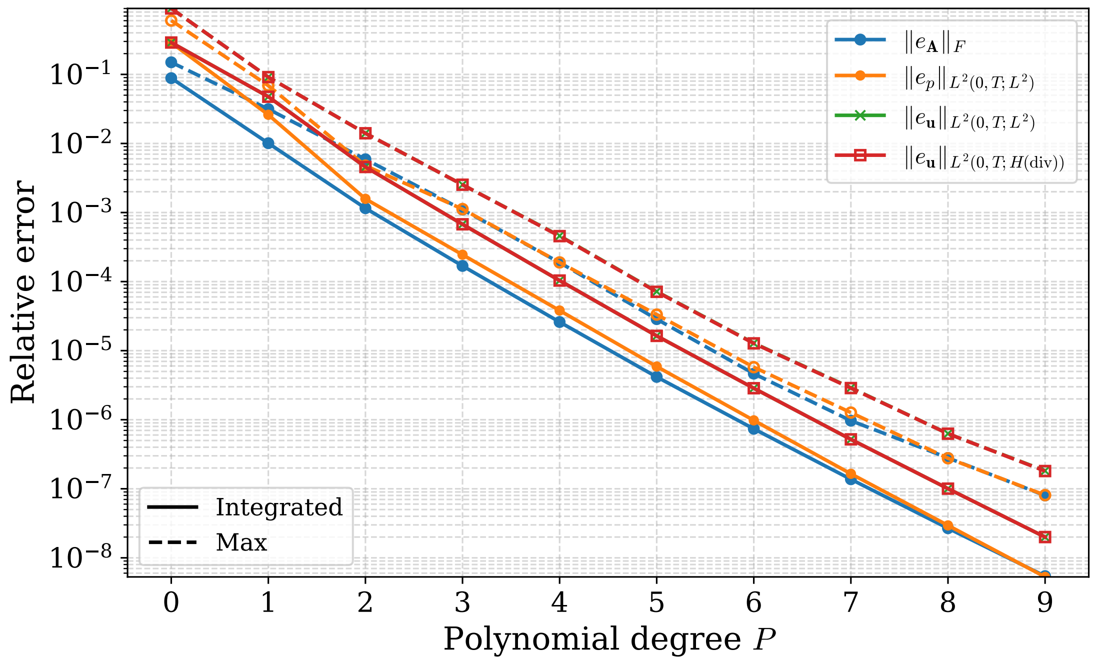

Figure 5.2.9: *Square with moving corner (transient).* Relative operator and solution errors obtained from the L^2-fit as functions of the polynomial degree P. Errors are computed using Gauss–Lobatto quadrature with 256 integration points over the parameter domain.

5.2.3 Transient

The transient results exhibit behavior similar to that observed in the stationary case, although slightly larger reduced dimensions are generally required to achieve comparable accuracy.

The relative L^2 fit errors shown in Figure 5.2.9 converge clearly with increasing polynomial degree P for both the operator and solution approximations.

The pressure and velocity errors presented in Figure 5.2.10 and Figure 5.2.11 exhibit the same coupling between polynomial degree P and total reduced dimension r as observed in the stationary case. Reducing the approximation error requires increasing both quantities. Compared to the stationary setting, slightly larger reduced dimensions are generally needed to obtain comparable error levels. Beyond approximately $P = 4$, little additional reduction in error is observed.

The relative discarded POD energy shown in Figure 5.2.12 decreases rapidly for small values of r, followed by a slower decay as additional modes are included.

Visual comparisons of the transient pressure and velocity solutions are presented in Figure 5.2.13 and Figure 5.2.14. The figures show the solutions at different time levels together with the corresponding pointwise error fields for different reduced dimensions. Increasing the total number of modes r leads to a clear reduction in the approximation error.

The computational speed-up relative to the FOM is shown in Figure 5.2.15. Compared to the stationary case (Figure 5.2.8), the dependence on polynomial degree P is less

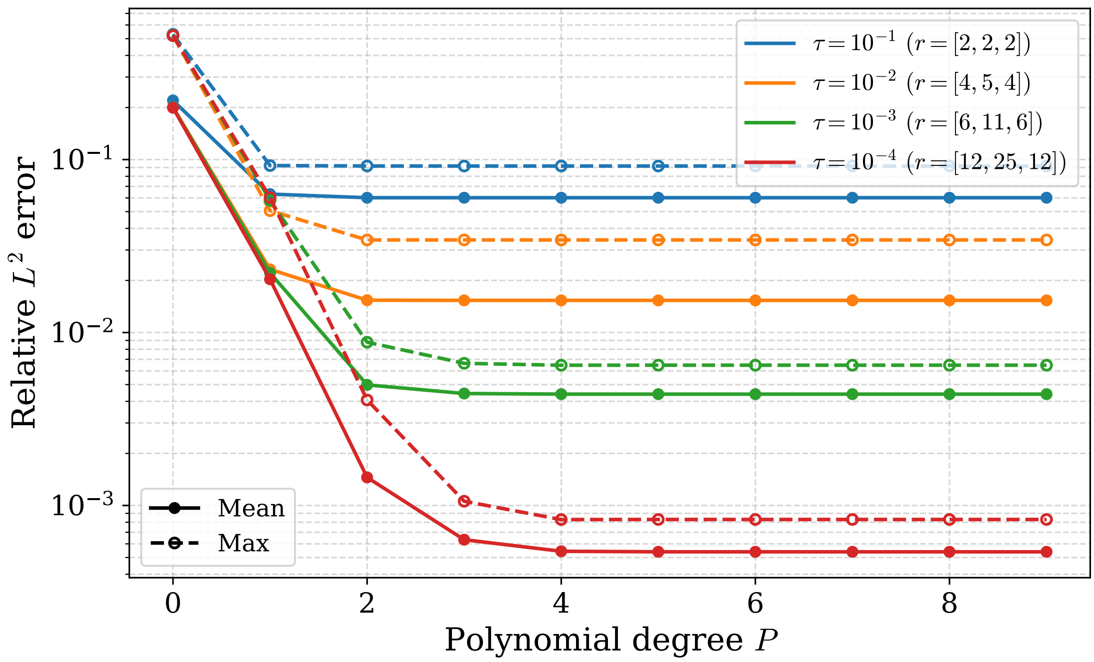

Figure 5.2.10: *Square with moving corner (transient).* Relative pressure error in the L^2 norm as a function of the polynomial degree P and reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 256 parameter samples and evaluated on 256 randomly sampled test points.

pronounced and the overall speed-up is reduced. This is likely because the affine operator assembly is performed once before the time-stepping loop, so the assembly cost is less dominant in the transient setting. Furthermore, for each parameter value $\boldsymbol{\mu}$, the full-order model performs only one system assembly and one sparse direct factorization (LU-type), and subsequent time steps reuse this factorization and are therefore relatively inexpensive.

Overall, the transient reduced-order model exhibits behavior similar to that observed in the stationary setting, although larger reduced dimensions are generally required to achieve comparable accuracy. The approximation errors decrease consistently with increasing polynomial degree and reduced dimension, while only limited improvements are observed beyond approximately $P = 4$. The numerical results further demonstrate that the transient reduced-order model is capable of accurately reproducing the dominant solution features throughout the time interval while still achieving substantial computational speed-ups relative to the full-order model.

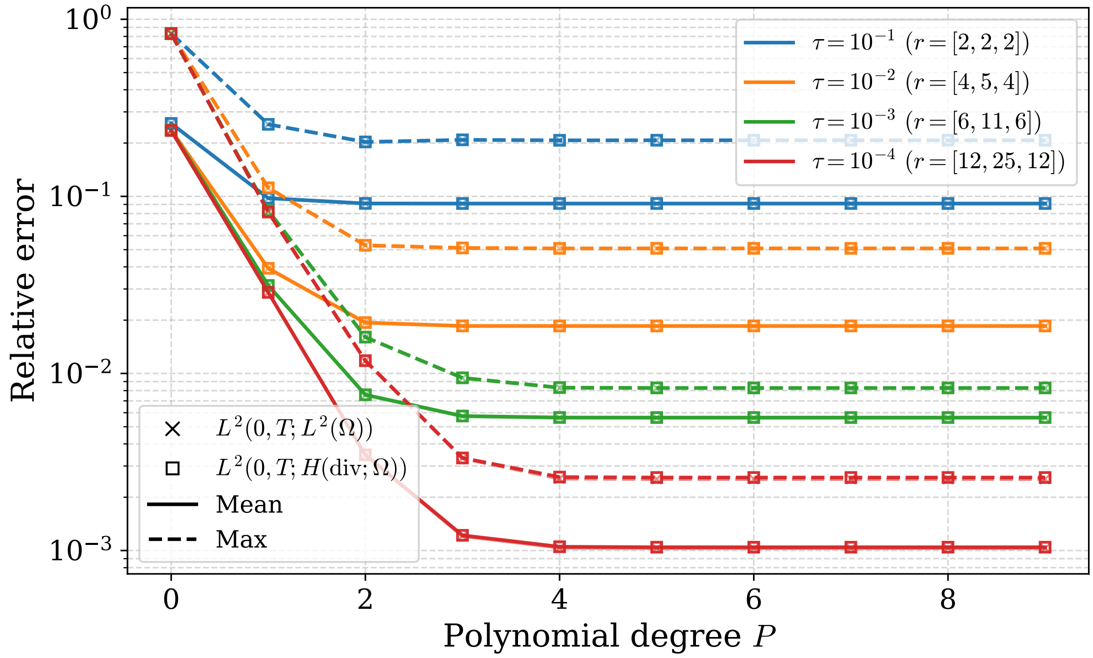

Figure 5.2.11: *Square with moving corner (transient).* Relative velocity error in the L^2 and $H(\text{div})$ norms as a function of the polynomial degree P and reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 256 parameter samples and evaluated on 256 randomly sampled test points. The L^2 and $H(\text{div})$ errors overlap almost exactly.

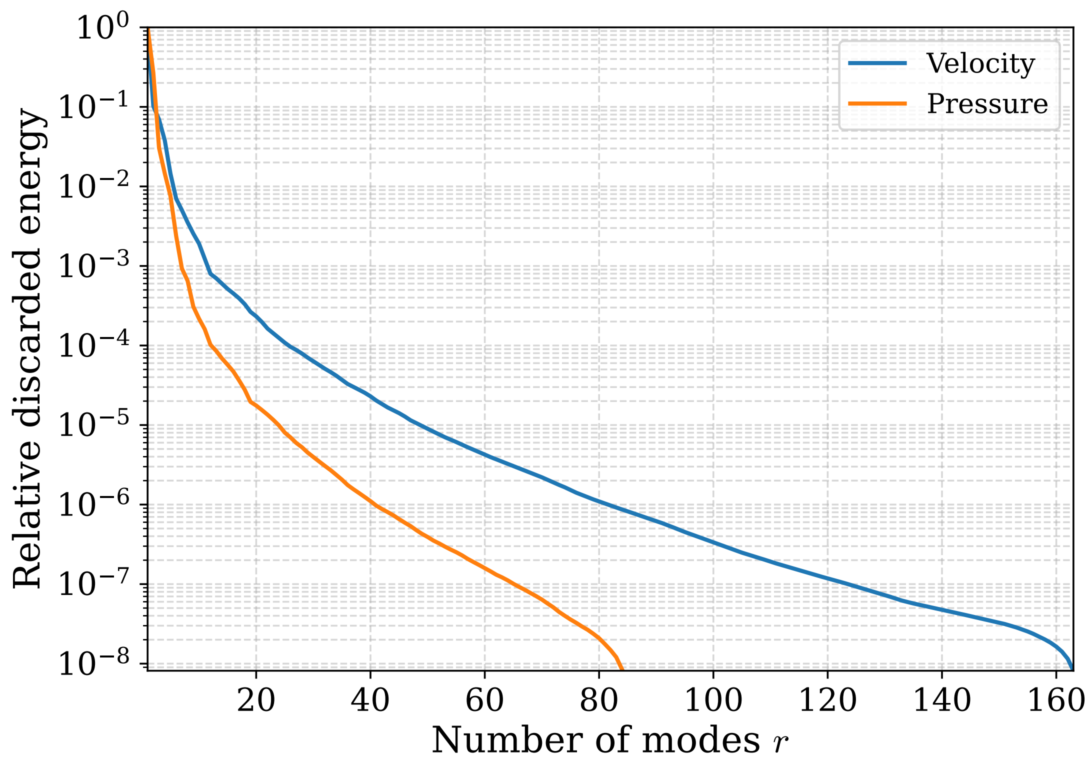

Figure 5.2.12: *Square with moving corner (transient).* Relative discarded POD energy as a function of the number of retained modes r, computed using 256 training samples.

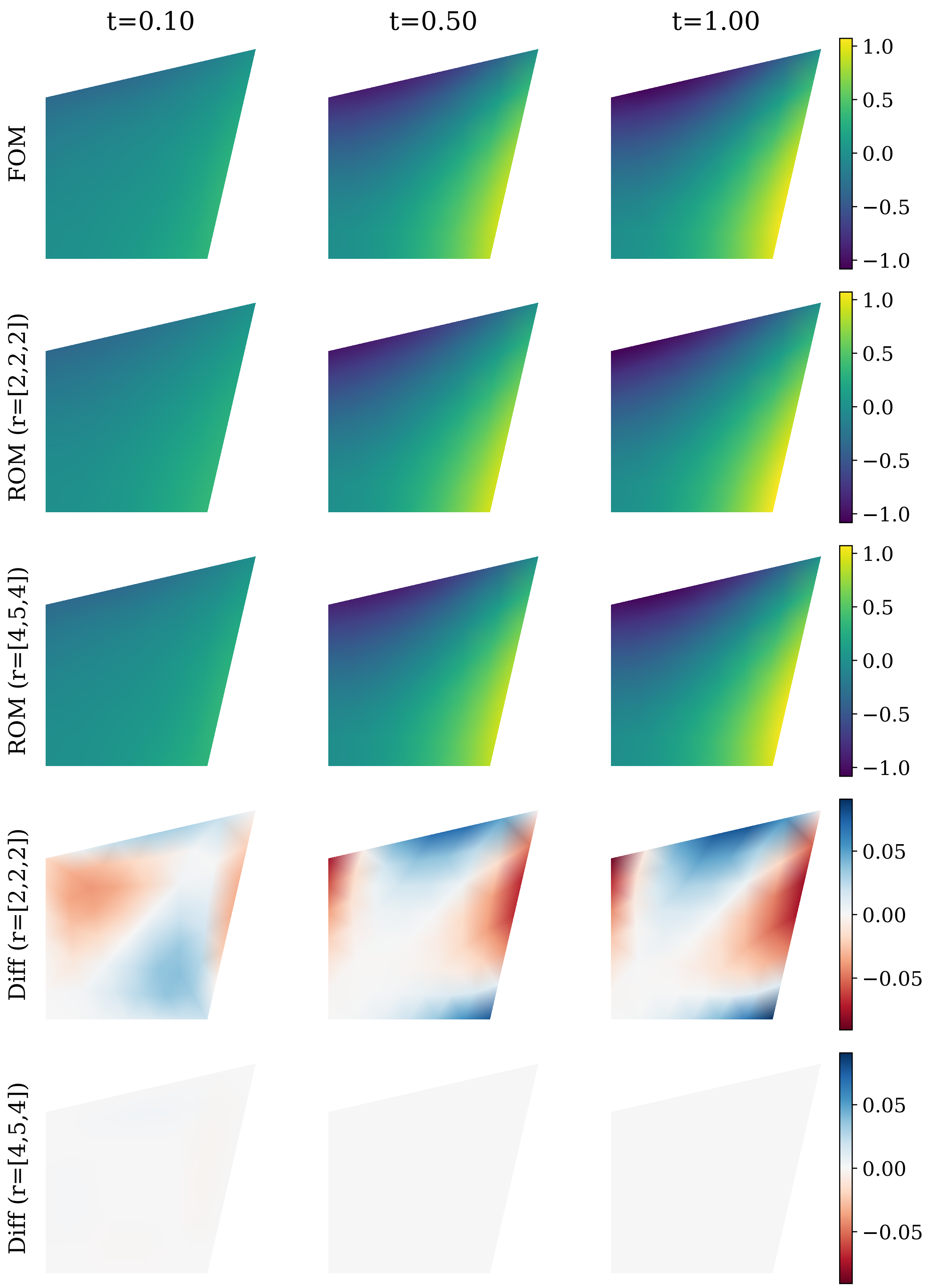

Figure 5.2.13: *Square with moving corner (transient).* Comparison between full-order and reduced-order pressure solutions at three selected time levels for $\boldsymbol{\mu} = (0.3, 0.3)$. Results are shown for different reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$ at polynomial degree $P = 3$.

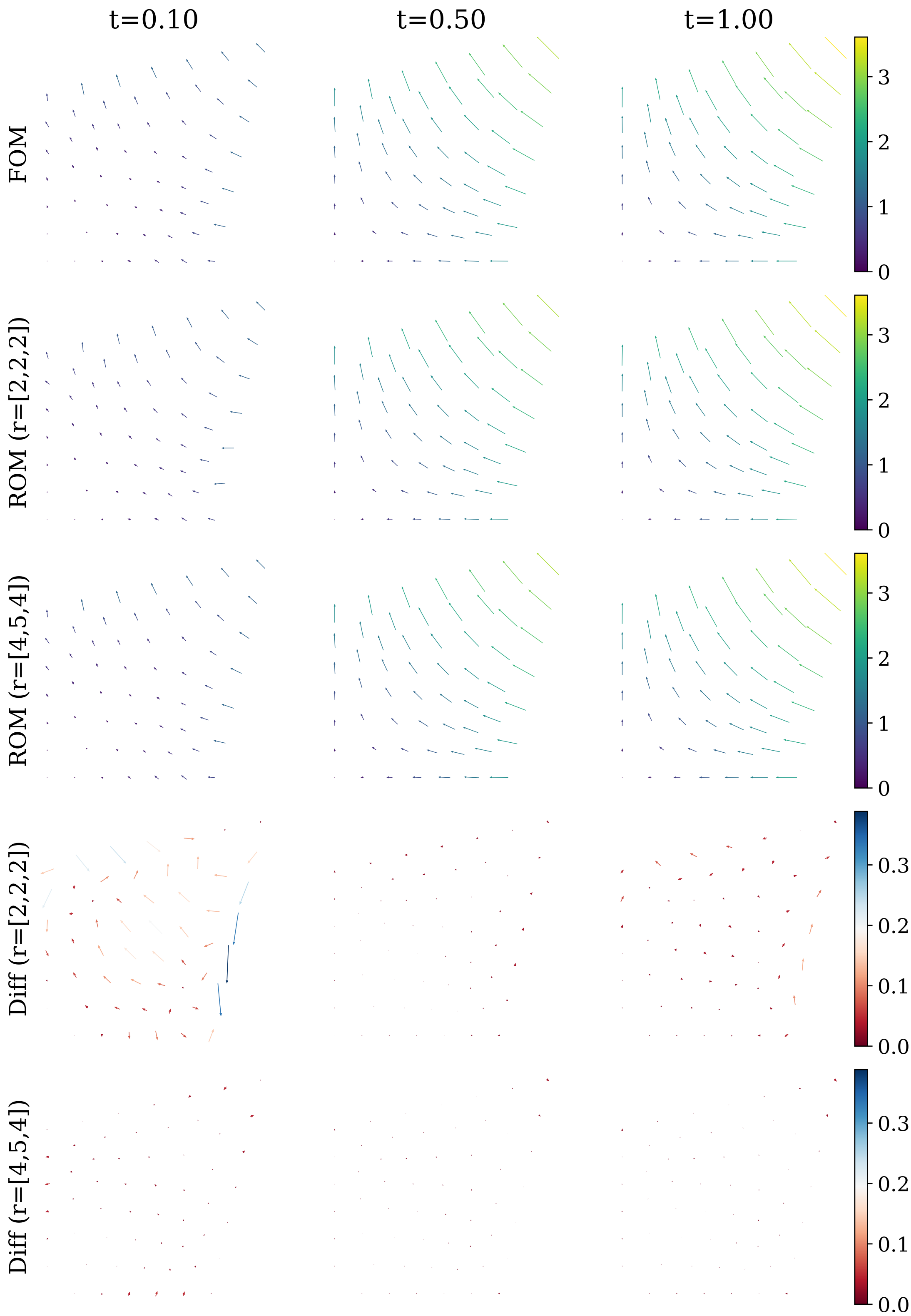

Figure 5.2.14: *Square with moving corner (transient).* Comparison between full-order and reduced-order velocity solutions at three selected time levels for $\boldsymbol{\mu} = (0.3, 0.3)$. Results are shown for different reduced dimensions $r = [r_p, r_{\mathbf{u}}, r_s]$ at polynomial degree $P = 3$.

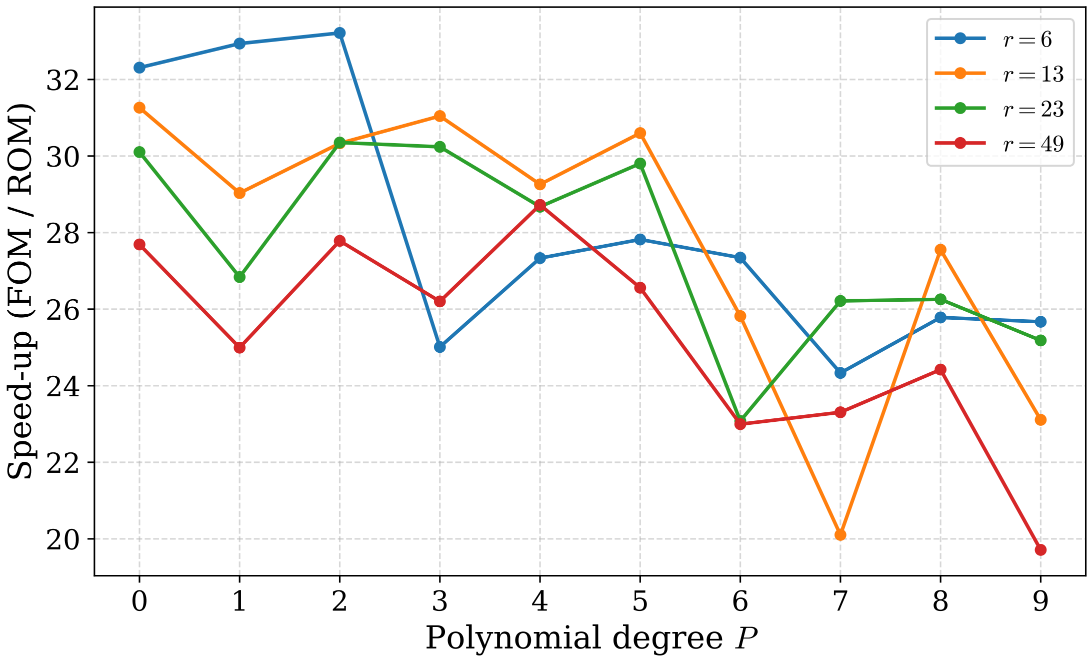

Figure 5.2.15: *Square with moving corner (transient).* Computational speed-up of the reduced-order model relative to the full-order model for varying polynomial degrees P and total reduced dimensions r.

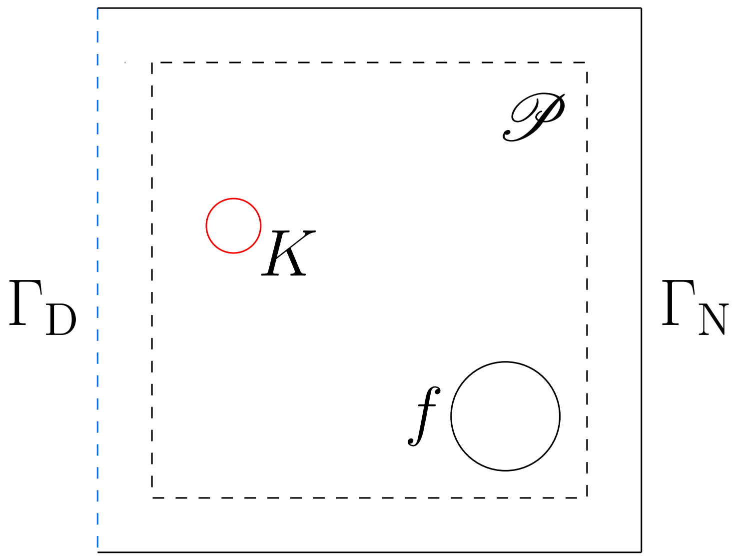

Figure 5.3.1: *Square with Gaussian permeability.* Illustration of the computational domain Ω with boundary conditions and parameterized permeability. The permeability field contains a localized Gaussian bump centered at $\boldsymbol{\mu}$, while the source term f is fixed in space. Dirichlet conditions are imposed on Γ_D and no-flow conditions on Γ_N.

5.3 Square with Gaussian permeability

In this example, we consider transient Darcy flow as described in (2.1) and (2.2), on a two-dimensional unit square domain with a parameter-dependent permeability field. The parameter $\boldsymbol{\mu} = (\mu_x, \mu_y)$ determines the spatial location of a localized high-permeability region within the domain. An illustration of the setup is shown in Figure 5.3.1.

The permeability field contains a Gaussian-shaped bump centered at $\boldsymbol{\mu}$, representing a localized region of increased permeability. The dotted rectangle indicates the parameter domain $\mathscr{P} = [0.2, 0.8]^2$, where $\boldsymbol{\mu} \in \mathscr{P}$. The source term f is also spatially localized, but its position remains fixed. Dirichlet boundary conditions are imposed on Γ_D, while the remaining boundary Γ_N is subject to no-flow conditions.

The permeability depends on the parameter $\boldsymbol{\mu} = (\mu_x, \mu_y)$ through

$$K(x, y; \boldsymbol{\mu}) = k_0 + \delta \exp\left(-\frac{(x-\mu_x)^2 + (y-\mu_y)^2}{2w^2} \right), \tag{5.1}$$

with $\boldsymbol{K}(x, y; \boldsymbol{\mu}) = K(x, y; \boldsymbol{\mu})\mathbf{I}$. This defines a localized Gaussian perturbation of the background permeability k_0, where $\boldsymbol{\mu}$ controls the center of the high-permeability region. In all experiments, the permeability parameters are chosen as

$$k_0 = 1, \qquad \delta = 10, \qquad w = 0.06.$$

The source term is defined in separable form as

$$f(x, y, t) = \alpha(t)\, g(x, y),$$

where the spatial profile is a Gaussian

$$g(x,y) = f_{\text{amp}} \exp\left(-\frac{(x-x_f)^2 + (y-y_f)^2}{2\sigma_f^2} \right),$$

and the temporal variation is given by the ramp function

$$\alpha(t) = 1 - e^{-t/t_r}.$$

The source term parameters are given by

$$f_{\text{amp}} = 10, \qquad (x_f, y_f) = (0.7, 0.4), \qquad \sigma_f = 0.1, \qquad t_r = 0.25.$$

For the transient problem, we use the final time $T = 1$ and storage coefficient $S = 1$.

The initial condition is given by

$$p(x,0) = 0 \qquad \text{in } \Omega,$$

and the boundary conditions are

$$\begin{aligned} \mathbf{u} \cdot \mathbf{n} &= 0 \qquad \text{on } \Gamma_{\mathrm{N}}, \\ p &= 0 \qquad \text{on } \Gamma_{\mathrm{D}}. \end{aligned}$$

The parameter dependence of the permeability introduces a non-affine dependence in the system matrix through the exponential term. In particular, the Gaussian expression in (5.1) cannot, in general, be represented exactly by a finite affine decomposition, which makes this example challenging for standard reduced-order modeling techniques that rely on affine parameter dependence. This example therefore serves as a representative test case for the proposed approximation strategy, where the non-affine dependence is approximated using polynomial fitting prior to model reduction.

This example is considered in both a stationary and a transient setting. In the stationary case, the storage coefficient is set to $S = 0$, the source term is taken to be time-independent (i.e., no ramp function is applied), and no temporal discretization is performed. In the transient case, the full time-dependent formulation described in the previous sections is used.

For the stationary problem, the CoSTA correction reduces to a purely algebraic correction of the solution, since no temporal evolution is involved. In contrast, for the transient case, CoSTA is applied at each time step, providing a correction to the time-dependent solution trajectory.

The ridge regularization parameter λ used in CoSTA is selected empirically based on numerical testing.

5.3.1 Finding discretization

To determine a suitable discretization for the Gaussian permeability problem, a discretization study was performed using the parameter value $\boldsymbol{\mu} = (0.5, 0.5)$. The resulting relative errors are reported in Table 5.3.1. Based on these results, we choose $h_{\max} = 0.08$ and $\Delta t = 0.02$, which yield relative errors below 1% in the considered norms. Note that this study is practical rather than fully rigorous: the discretization is assessed at a single parameter value, and the selected $(h_{\max}, \Delta t)$ is a pragmatic accuracy–cost compromise.

Table 5.3.1: *Square with Gaussian permeability.* Relative discretization errors for different spatial and temporal resolutions, computed with respect to a reference solution obtained using a sufficiently fine discretization. The comparison is performed for the parameter value $\boldsymbol{\mu} = (0.5, 0.5)$.

$h_{\max}$	Δt	$\|e_p\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;L^2(\Omega))}$	$\|e_{\mathbf{u}}\|_{L^2(0,T;H(\mathrm{div};\Omega))}$
0.32	0.04	2.8×10^{-2}	3.1×10^{-2}	1.1×10^{-1}
0.32	0.02	2.5×10^{-2}	2.9×10^{-2}	1.1×10^{-1}
0.32	0.01	2.4×10^{-2}	2.8×10^{-2}	1.1×10^{-1}
0.32	0.005	2.4×10^{-2}	2.8×10^{-2}	1.1×10^{-1}
0.16	0.04	1.5×10^{-2}	1.4×10^{-2}	2.3×10^{-2}
0.16	0.02	9.0×10^{-3}	7.8×10^{-3}	2.2×10^{-2}
0.16	0.01	6.3×10^{-3}	5.1×10^{-3}	2.1×10^{-2}
0.16	0.005	5.5×10^{-3}	4.2×10^{-3}	2.1×10^{-2}
0.08	0.04	1.5×10^{-2}	1.3×10^{-2}	9.2×10^{-3}
0.08	0.02	7.5×10^{-3}	6.7×10^{-3}	6.7×10^{-3}
0.08	0.01	3.7×10^{-3}	3.3×10^{-3}	5.7×10^{-3}
0.08	0.005	2.0×10^{-3}	1.5×10^{-3}	5.5×10^{-3}

5.3.2 Stationary

In this section, we present numerical results for stationary Darcy flow with Gaussian permeability. The reduced-order models are trained using a 48×48 equispaced sampling of the parameter domain.

For parameter-domain splitting, separate ROMs are constructed for each subdomain, which generally results in different reduced dimensions r across the local models. Unless otherwise stated, the reported values of r therefore denote the mean reduced dimension over all subdomains. For the 8×8 parameter-domain splitting, results are shown only up to polynomial degree $P = 5$. Beyond this degree, the approximation becomes susceptible to overfitting due to the limited number of training samples available within each subdomain, with only 36 parameter points per local model. Furthermore, the term *CoSTA* refers to a reduced-order model augmented with the corrective source term approach described in chapter 4.

The relative L^2-fit errors for different parameter-domain splittings are presented in Figure 5.3.2 as functions of the polynomial degree P. All domain decompositions exhibit convergence as P increases. The single-domain approximation shows the slowest convergence, while increasing the number of subdomains generally improves the convergence rate. Similar behavior is observed in several of the subsequent experiments.

The relative L^2 pressure error for the ROM under different parameter-domain splittings is shown in Figure 5.3.3. The reduced dimensions r corresponding to the prescribed POD tolerances τ decrease as the number of subdomains increases. Despite this reduction in basis size, the obtained errors remain comparable across the different splittings. Higher numbers of subdomains generally require lower polynomial degrees before the error stagnates. All four domain splittings achieve pressure errors below 1% for sufficiently large values of P and r. In particular, the 8×8 splitting attains errors below 1% already for $P = 2$ with approximately $r \approx 16$ reduced modes.

The relative L^2 velocity error for the baseline ROM is presented in Figure 5.3.4. In contrast to the pressure approximation, the velocity errors are generally larger and converge more slowly with respect to both polynomial degree and reduced dimension. The single-domain ROM does not achieve errors below 1%, whereas all parameter-domain splittings attain errors at or slightly below this threshold for sufficiently large values of P and r. However, the improvement is relatively modest, with several configurations only marginally satisfying the 1% criterion.

The relative velocity error measured in the energy norm is shown in Figure 5.3.5. The errors are generally larger than those observed in the L^2 norm, indicating that accurately capturing divergence-related behavior is more challenging for the reduced-order model. Increasing the polynomial degree beyond relatively small values provides only limited additional error reduction.

The relative discarded POD energy for the pressure and velocity bases is shown in Figure 5.3.6. The discarded energy decreases substantially faster than the corresponding ROM solution errors. Consequently, the relative discarded energy does not appear to provide a reliable quantitative indicator for the actual reduced-order approximation error in this example.

The relative L^2 pressure error for the ROM augmented with CoSTA is presented in Figure 5.3.7. Overall, the corrective model reduces the pressure errors moderately, while the qualitative trends with respect to polynomial degree and parameter-domain splitting remain similar to those observed for the baseline ROM.

The relative L^2 velocity error for the ROM with CoSTA correction is shown in Figure 5.3.8. Compared with the baseline ROM, the errors are reduced significantly across most configurations, demonstrating that the corrective model is particularly effective for the velocity approximation.

The velocity error measured in the energy norm for the ROM with CoSTA correction is presented in Figure 5.3.9. Among the considered error measures, this is where the largest improvements obtained from CoSTA are observed.

Visual comparisons between the FOM, baseline ROM, and CoSTA-corrected ROM solutions for the pressure and velocity fields are presented in Figure 5.3.10 and Figure 5.3.11, respectively. In both cases, the approximation errors are primarily localized near the parameter-dependent Gaussian permeability feature. For the pressure field, the improvement obtained using CoSTA is limited for $r = 37$, whereas a clearer improvement is observed for $r = 63$. In contrast, the CoSTA correction provides a visible improvement in the velocity approximation for both reduced dimensions.

The improvement factor of the mean errors is shown in Figure 5.3.12. Overall, CoSTA provides substantial improvements for the velocity approximation, with error reductions reaching up to two orders of magnitude in both the L^2 and energy norms. In contrast, the pressure approximation exhibits only modest improvements. The results further indicate a tendency for smaller POD tolerances τ to yield larger improvement factors. Since smaller tolerances correspond to larger reduced spaces, this suggests that CoSTA benefits from an already reasonably accurate reduced-order approximation.

The computational speed-up obtained using the baseline ROM and the ROM with CoSTA correction is shown in Figure 5.3.13 and Figure 5.3.14, respectively. In both cases, a clear relationship between the reduced dimension r and the polynomial degree P is observed. The CoSTA-corrected ROM generally exhibits a modest reduction in computational speed-up compared with the baseline ROM. This additional computational cost is primarily associated with evaluating the correction model $\mathcal{M}$ and performing an additional reduced solve.

The speed-up–error trade-off based on the mean relative velocity error in the energy norm is shown in Figure 5.3.15. Since the energy norm generally proved to be the most challenging error measure for this problem, it provides a representative indicator of the overall reduced-order approximation quality. Since the energy norm generally proved to be the most challenging error measure for this problem, it provides a representative indicator of the overall reduced-order approximation quality. Across all parameter-domain splits, the ROM with CoSTA points (hollow markers) are, in most cases, located below the corresponding baseline ROM points (filled markers) of the same color/shape, i.e., at lower relative energy error for comparable speed-up. This indicates that CoSTA improves accuracy while only marginally affecting online efficiency. Note that in some cases, the CoSTA speed-up is slightly higher than the corresponding baseline ROM speed-up. This is likely due to run-to-run timing variability in the computational environment.

Overall, the Gaussian permeability example proved significantly more challenging for the reduced-order approximation than the moving-corner problem. The localized and strongly non-affine parameter dependence generally required larger reduced dimensions in order to obtain satisfactory accuracy, particularly for the velocity-related error measures. Parameter-domain splitting substantially improved the approximation quality and allowed accurate reduced-order models to be obtained using lower polynomial degrees. Furthermore, CoSTA provided significant additional improvements for the velocity approximation, while the pressure approximation benefited more moderately. The additional computational cost associated with CoSTA was generally modest, such that the overall online speed-up relative to the full-order model remained substantial.

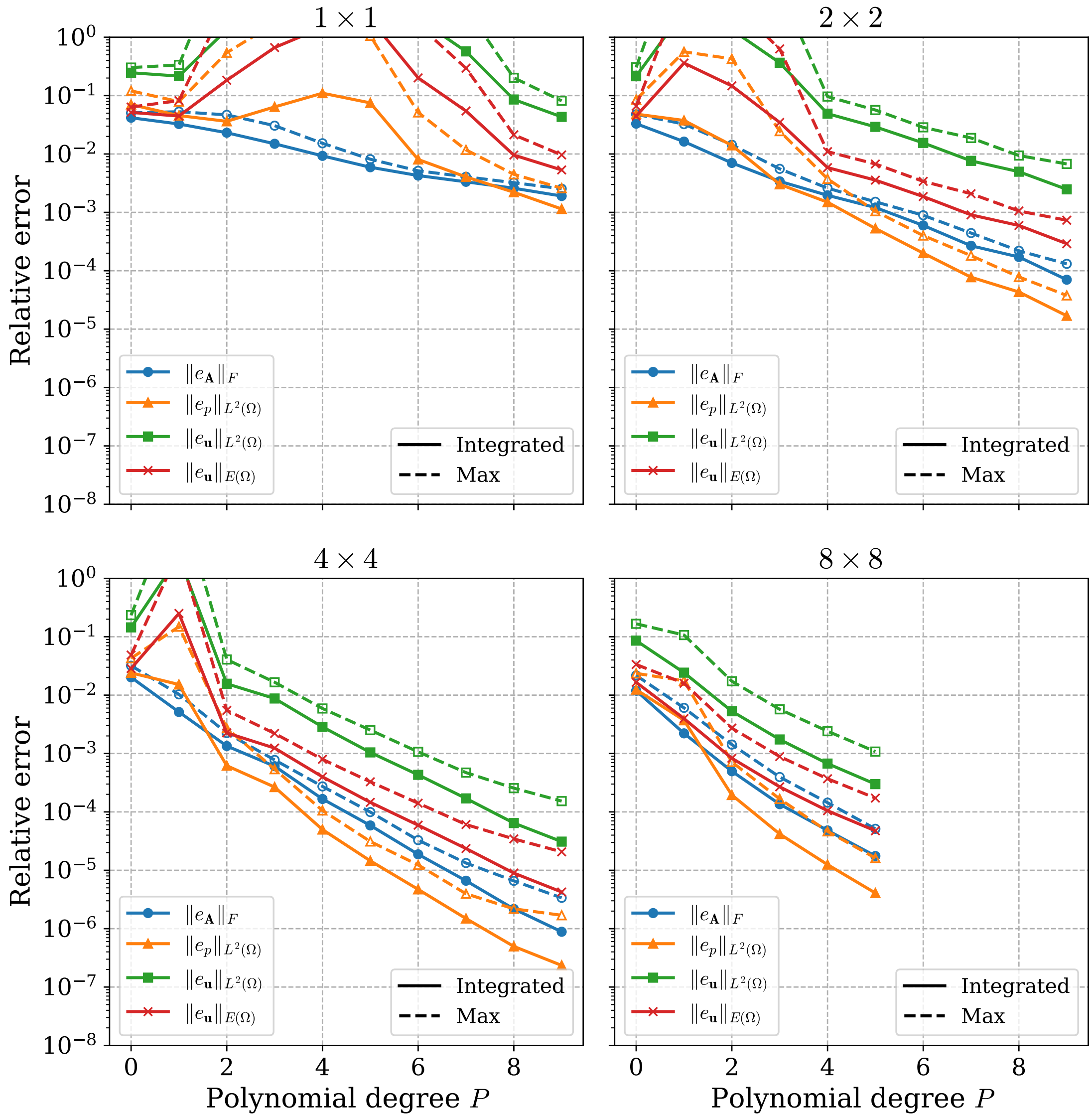

Figure 5.3.2: *Square with Gaussian permeability (stationary).* Relative operator and solution errors obtained from the L^2-fit as functions of the polynomial degree P for different parameter-domain splittings. Errors are computed using Gauss–Lobatto quadrature with 256 integration points over the parameter domain.

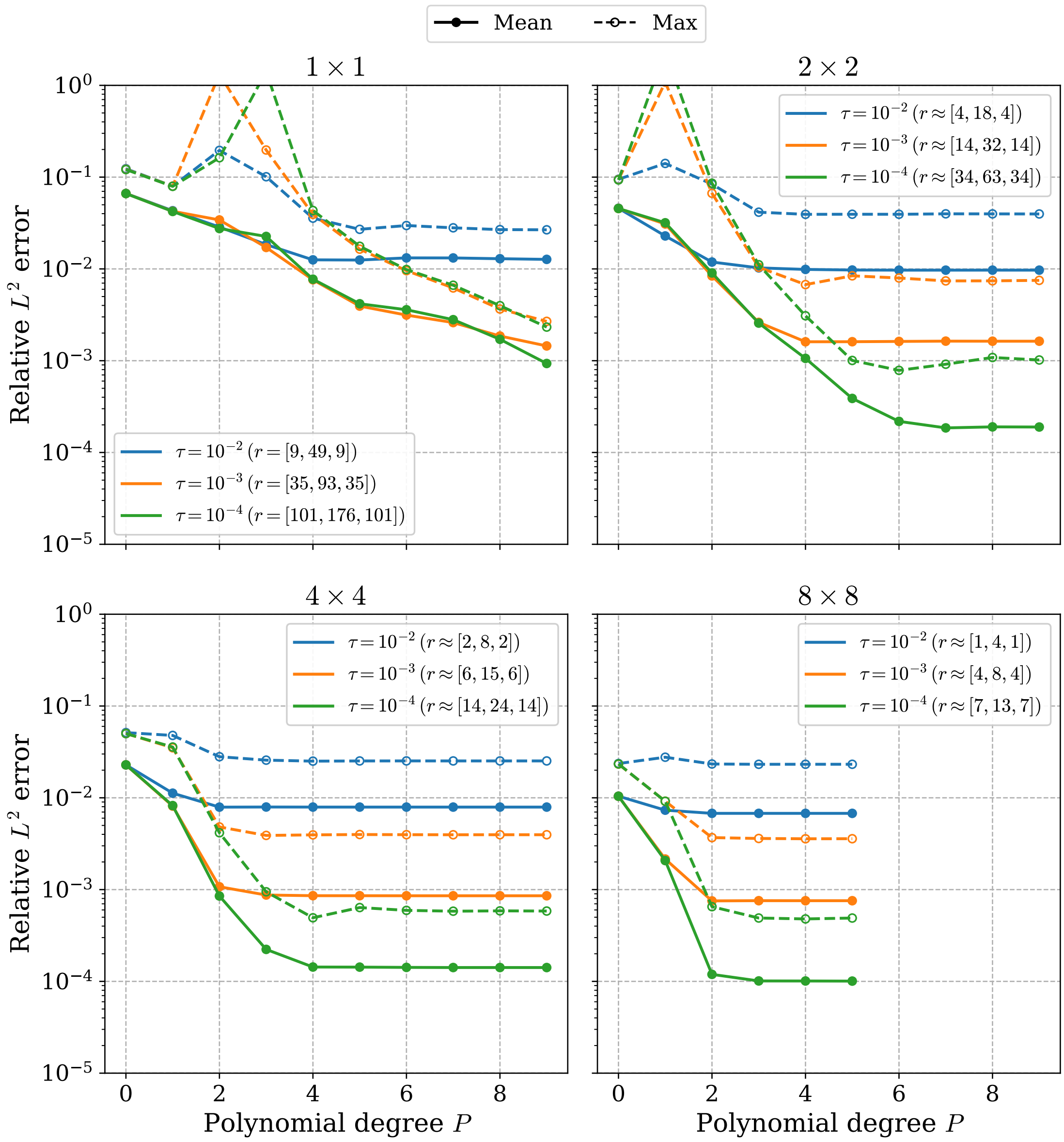

Figure 5.3.3: *Square with Gaussian permeability (stationary).* Relative L^2 error of the pressure from the ROM as a function of polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The model is trained using 2304 training points and evaluated on 256 randomly sampled test points. The figure shows four different parameter domain splittings, in equispaced squares.

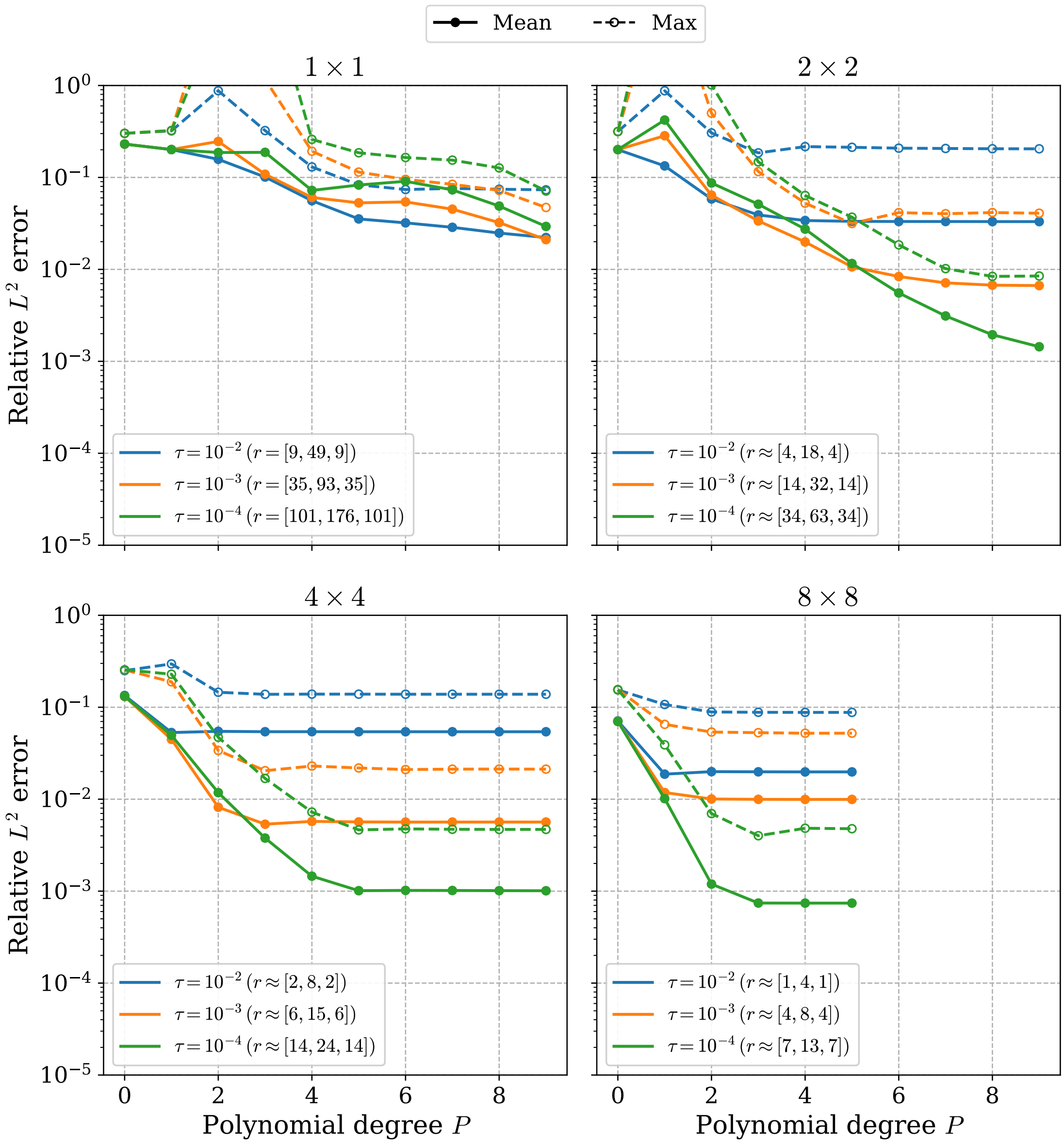

Figure 5.3.4: *Square with Gaussian permeability (stationary).* Relative pressure error in the L^2 norm for the ROM as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. Results are shown for four parameter-domain splittings using equispaced square partitions.

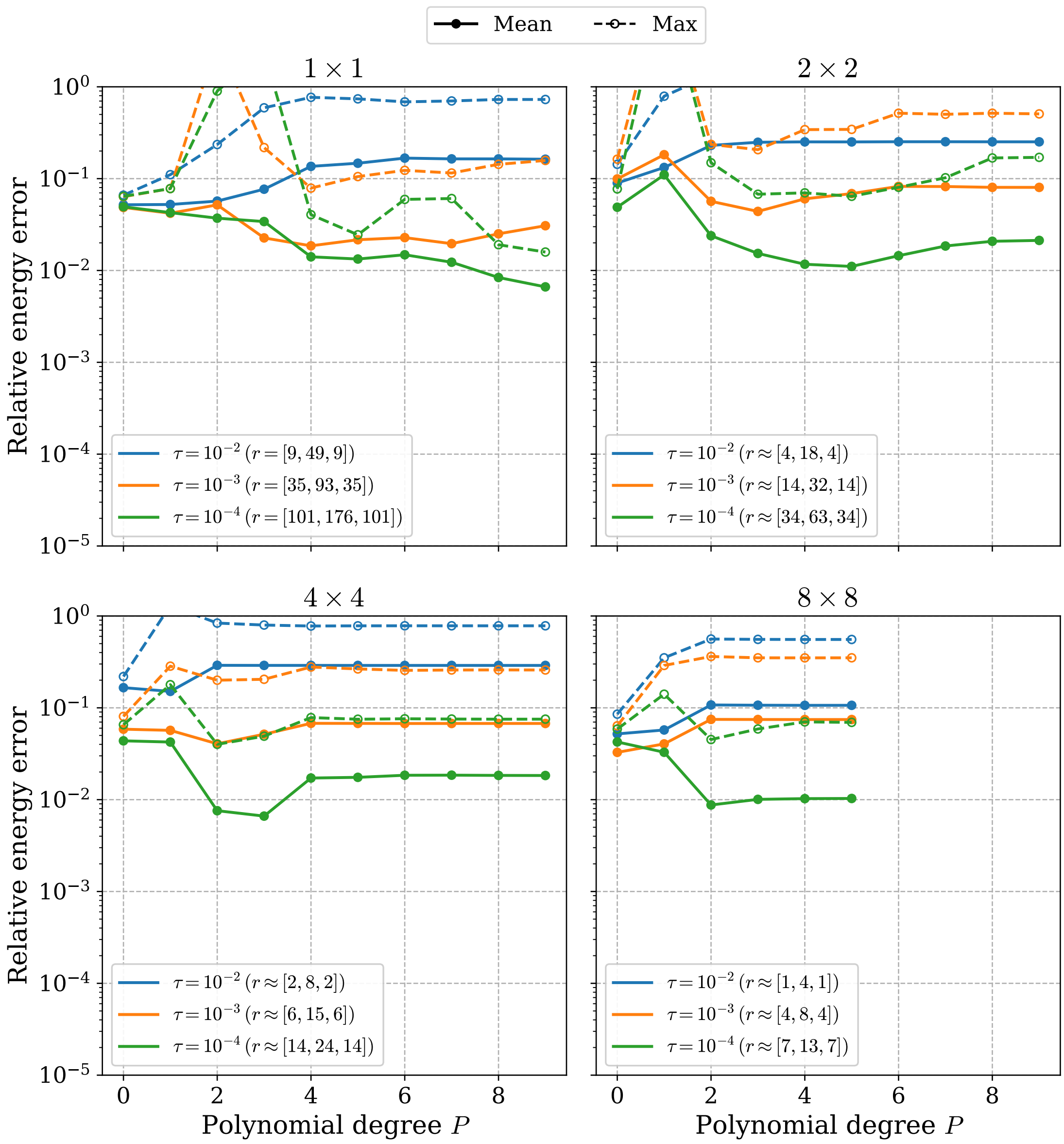

Figure 5.3.5: *Square with Gaussian permeability (stationary).* Relative velocity error in the energy norm for the ROM as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. Results are shown for four parameter-domain splittings using equispaced square partitions.

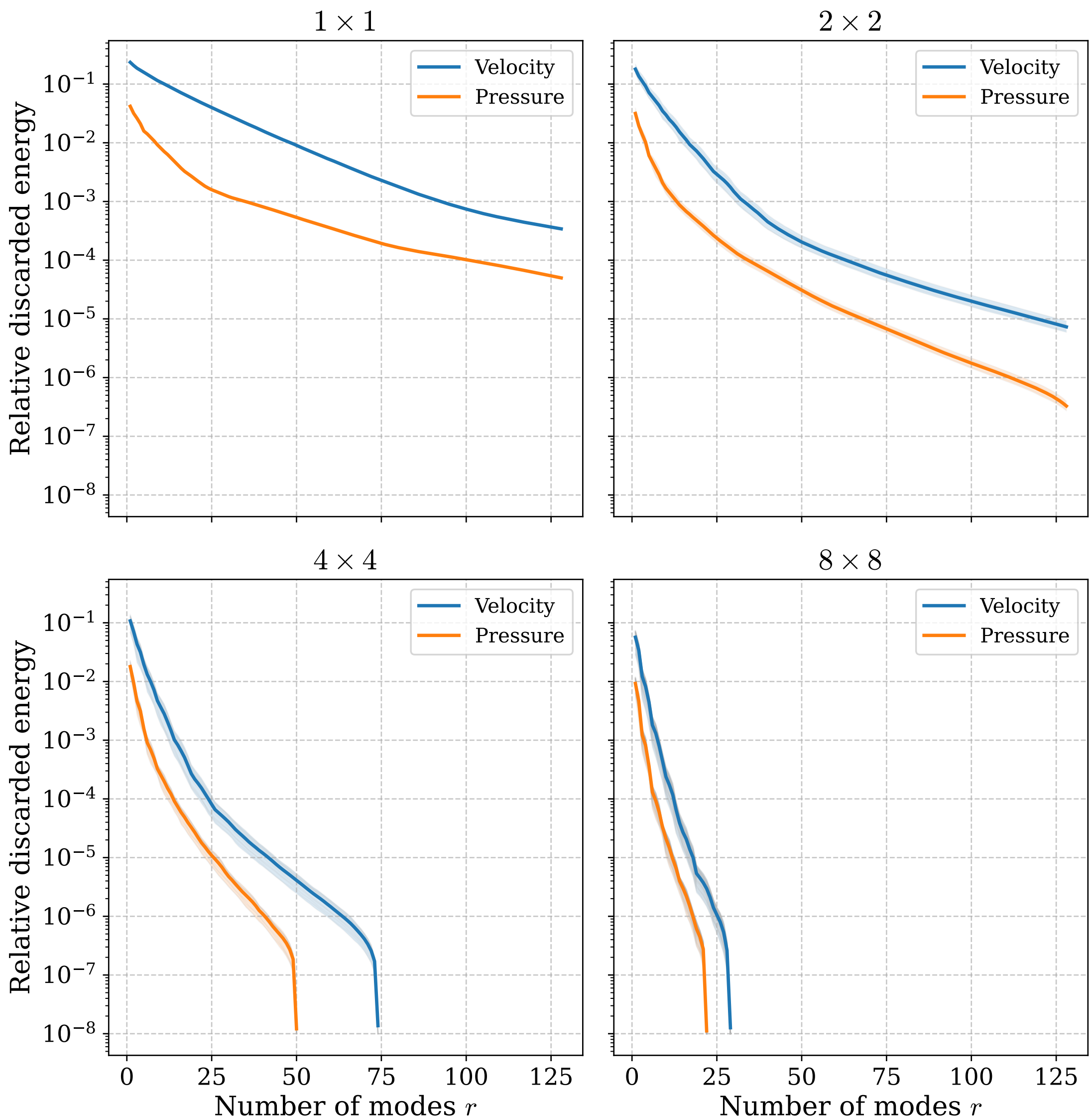

Figure 5.3.6: *Square with Gaussian permeability (stationary).* Relative discarded POD energy for the pressure and velocity reduced bases. Results are shown for four parameter-domain splittings using equispaced square partitions. For splittings with multiple subdomains, the shaded envelopes represent the range of discarded energies obtained from the corresponding local reduced bases.

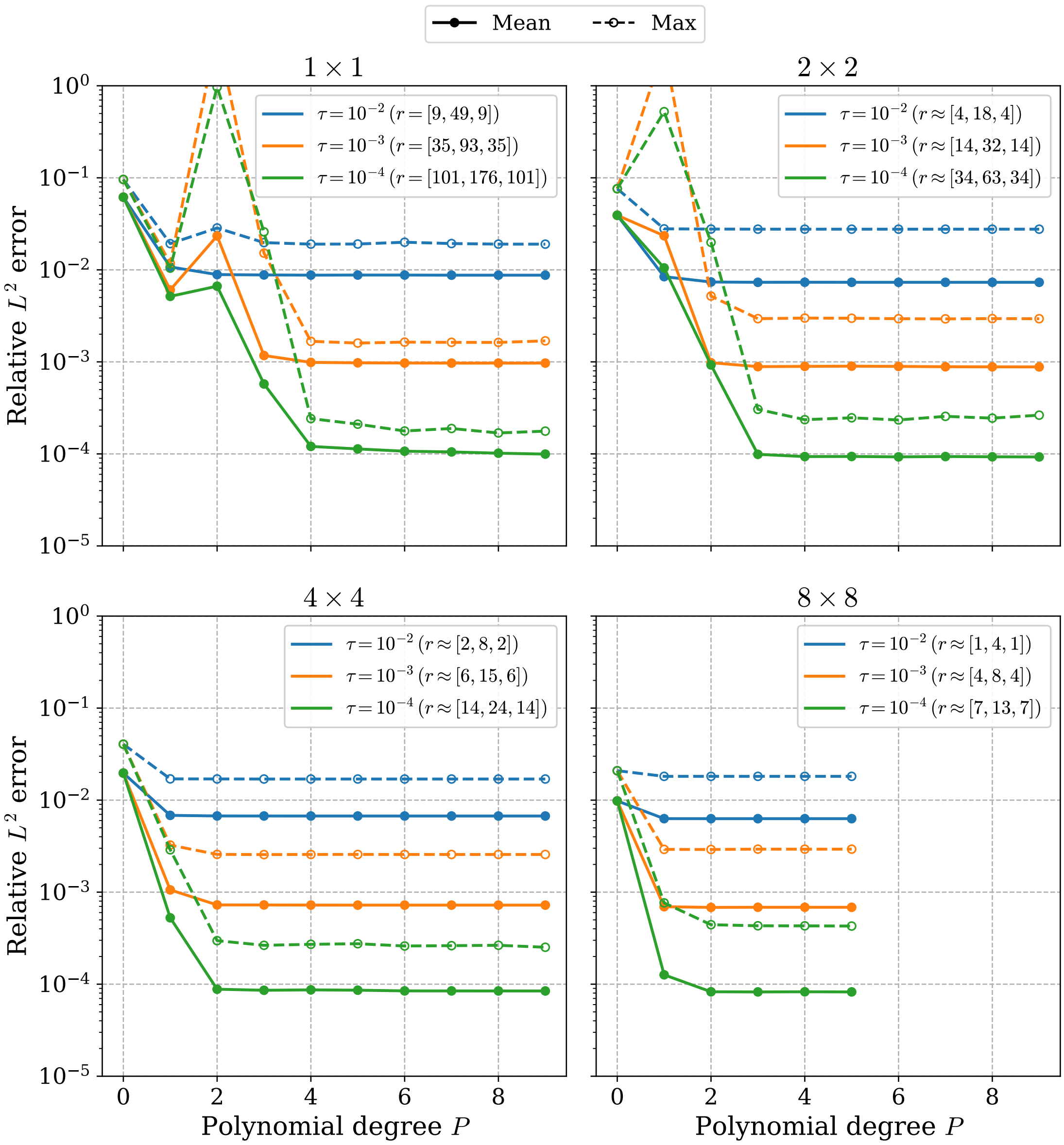

Figure 5.3.7: *Square with Gaussian permeability (stationary).* Relative pressure error in the L^2 norm for the ROM with CoSTA as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. The CoSTA correction is computed using ridge regression with $\lambda = 0.1$. Results are shown for four parameter-domain splittings using equispaced square partitions.

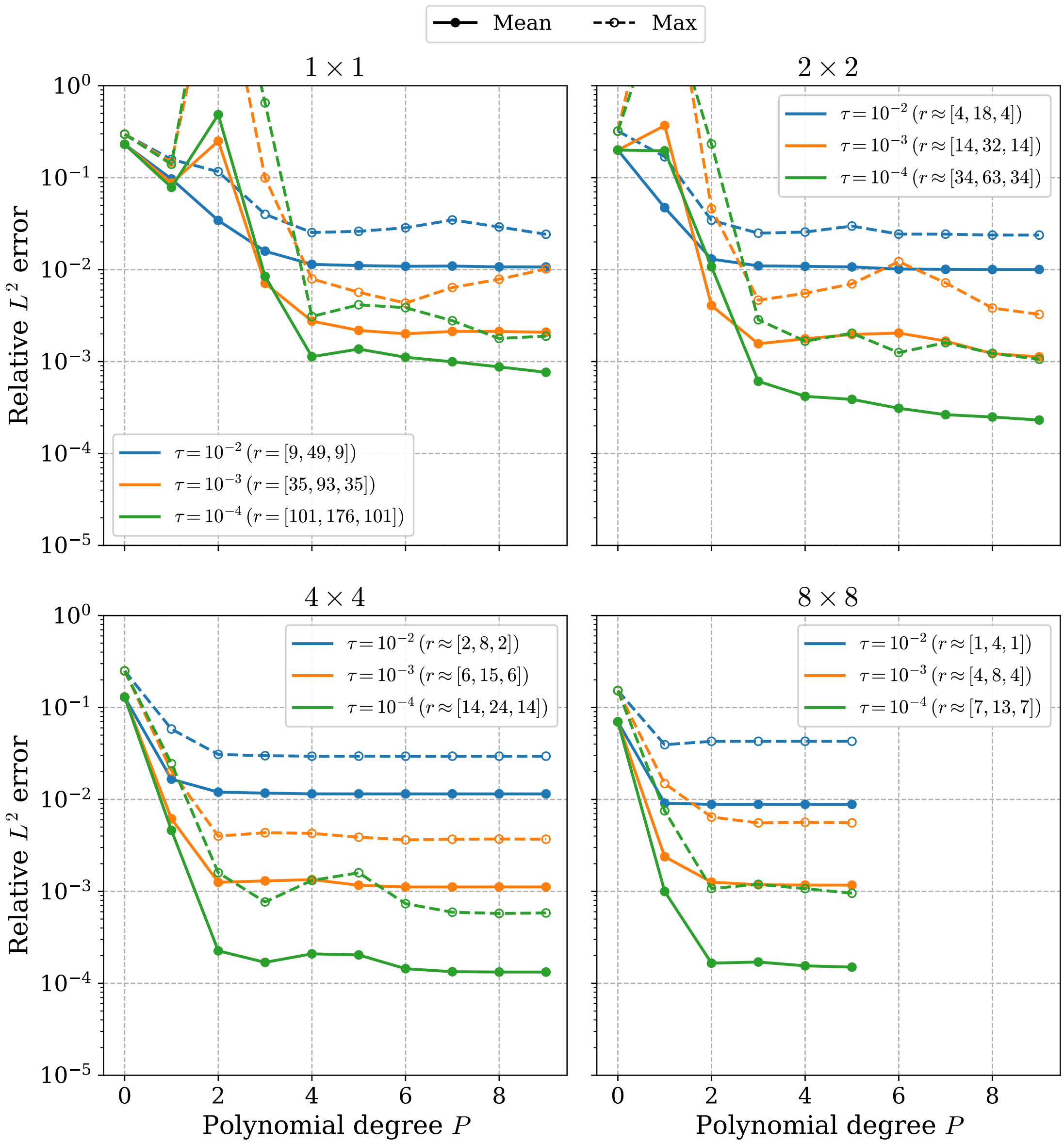

Figure 5.3.8: *Square with Gaussian permeability (stationary).* Relative velocity error in the L^2 norm for the ROM with CoSTA as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. The CoSTA correction is computed using ridge regression with $\lambda = 0.1$. Results are shown for four parameter-domain splittings using equispaced square partitions.

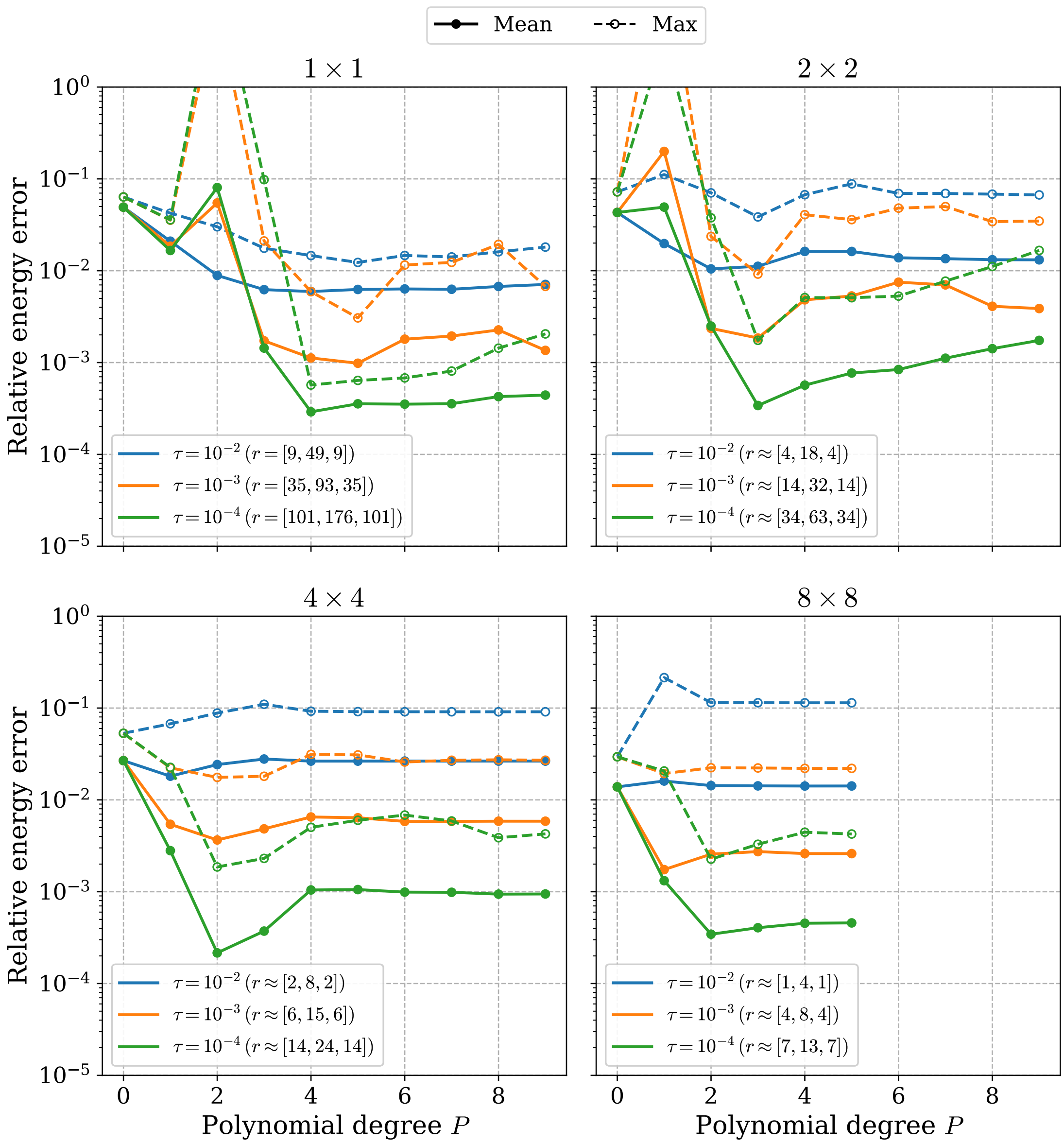

Figure 5.3.9: *Square with Gaussian permeability (stationary).* Relative velocity error in the energy norm for the ROM with CoSTA as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. The CoSTA correction is computed using ridge regression with $\lambda = 0.1$. Results are shown for four parameter-domain splittings using equispaced square partitions.

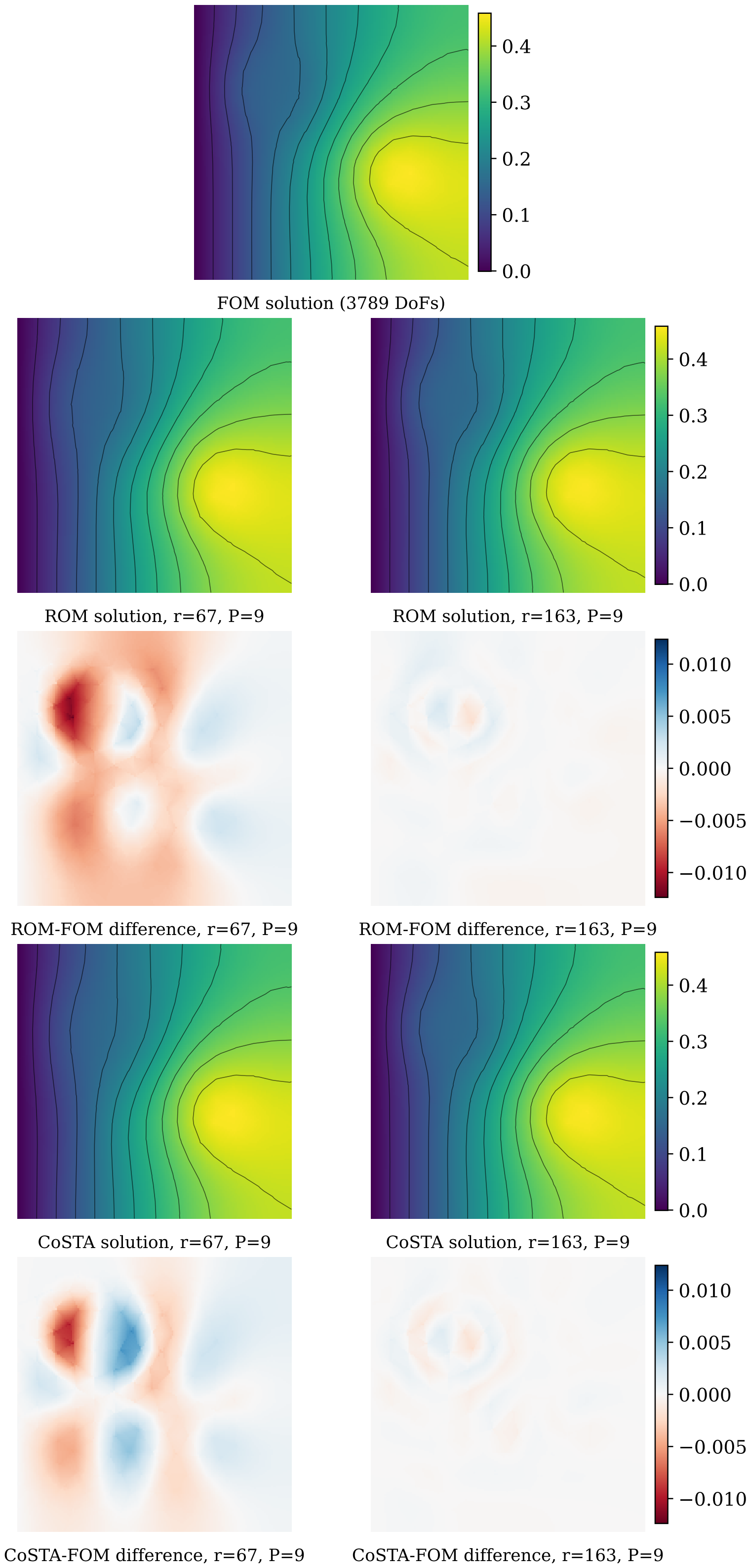

Figure 5.3.10: *Square with Gaussian permeability (stationary).* Visual comparison of pressure solutions for $\boldsymbol{\mu} = (0.3, 0.7)$. The rows show the full-order solution, baseline ROM solution, pointwise ROM error, CoSTA-corrected ROM solution, and pointwise CoSTA error, respectively.

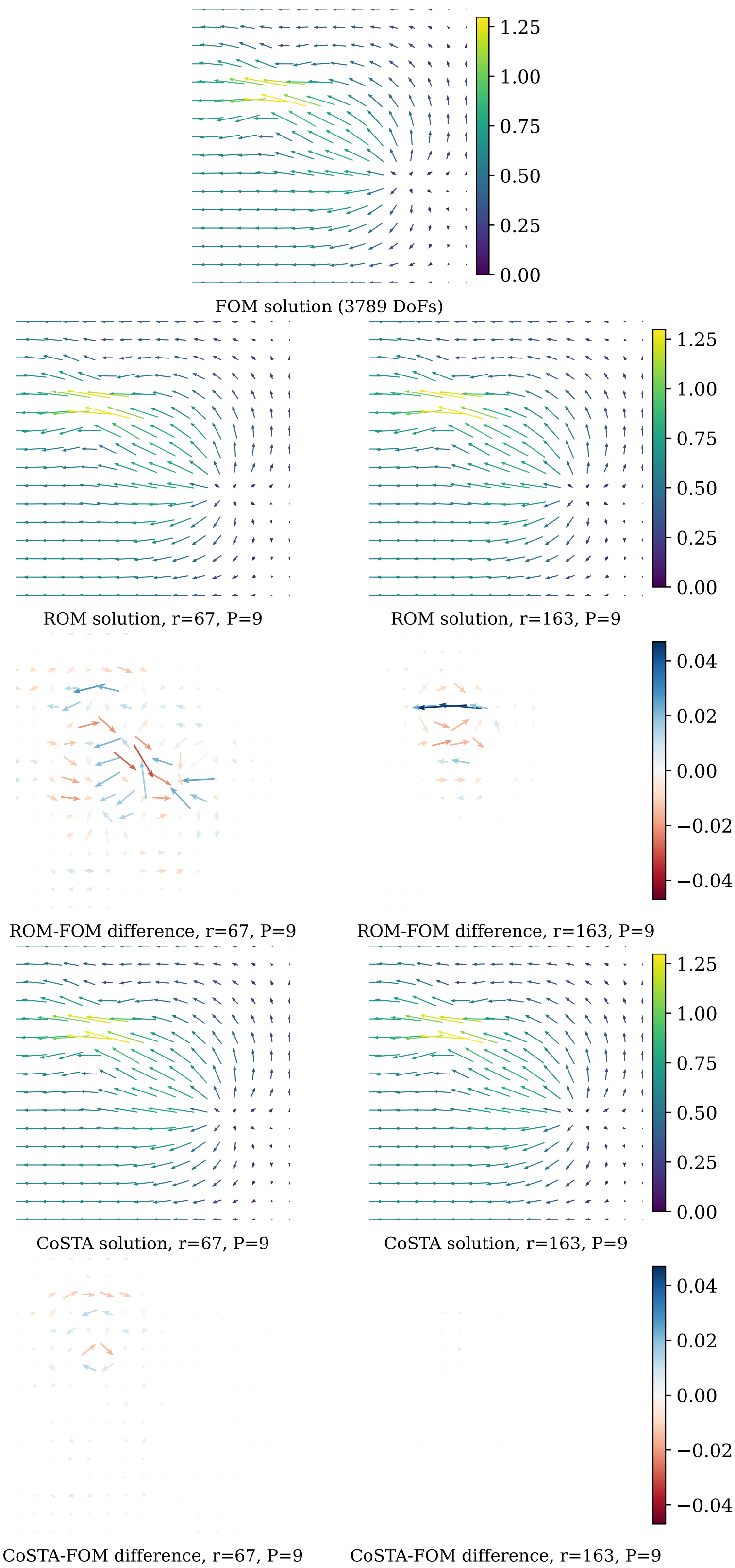

Figure 5.3.11: *Square with Gaussian permeability (stationary).* Visual comparison of velocity solutions for $\boldsymbol{\mu} = (0.3, 0.7)$. The rows show the full-order solution, baseline ROM solution, pointwise ROM error, CoSTA-corrected ROM solution, and pointwise CoSTA error, respectively.

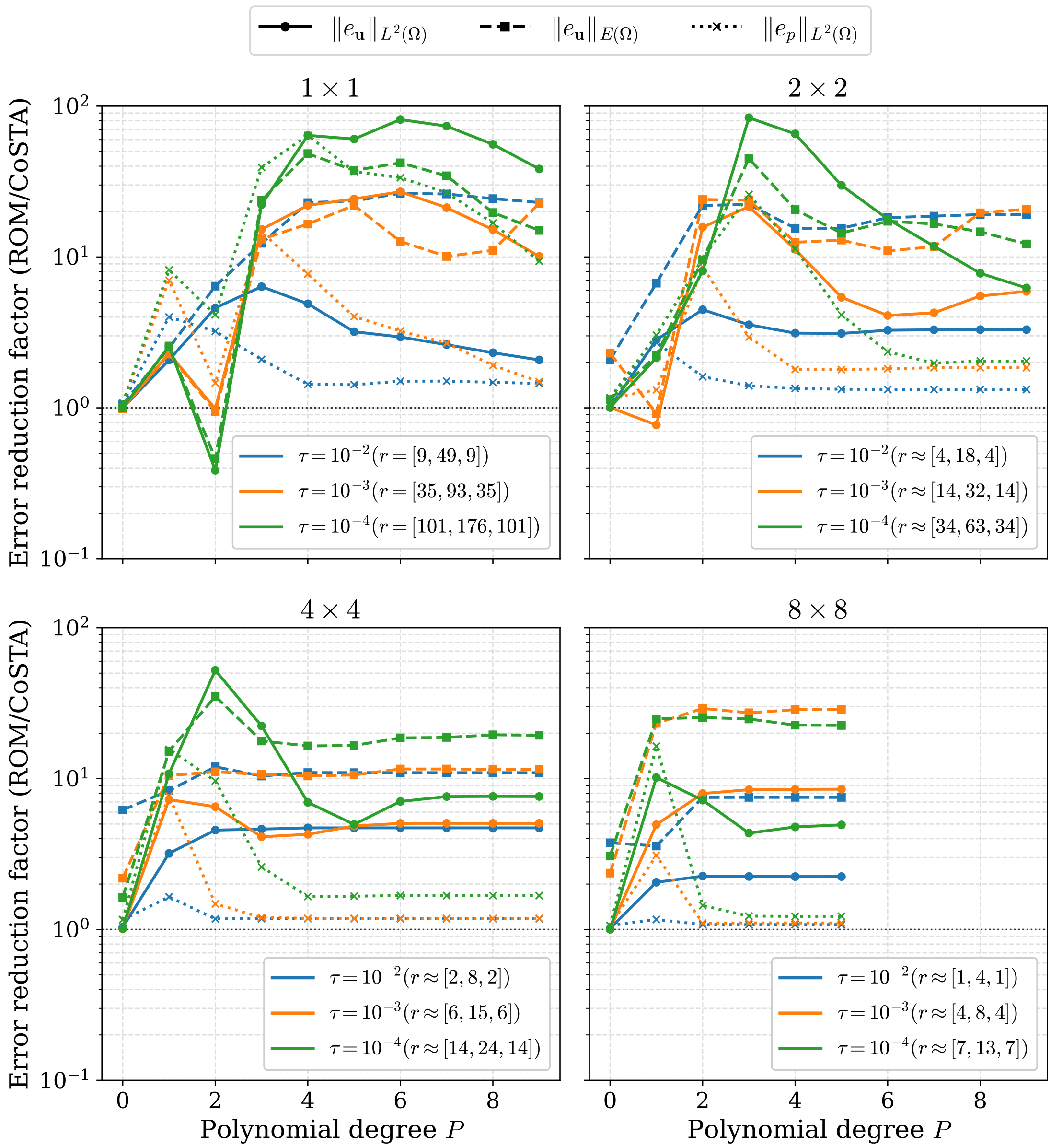

Figure 5.3.12: *Square with Gaussian permeability (stationary).* Reduction factor of the mean error of the ROM obtained with CoSTA as a function of polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. Values above 10^0 correspond to improved accuracy compared with the baseline reduced-order model.

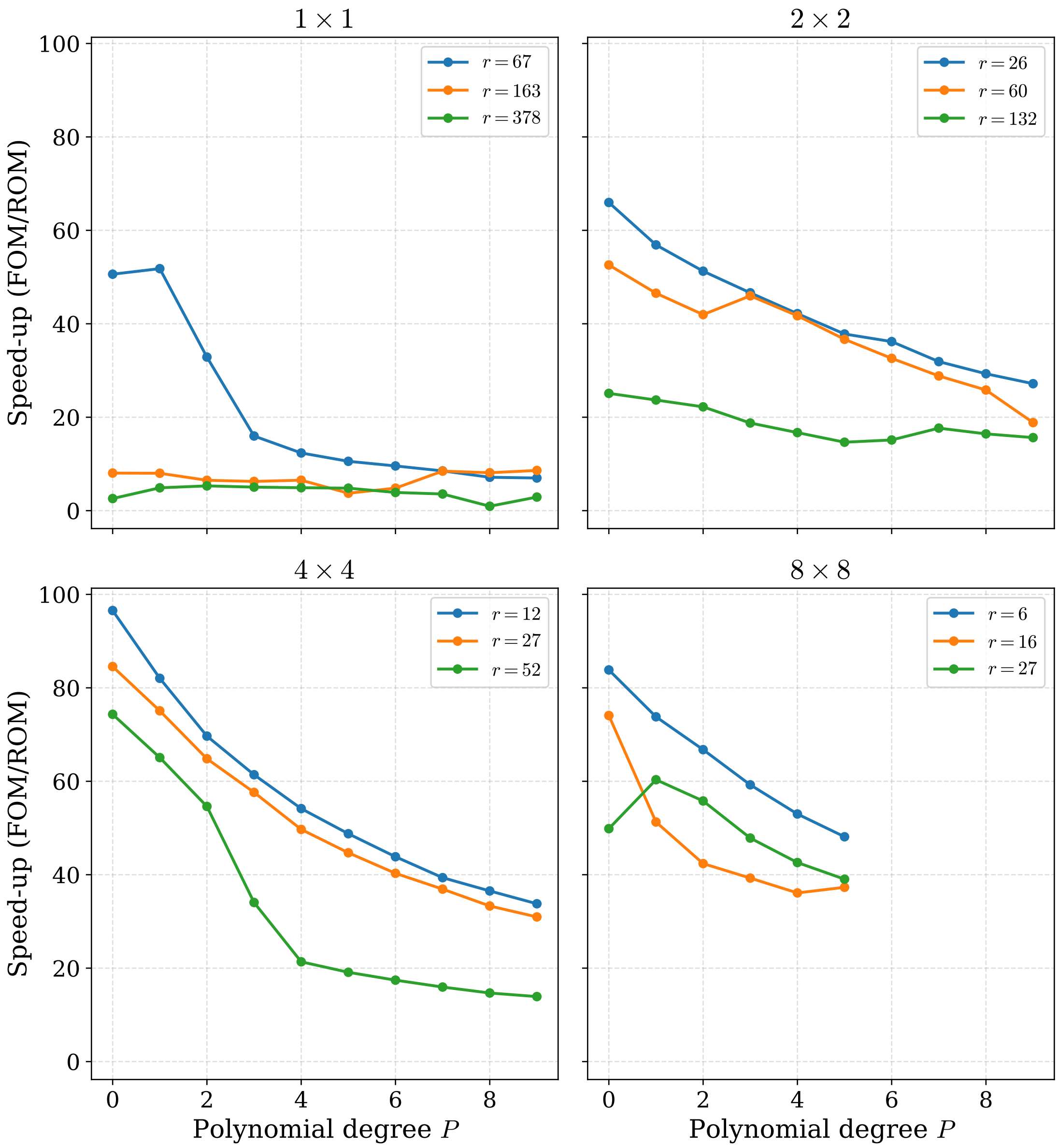

Figure 5.3.13: *Square with Gaussian permeability (stationary).* Computational speed-up of the ROM as a function of the polynomial degree P and total reduced dimension r. Results are shown for four parameter-domain splittings using equispaced square partitions.

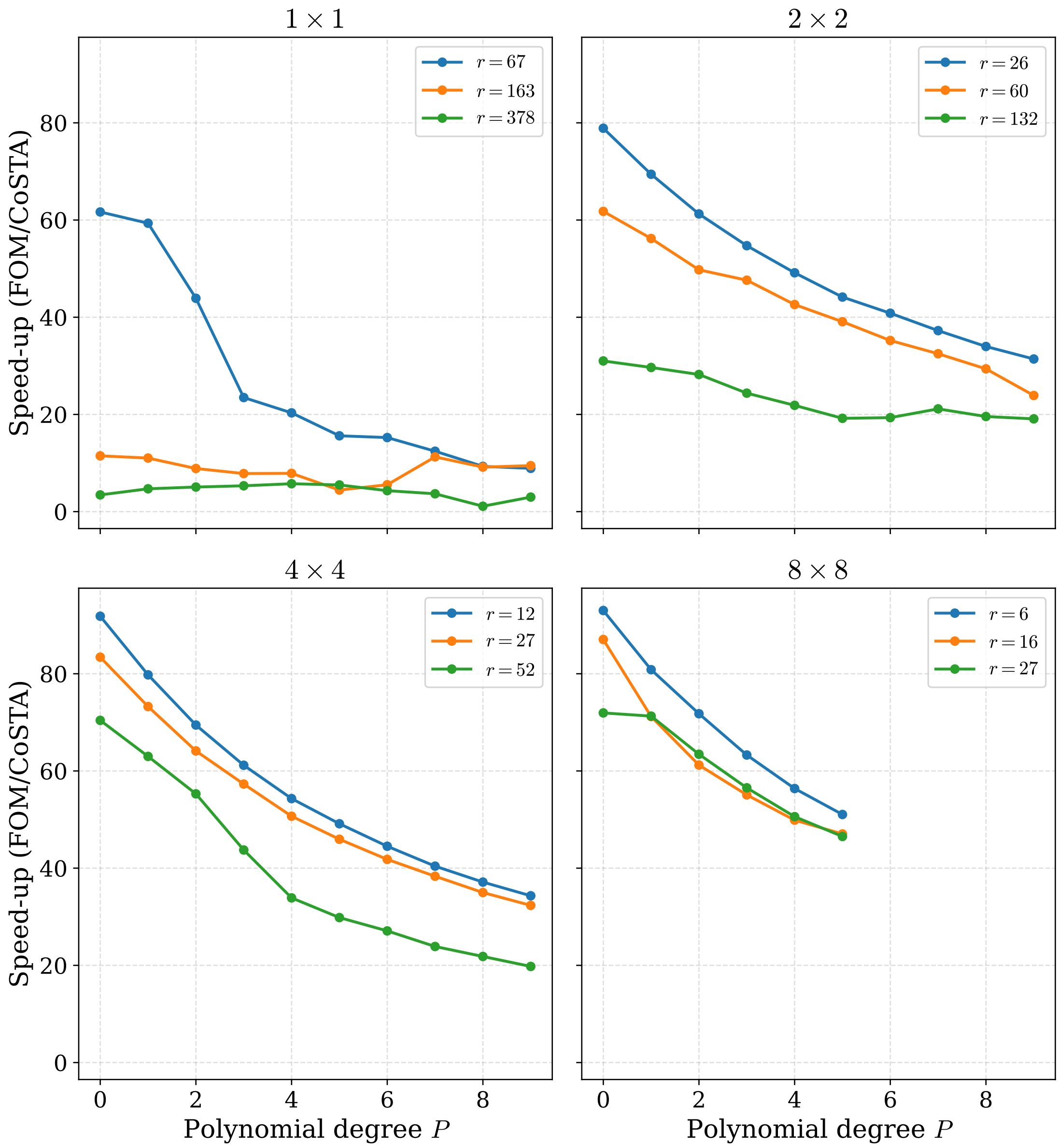

Figure 5.3.14: *Square with Gaussian permeability (stationary).* Computational speed-up of the ROM with CoSTA as a function of the polynomial degree P and total reduced dimension r. Results are shown for four parameter-domain splittings using equispaced square partitions.

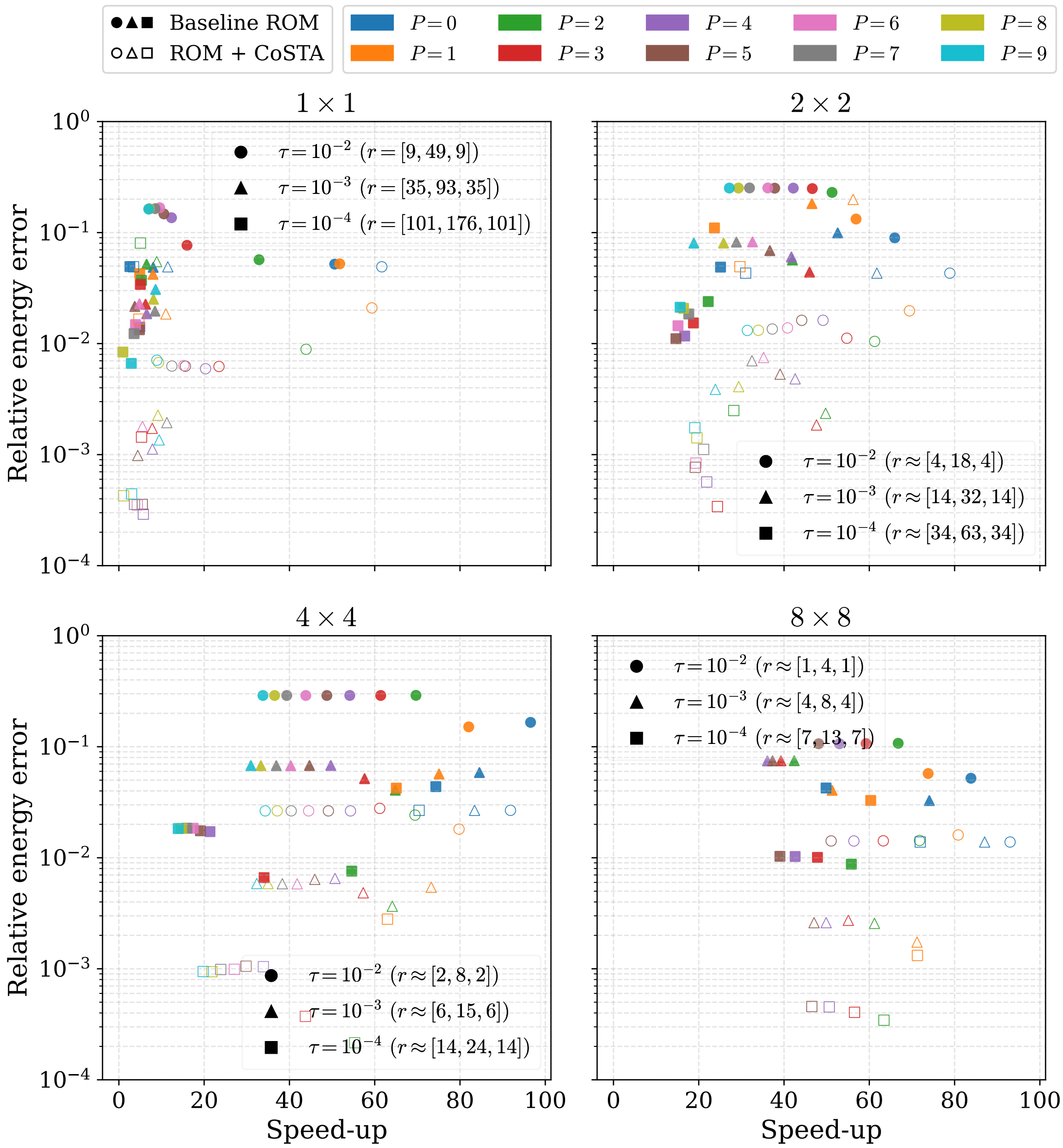

Figure 5.3.15: *Square with Gaussian permeability (stationary).* Speed-up–error scatter plot for baseline ROM (filled markers) and ROM with CoSTA (hollow markers), with color indicating polynomial degree P and marker shape indicating τ. The x-axis shows online speed-up relative to FOM, while the y-axis shows the relative velocity error measured in the energy norm. Points lower and further to the right indicate a better accuracy–efficiency trade-off.

5.3.3 Transient

The transient results exhibit behavior largely consistent with the stationary case. In general, the same trends with respect to parameter-domain splitting, polynomial degree P, and reduced dimension r are observed. The primary differences are slightly larger approximation errors and an increased number of reduced modes required to achieve comparable accuracy.

The transient L^2-fit errors shown in Figure 5.3.16 are qualitatively similar to those observed in the stationary case. However, all parameter-domain splittings except the 8×8 decomposition become unstable at lower polynomial degrees, indicating increased sensitivity to overfitting in the transient setting.

The baseline ROM errors for pressure, velocity, and the energy norm are shown in Figure 5.3.17, Figure 5.3.18, and Figure 5.3.19. Overall, the transient results follow trends similar to those of the stationary case, although slightly larger reduced dimensions are generally required to achieve comparable error levels. In particular, the velocity and energy errors remain significantly larger than the corresponding pressure errors. We also observe similar overfitting effects as in the L^2-fit errors.

The relative discarded POD energy shown in Figure 5.3.20 is also comparable to the stationary case, although the decay is generally somewhat slower.

The CoSTA-corrected ROM errors are presented in Figure 5.3.21, Figure 5.3.22, and Figure 5.3.23. As in the stationary case, CoSTA provides the largest improvements for the velocity-related error measures, while the pressure errors are reduced more moderately. Overall, the transient setting produces slightly larger errors than those observed for the stationary problem.

Visual comparisons of the transient pressure and velocity solutions are shown in Figure 5.3.24 and Figure 5.3.25. For both variables, the approximation errors are primarily localized around the parameter-dependent Gaussian permeability feature and generally increase over time. For the pressure field, the addition of CoSTA does not appear to significantly reduce the maximum error, although the overall error magnitude becomes smaller. The velocity results exhibit similar behavior, with CoSTA providing clearer overall improvements.

The reduction factor of the mean errors is shown in Figure 5.3.26. Overall, the transient case exhibits trends similar to those observed in the stationary setting, although the improvements obtained with CoSTA are generally smaller. Substantial improvements are still observed for the velocity approximation in both norms, while the pressure approximation again exhibits more modest improvements. A few extreme outliers are present for certain polynomial degrees. These correspond to cases where the underlying L^2-fit approximation becomes unstable, leading to very large baseline errors. Consequently, these points are not representative of the general behavior of the method.

The computational speed-up results shown in Figure 5.3.27 and Figure 5.3.28 exhibit trends similar to those observed in the stationary case. The dependence on the reduced dimension r is particularly pronounced, while the influence of the polynomial degree P is somewhat less distinct in the transient setting.

The speed-up–error trade-off based on the mean relative velocity error in the energy norm is shown in Figure 5.3.29. Since the energy norm generally proved to be the most challenging error measure for this problem, it provides a representative indicator of the overall reduced-order approximation quality. In the transient case, the ROM with CoSTA points are generally shifted down and to the left relative to the corresponding baseline ROM points. This indicates a clear trade-off when using CoSTA: improved accuracy at the cost of reduced speed-up.

Overall, the transient Gaussian permeability problem exhibited behavior largely consistent with the stationary setting, although larger reduced dimensions were generally required to achieve comparable accuracy. The transient setting also showed increased sensitivity to overfitting in the polynomial approximation, particularly for higher polynomial degrees and finer parameter-domain splittings. Parameter-domain splitting again improved the reduced-order approximation substantially, especially for the localized permeability variations. CoSTA provided clear improvements for the velocity-related error measures, although the improvements were generally smaller than those observed in the stationary case and came at the cost of reduced online speed-up. Despite the increased complexity of the problem, the reduced-order models still provided substantial computational speed-ups while maintaining satisfactory approximation accuracy.

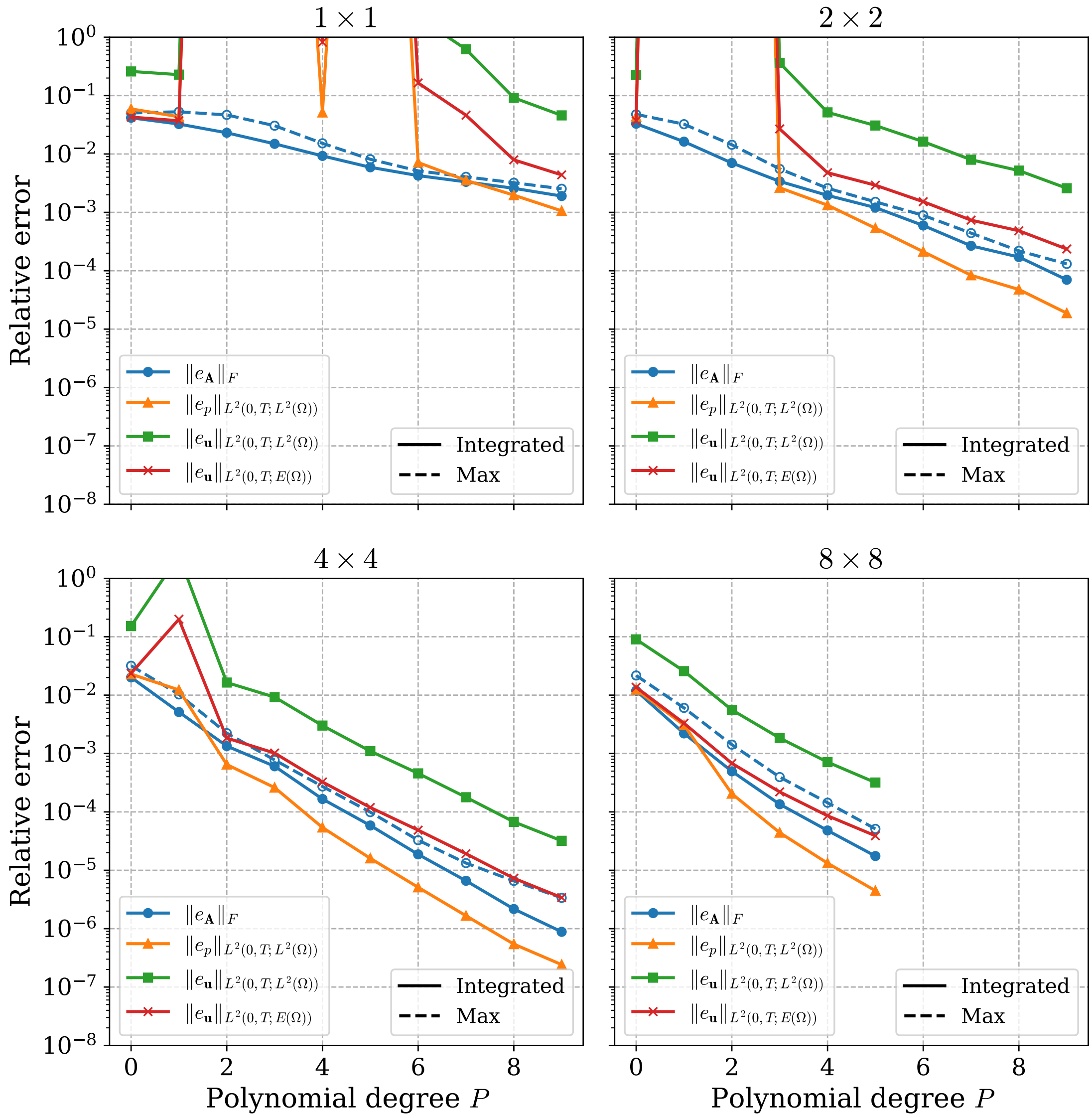

Figure 5.3.16: *Square with Gaussian permeability (transient).* Relative operator and solution errors obtained from the L^2-fit as functions of the polynomial degree P for different parameter-domain splittings. Errors are computed using Gauss–Lobatto quadrature with 256 integration points over the parameter domain.

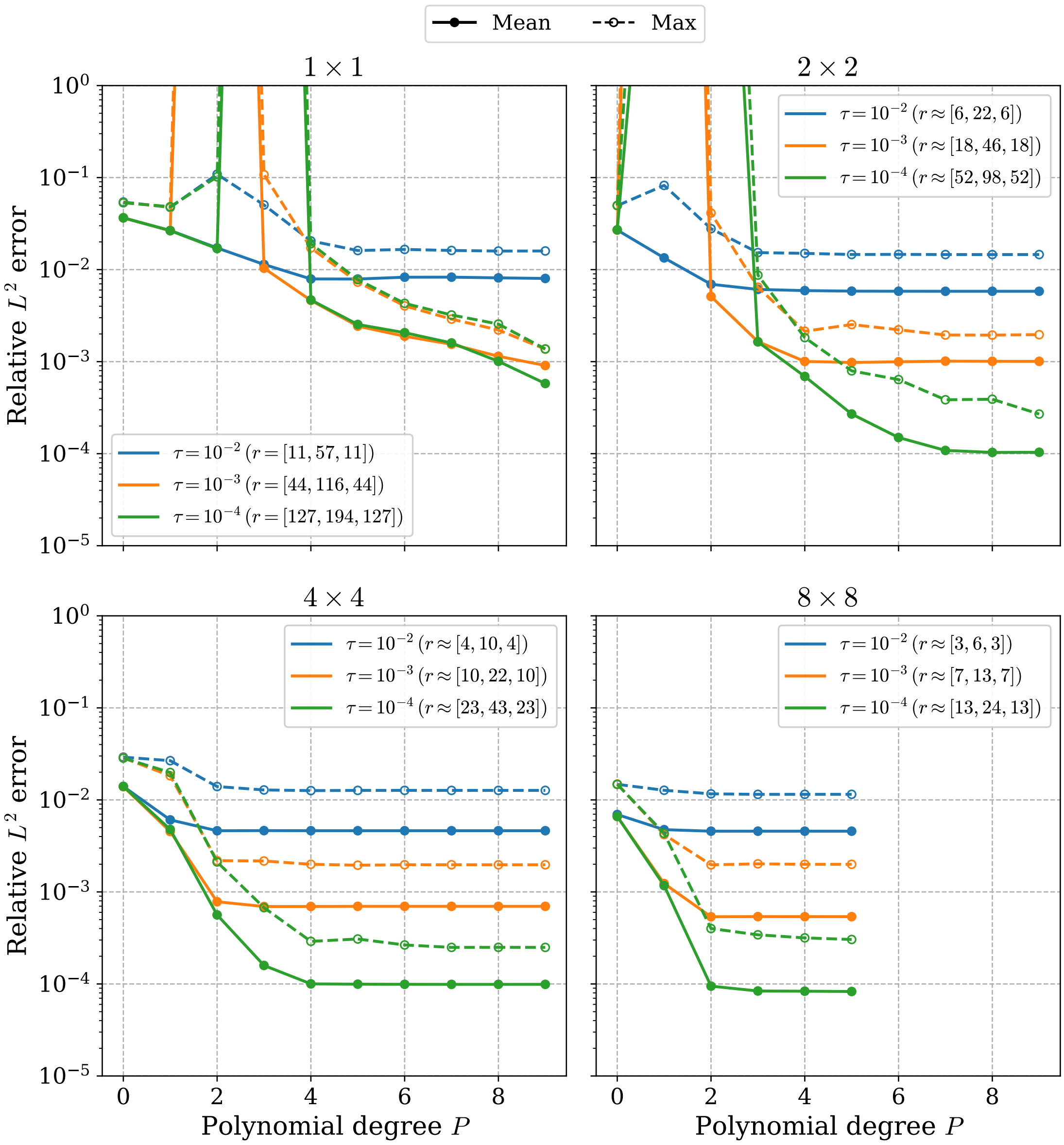

Figure 5.3.17: *Square with Gaussian permeability (transient).* Relative pressure error in the $L^2(0, T; L^2(\Omega))$ norm for the ROM as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. Results are shown for four parameter-domain splittings using equispaced square partitions.

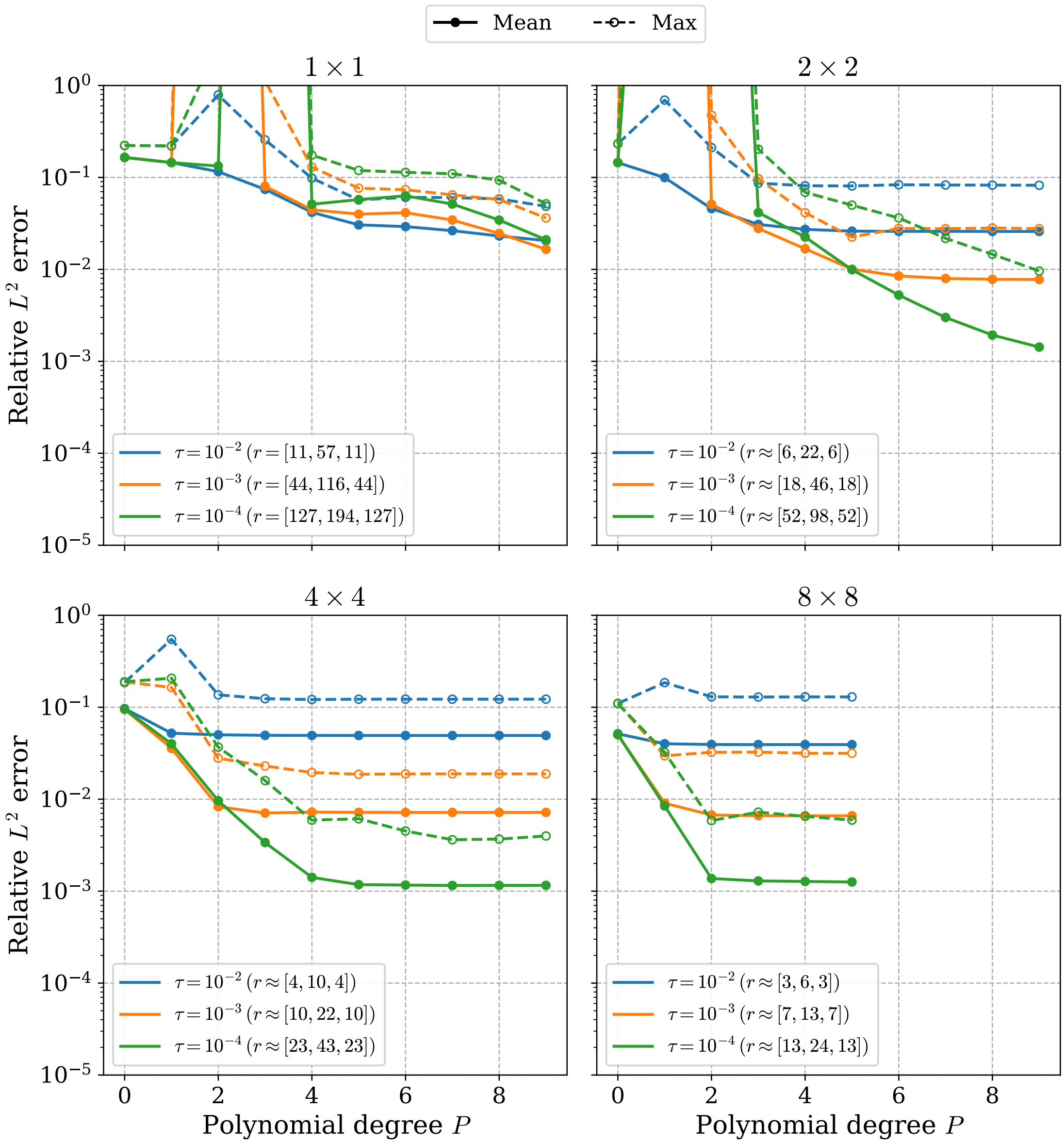

Figure 5.3.18: *Square with Gaussian permeability (transient).* Relative velocity error in the $L^2(0, T; L^2(\Omega))$ norm for the ROM as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. Results are shown for four parameter-domain splittings using equispaced square partitions.

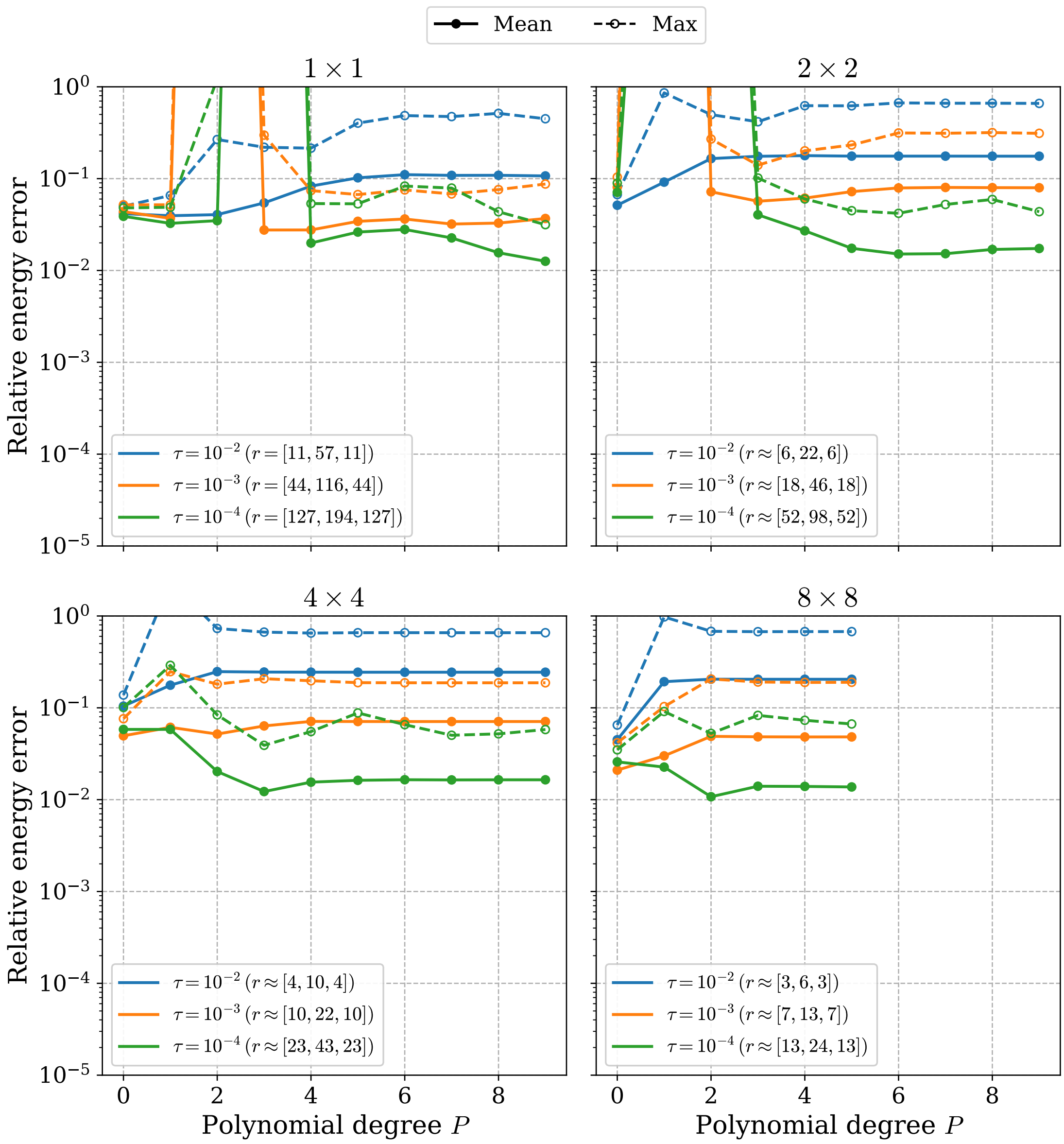

Figure 5.3.19: *Square with Gaussian permeability (transient).* Relative velocity error in the $L^2(0,T;E(\Omega))$ norm for the ROM as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. Results are shown for four parameter-domain splittings using equispaced square partitions.

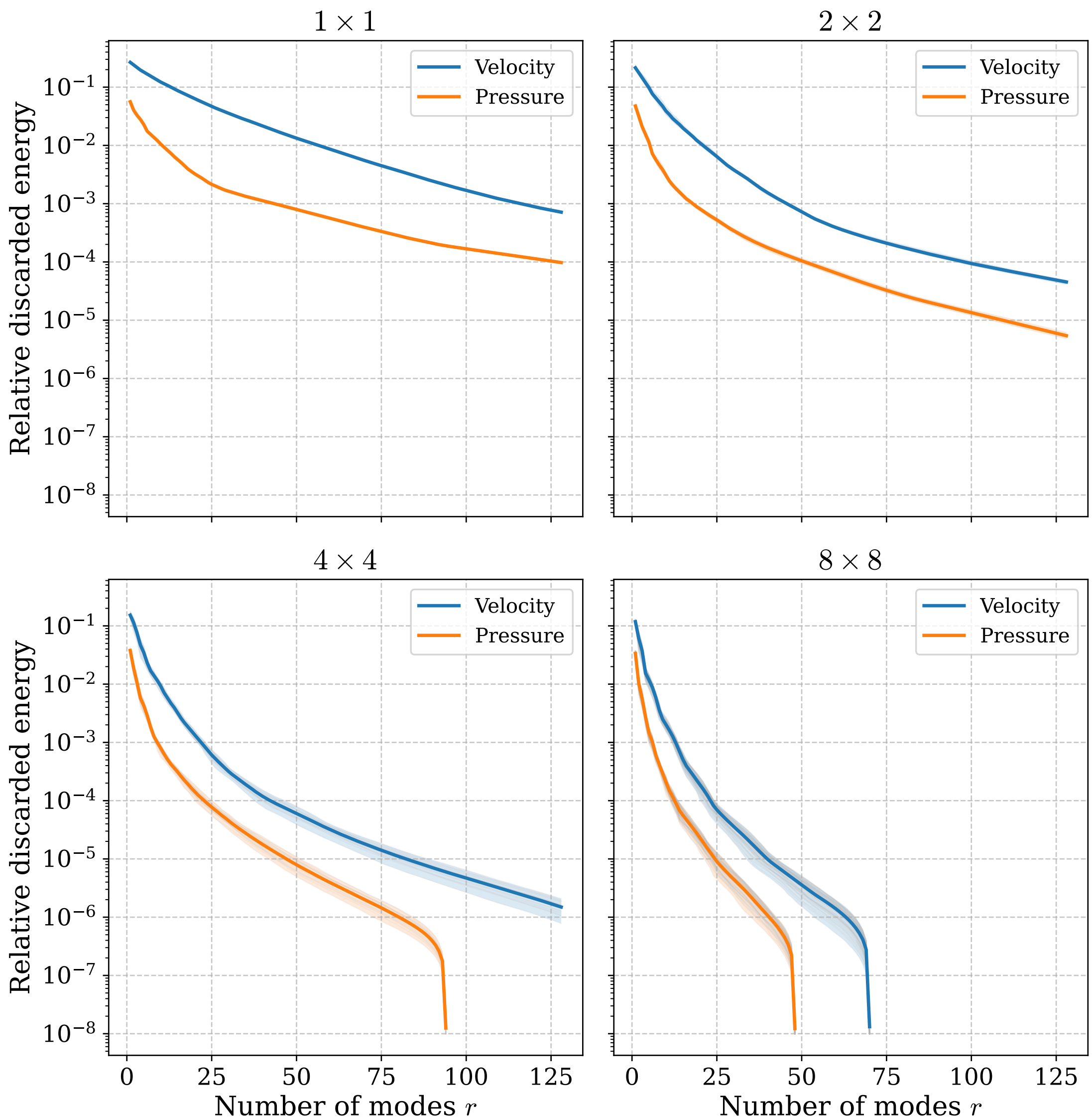

Figure 5.3.20: *Square with Gaussian permeability (transient).* Relative discarded POD energy for the pressure and velocity reduced bases. Results are shown for four parameter-domain splittings using equispaced square partitions. For splittings with multiple subdomains, the shaded envelopes represent the range of discarded energies obtained from the corresponding local reduced bases.

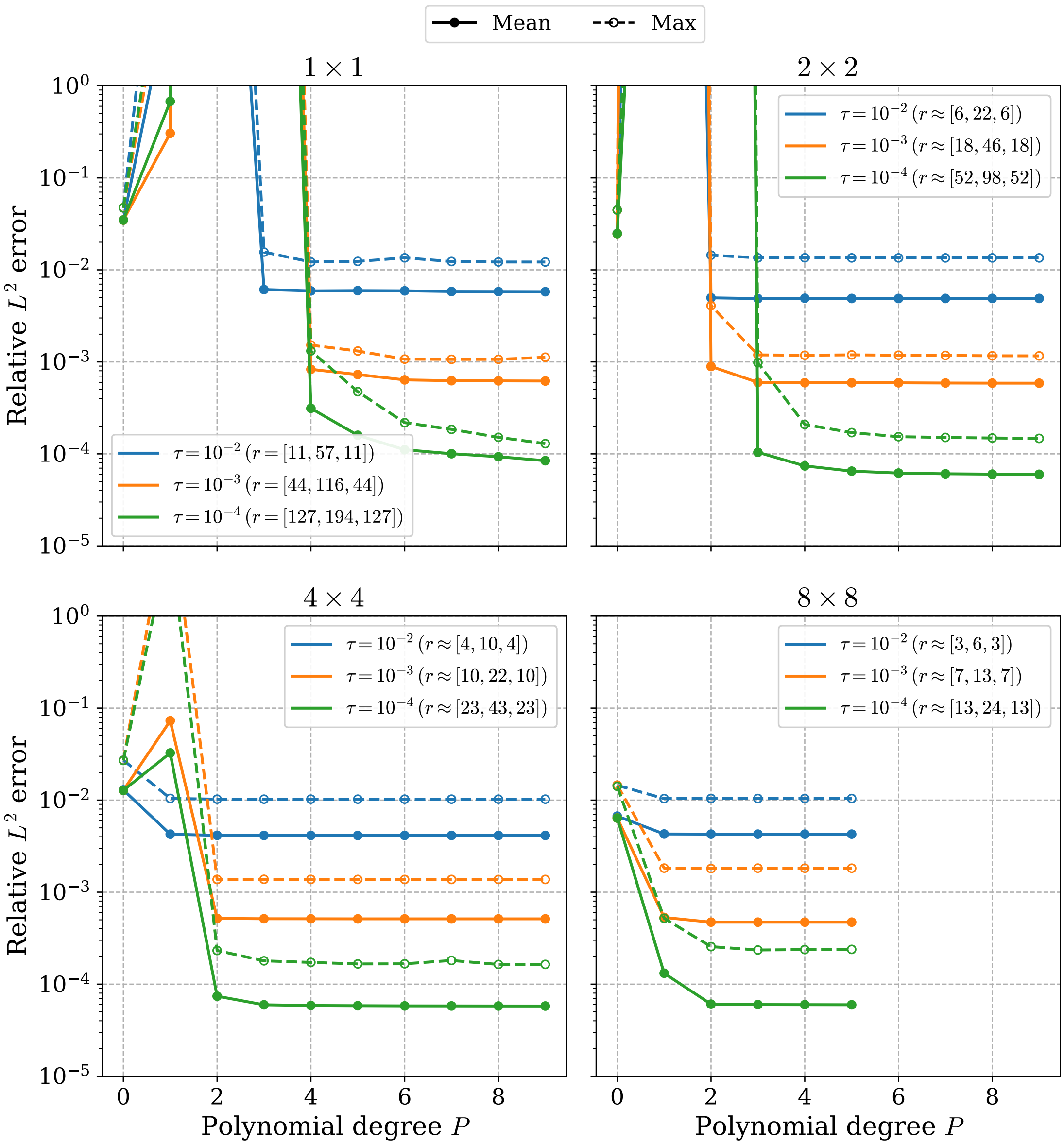

Figure 5.3.21: *Square with Gaussian permeability (transient).* Relative pressure error in the $L^2(0, T; L^2(\Omega))$ norm for the ROM with CoSTA as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. The CoSTA correction is computed using ridge regression with $\lambda = 100$. Results are shown for four parameter-domain splittings using equispaced square partitions.

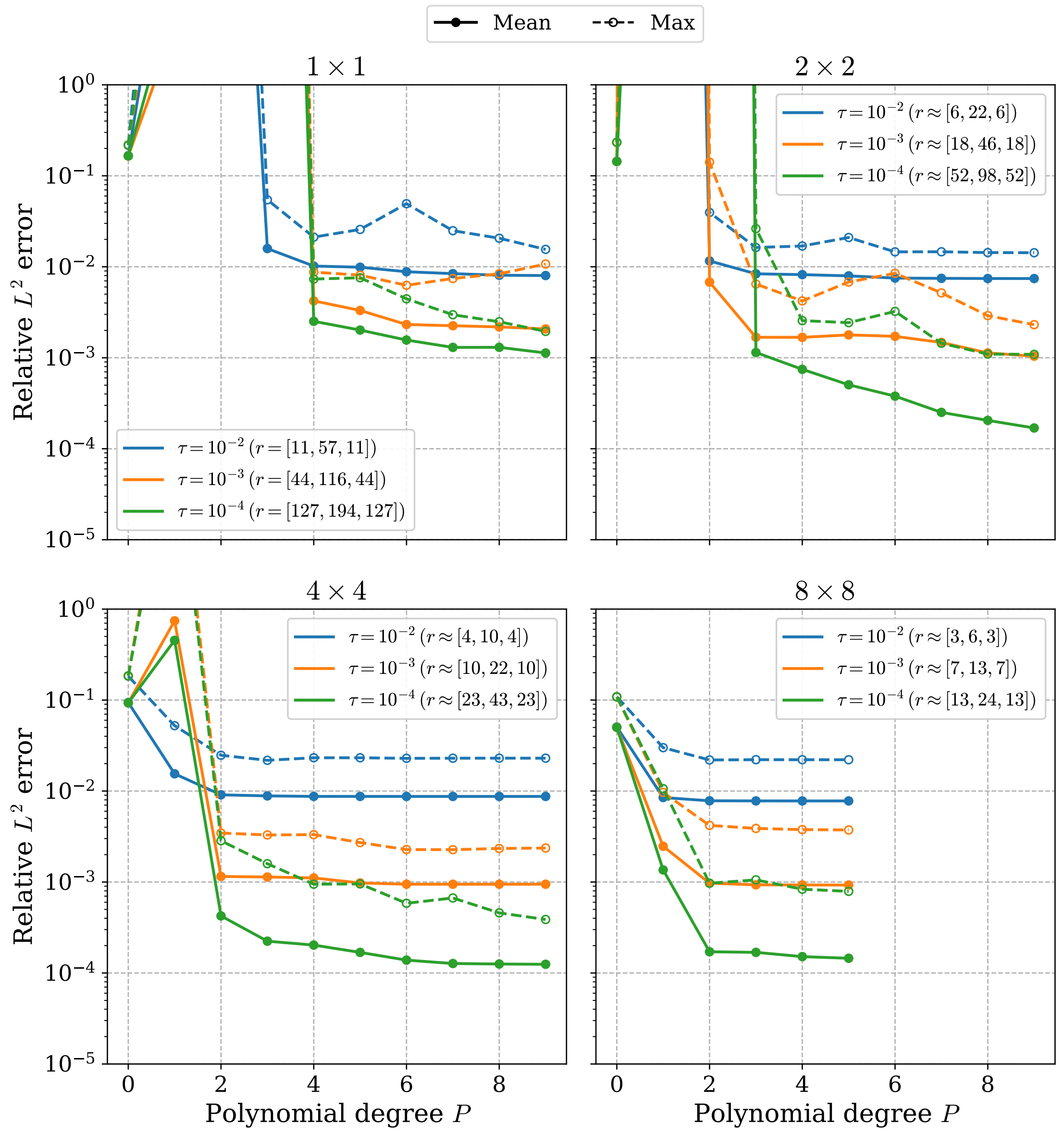

Figure 5.3.22: *Square with Gaussian permeability (transient).* Relative velocity error in the $L^2(0, T; L^2(\Omega))$ norm for the ROM with CoSTA as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. The CoSTA correction is computed using ridge regression with $\lambda = 100$. Results are shown for four parameter-domain splittings using equispaced square partitions.

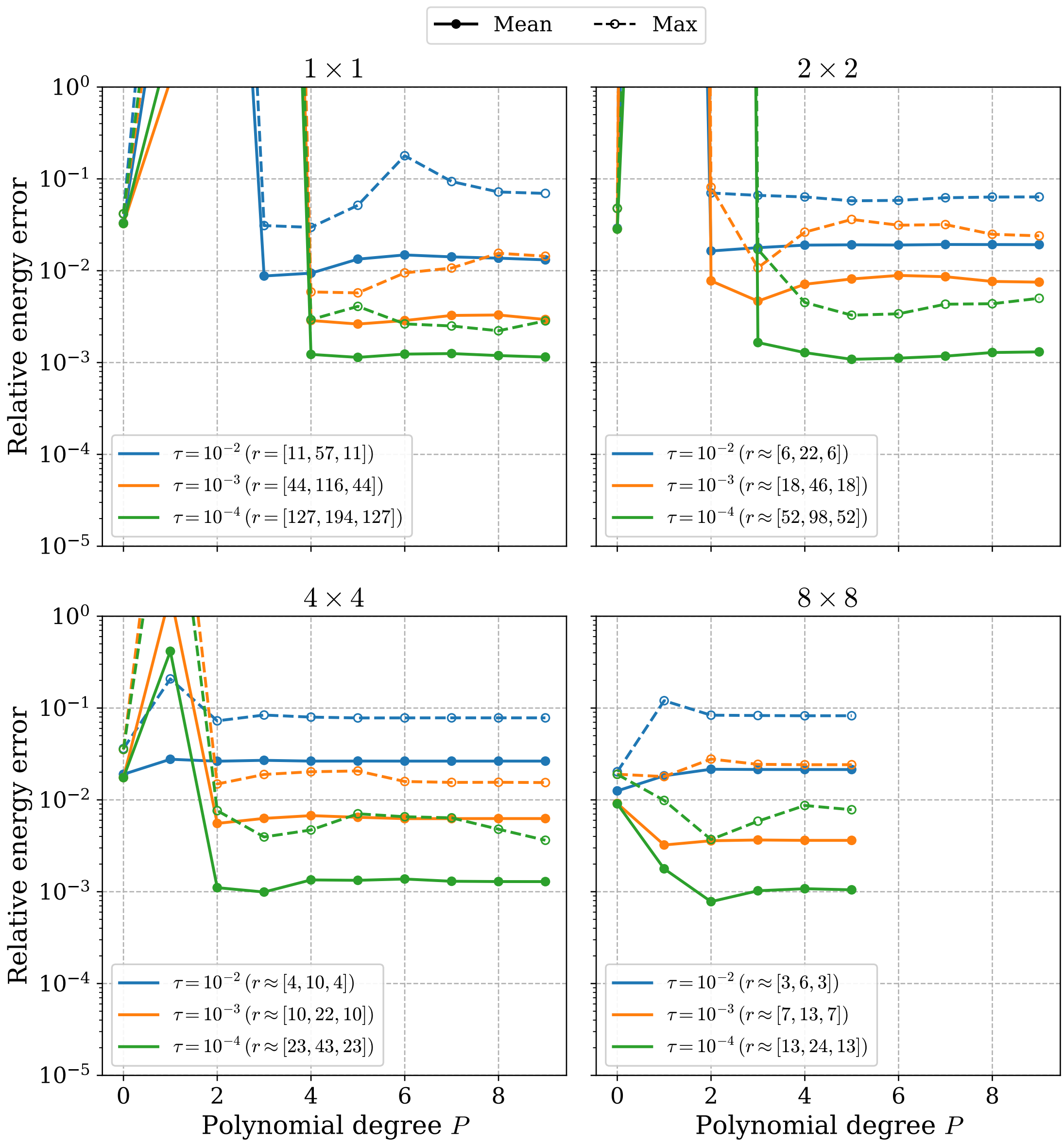

Figure 5.3.23: *Square with Gaussian permeability (transient).* Relative velocity error in the $L^2(0, T; E(\Omega))$ norm for the ROM with CoSTA as a function of the polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The ROM is trained using 2304 parameter samples and evaluated on 256 randomly sampled test points. The CoSTA correction is computed using ridge regression with $\lambda = 100$. Results are shown for four parameter-domain splittings using equispaced square partitions.

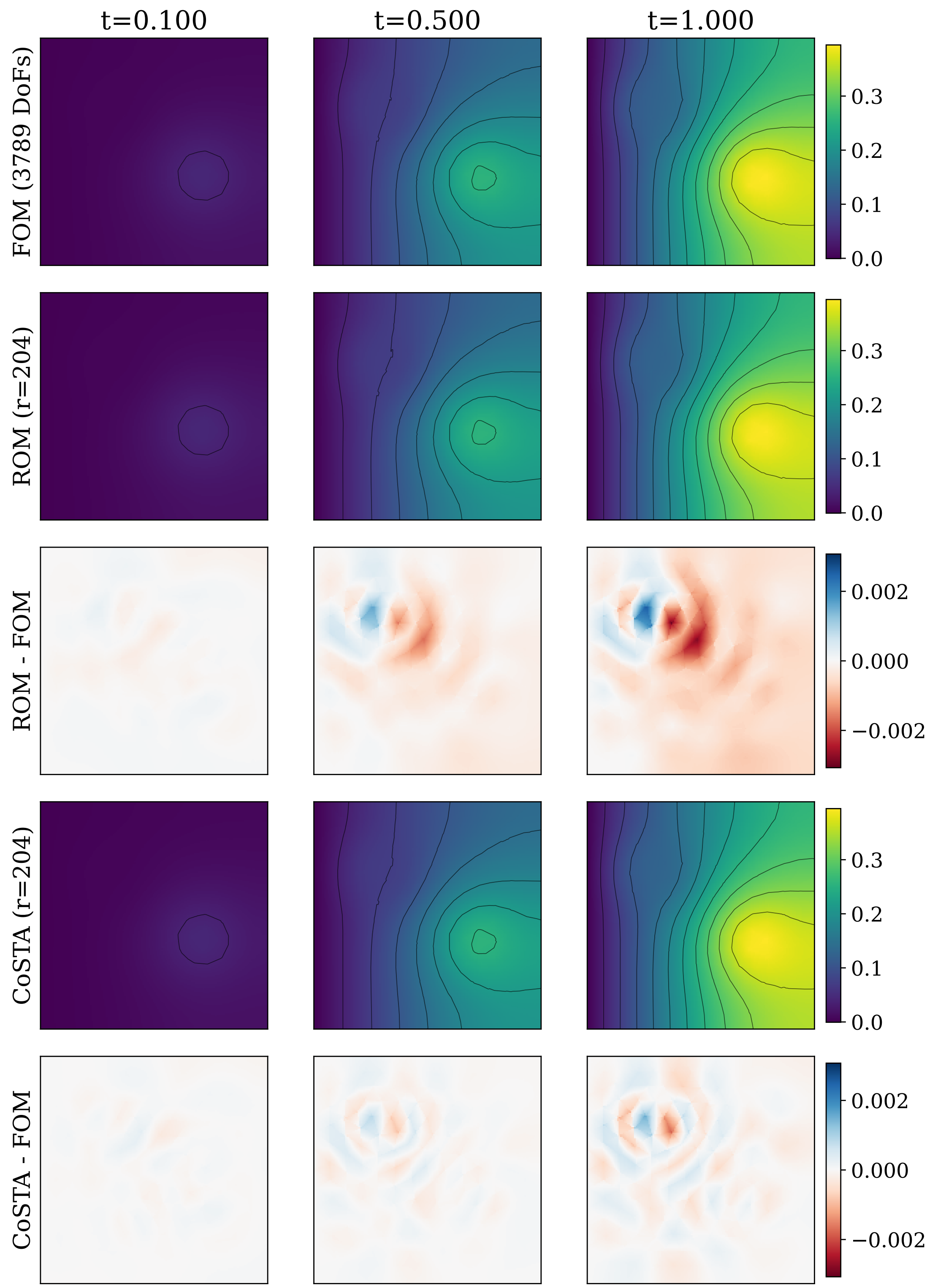

Figure 5.3.24: *Square with Gaussian permeability (transient).* Visual comparison of the full-order, baseline ROM, and CoSTA-corrected pressure solutions for $\boldsymbol{\mu} = (0.3, 0.7)$, together with the corresponding pointwise error fields.

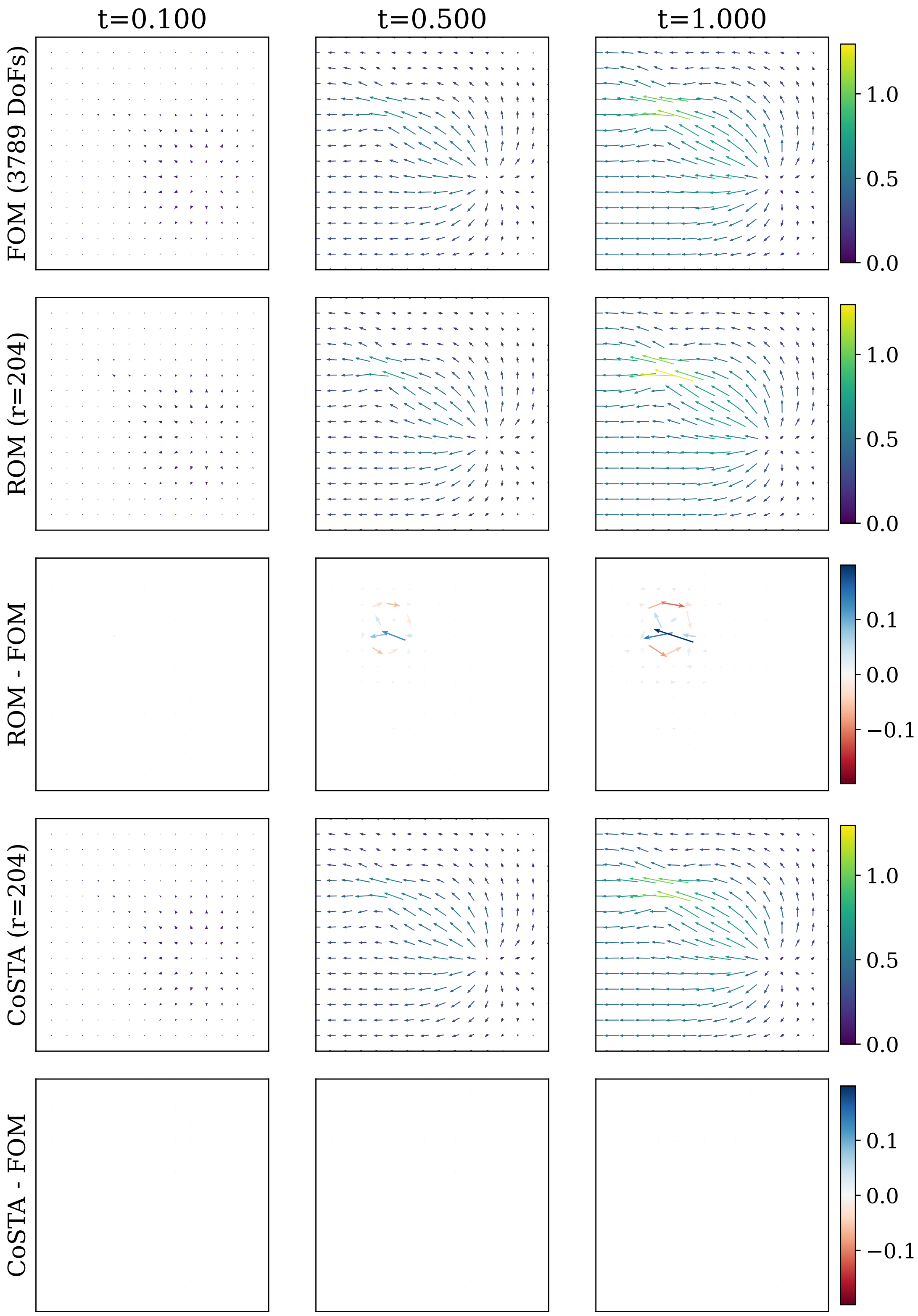

Figure 5.3.25: *Square with Gaussian permeability (transient).* Visual comparison of the full-order, baseline ROM, and CoSTA-corrected velocity solutions for $\boldsymbol{\mu} = (0.3, 0.7)$, together with the corresponding pointwise error fields.

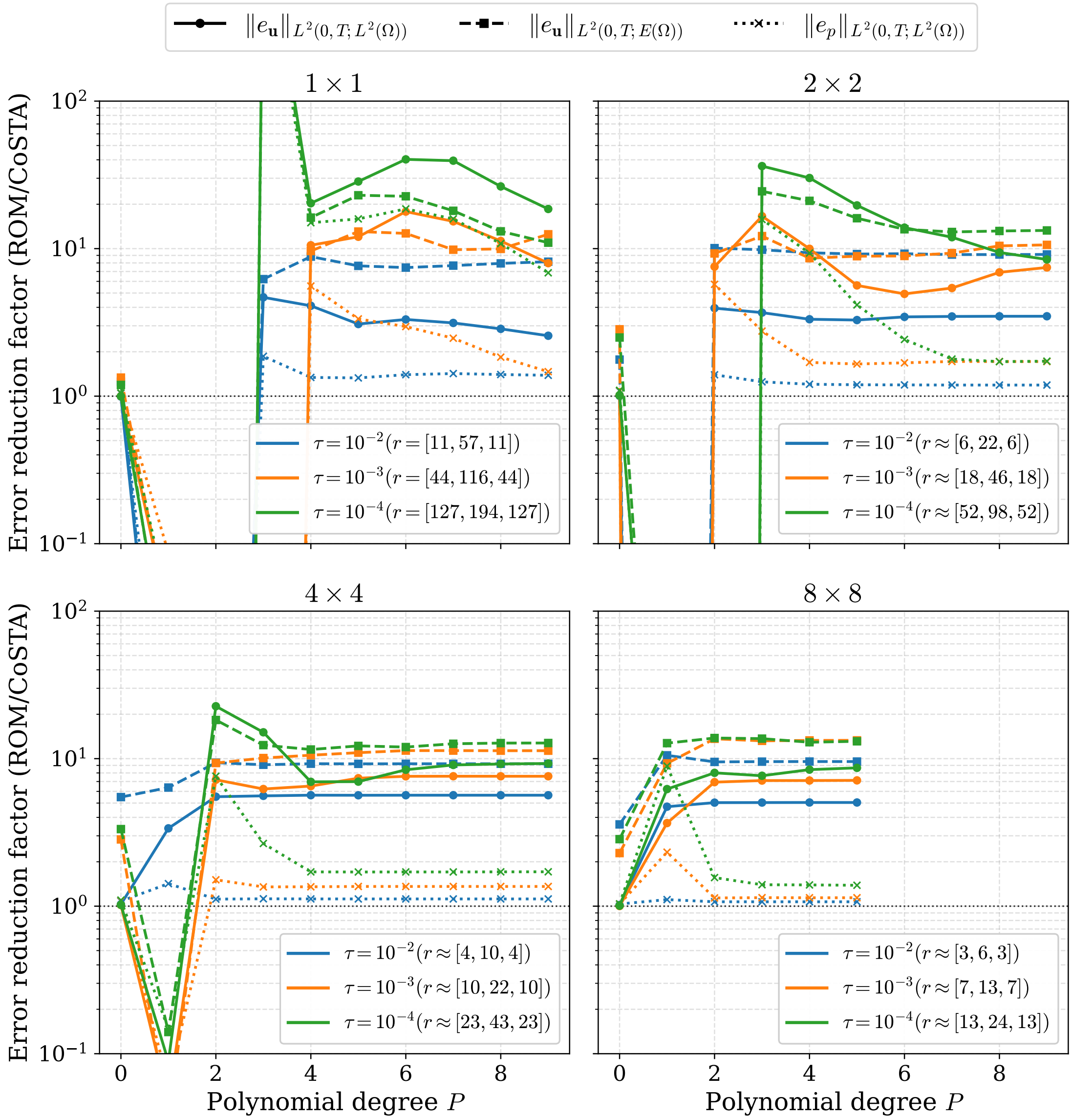

Figure 5.3.26: *Square with Gaussian permeability (transient).* Reduction factor of the mean ROM error obtained with CoSTA as a function of polynomial degree P and reduced dimension $r = [r_p, r_{\mathbf{u}}, r_s]$. The reduction factor is computed as the ratio between the mean baseline error and the mean CoSTA error. Values above 10^0 indicate improved accuracy.

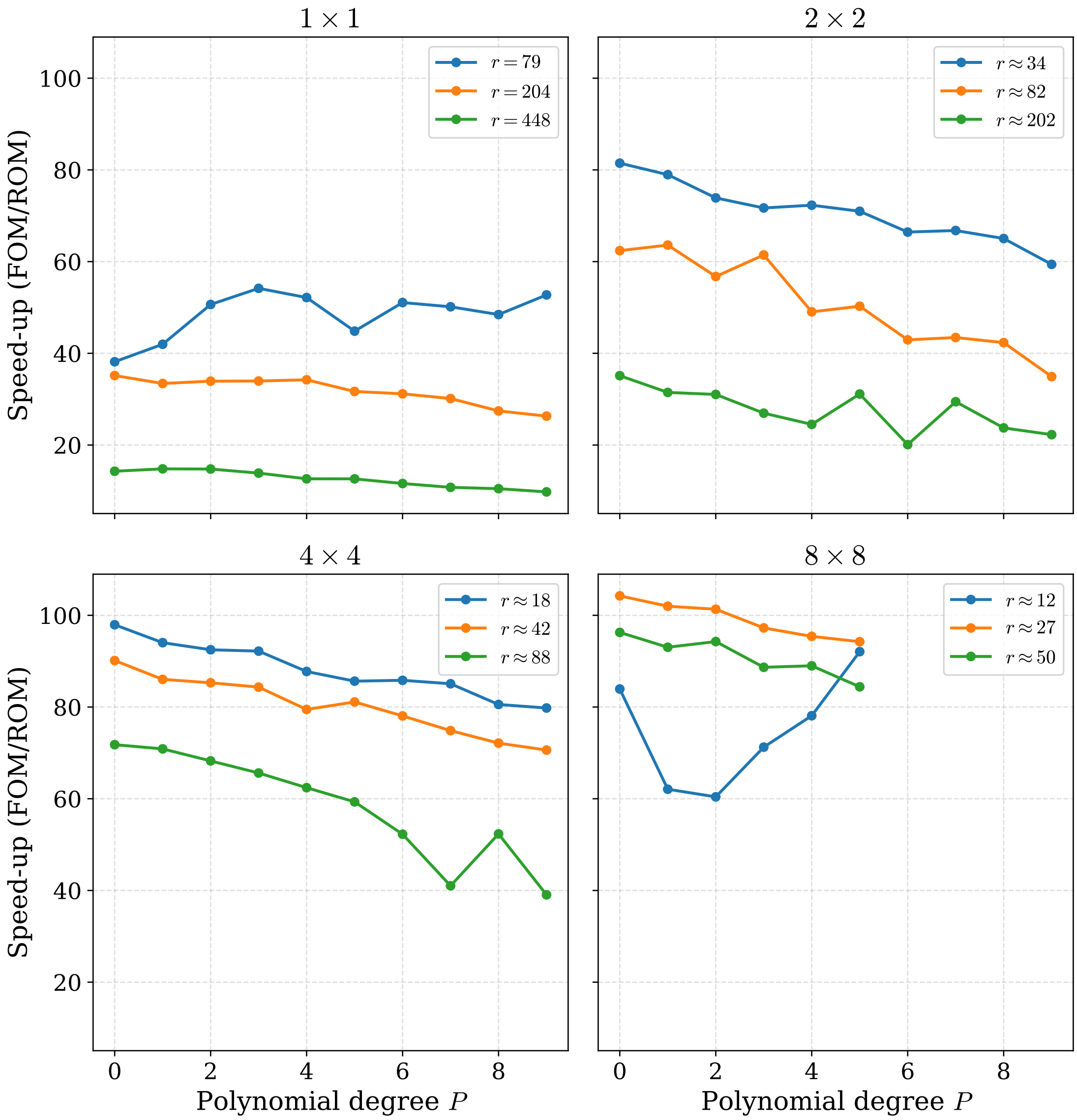

Figure 5.3.27: *Square with Gaussian permeability (transient).* Computational speed-up of the baseline ROM as a function of the polynomial degree P and total reduced dimension r. Results are shown for four parameter-domain splittings using equispaced square partitions.

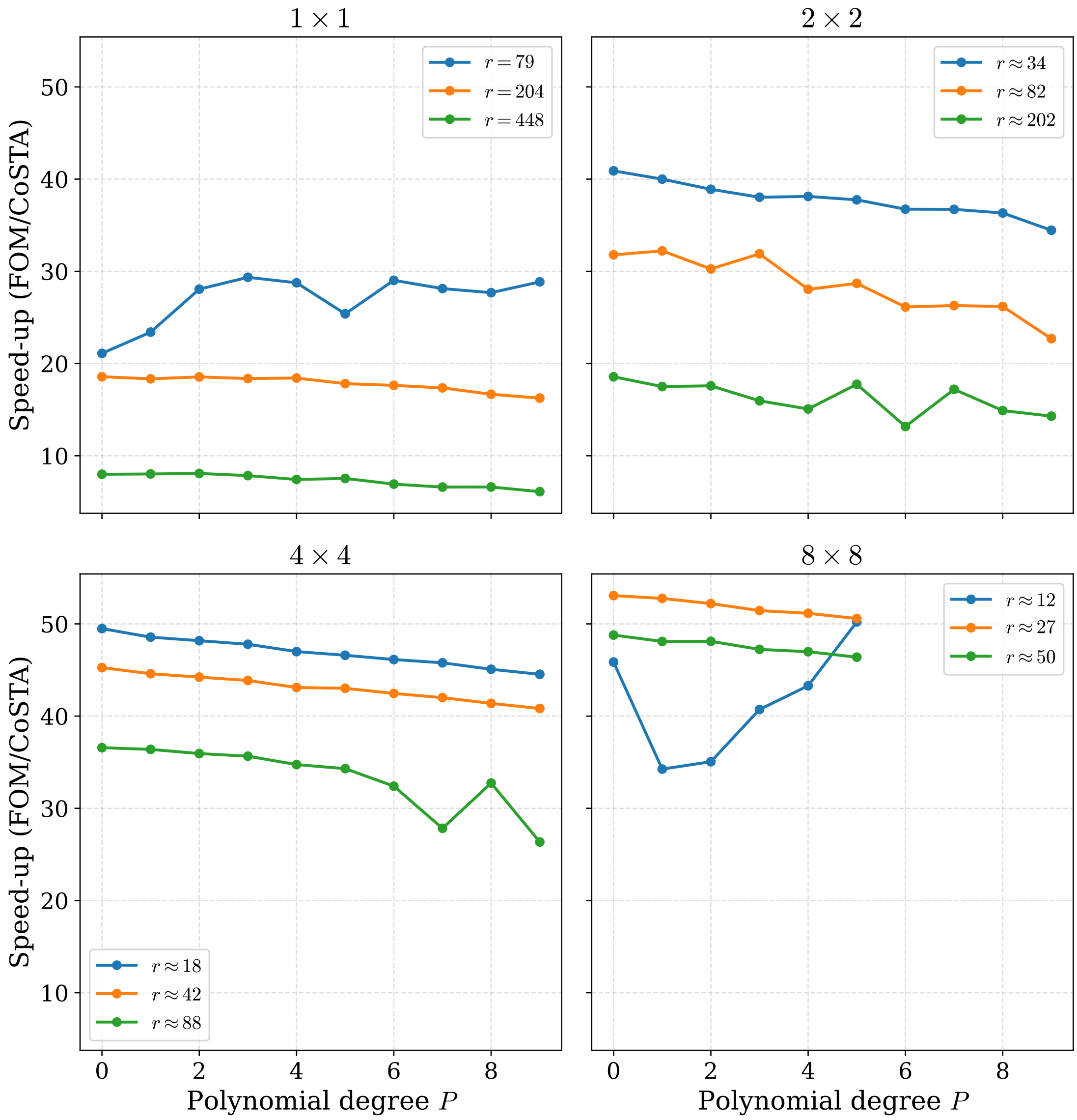

Figure 5.3.28: *Square with Gaussian permeability (transient).* Computational speed-up of the ROM with CoSTA as a function of the polynomial degree P and total reduced dimension r. Results are shown for four parameter-domain splittings using equispaced square partitions.

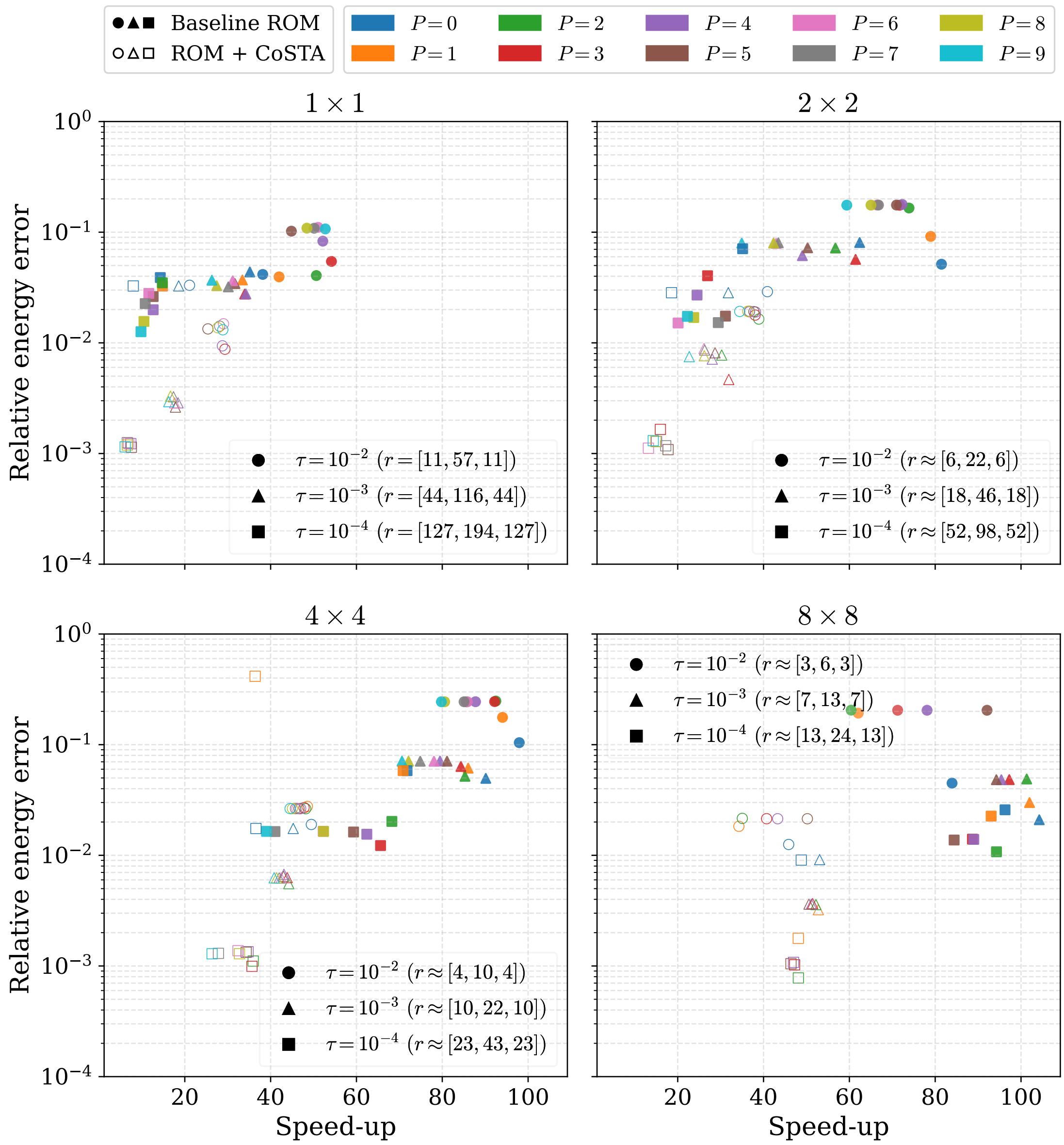

Figure 5.3.29: *Square with Gaussian permeability (transient).* Speed-up–error scatter plot for baseline ROM (filled markers) and ROM with CoSTA (hollow markers), with color indicating polynomial degree P and marker shape indicating τ. The x-axis shows online speed-up relative to FOM, while the y-axis shows the relative velocity error measured in the energy norm. Points lower and further to the right indicate a better accuracy–efficiency trade-off.

CHAPTER

SIX

CONCLUSIONS

In this thesis, reduced-order modeling techniques for parameterized transient Darcy flow problems have been investigated. The finite element method was employed as the high-fidelity model, and reduced basis methods based on Proper Orthogonal Decomposition (POD) and Galerkin projection were developed in order to construct computationally efficient surrogate models.

Particular attention was devoted to problems with non-affine parameter dependence. To address this challenge, a lightly intrusive polynomial approximation procedure based on weighted least-squares fitting was employed to recover an approximate affine decomposition of the system operators. In addition, a simple grid-based parameter-domain splitting strategy was investigated in order to improve approximation quality in parameter regions with strongly varying solution behavior.

Two parameterized Darcy flow problems were studied numerically: a geometrically parameterized moving-corner problem and a problem involving localized Gaussian permeability variations. Both stationary and transient settings were considered. The numerical experiments demonstrated that the reduced-order models achieved substantial reductions in computational cost while maintaining good approximation accuracy. The results further showed that parameter-domain splitting significantly improved the approximation quality for strongly non-affine problems.

For both numerical examples, the transient setting generally required a larger number of reduced modes in order to achieve approximation errors comparable to those obtained in the stationary case. This is expected, since the reduced basis must capture both temporal and parameter-dependent solution variations simultaneously.

The Gaussian permeability example proved significantly more challenging for the reduced-order approximation than the moving-corner problem. In particular, substantially larger reduced dimensions were required in order to obtain satisfactory accuracy. A likely explanation is the highly localized nature of the parameter dependence, which leads to solution features that are more difficult to represent efficiently using a global reduced basis.

The thesis also investigated the Corrective Source Term Approach (CoSTA), which combines physics-based reduced-order models with data-driven corrections. The numerical experiments indicated that CoSTA-based corrections can significantly reduce reduced-order errors, particularly in parameter regimes where the reduced basis approximation becomes less accurate.

Several directions for future work remain. In particular, more advanced parameter-domain partitioning strategies and adaptive sampling procedures could be investigated. It would also be of interest to extend the framework to more complex flow problems, higher-dimensional parameter spaces, and experimental or measurement-based training data within the CoSTA framework.

Use of Artificial Intelligence Tools*

In the work on this thesis, I have used artificial intelligence (AI) as a tool in the following way: I used ChatGPT to assist with idea development and structuring of the text. ChatGPT was also used to support literature search by helping to identify relevant topics, keywords, and potential sources. In addition, ChatGPT was used to generate parts of computer code, including plotting code, and to assist with the restructuring and refactoring of existing code. ChatGPT was further used to suggest formulations and provide linguistic improvements. All content, including text, code, and referenced literature, has been reviewed, edited, verified, and academically assessed by me. ChatGPT was not used to generate the project in its entirety or to independently solve the tasks. I have not uploaded any personal data or sensitive information to the AI tool. I am solely responsible for the academic content of this thesis.

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APPENDICES